

REHABILITATE CLIMBING FACILITIES AT CAMP MUIR

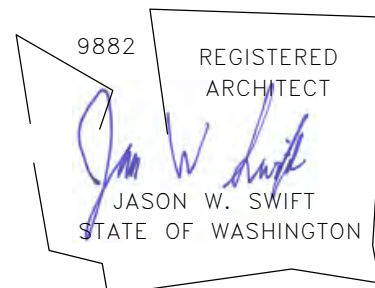
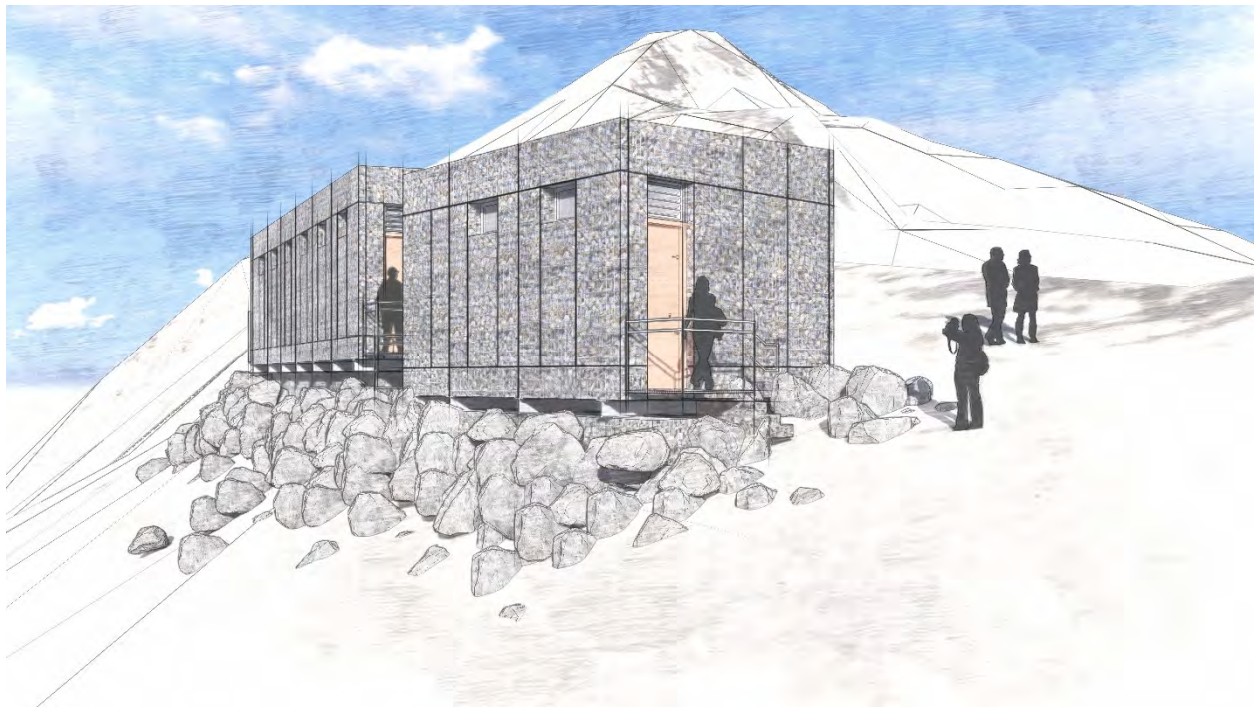
TITLE II DESIGN DEVELOPMENT AND CONSTRUCTION DOCUMENTS SERVICES
MOUNT RAINIER NATIONAL PARK, PIERCE COUNTY, WASHINGTON

100% CONSTRUCTION DOCUMENTS - SPECIFICATION

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for double-sided printing

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END OF SECTION

**SECTION 011000
SUMMARY**

PART 1 GENERAL

1.01 PROJECT

- A. Project Name: Rehabilitate Climbing Facilities at Camp Muir
- B. Owner's Name: National Park Service.

1.02 DESCRIPTION OF ALTERATIONS WORK

- A. Scope generally includes removal of an existing modular building and replacement with two new climbing shelters, plus associated site stabilization work.
- B. Scope of demolition and removal work is indicated on drawings and specified in Section 024100.

1.03 CONTRACTOR USE OF SITE

- A. Construction Operations: Limited to areas noted on Drawings.
 - 1. Locate and conduct construction activities in ways that will limit disturbance to site.
- B. Arrange use of site to allow:
 - 1. Access through the site by guided climbing companies and members of the public.
- C. Provide access to and from site as required by law and by Owner:
 - 1. Do not obstruct roadways, sidewalks, or other public ways without permit.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

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**SECTION 012500
SUBSTITUTION PROCEDURES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Procedural requirements for proposed substitutions.

1.02 DEFINITIONS

- A. Substitutions: Changes from Contract Documents requirements proposed by Contractor to materials, products, assemblies, and equipment.
 - 1. Substitutions for Cause: Proposed due to changed Project circumstances beyond Contractor's control.
 - a. Unavailability.
 - b. Regulatory changes.
 - 2. Substitutions for Convenience: Proposed due to possibility of offering substantial advantage to the Project.
 - a. Substitution requests offering advantages solely to the Contractor will not be considered.

1.03 REFERENCE STANDARDS

- A. CSI/CSC Form 1.5C - Substitution Request (During the Bidding/Negotiating Stage); Current Edition.
- B. CSI/CSC Form 13.1A - Substitution Request (After the Bidding/Negotiating Phase); Current Edition.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 GENERAL REQUIREMENTS

- A. A Substitution Request for products, assemblies, materials, and equipment constitutes a representation that the submitter:
 - 1. Has investigated proposed product and determined that it meets or exceeds the quality level of the specified product, equipment, assembly, or system.
 - 2. Agrees to provide the same warranty for the substitution as for the specified product.
 - 3. Agrees to coordinate installation and make changes to other work that may be required for the work to be complete, with no additional cost to Owner.
 - 4. Waives claims for additional costs or time extension that may subsequently become apparent.
- B. Document each request with complete data substantiating compliance of proposed substitution with Contract Documents. Burden of proof is on proposer.
 - 1. Note explicitly any non-compliant characteristics.
- C. Content: Include information necessary for tracking the status of each Substitution Request, and information necessary to provide an actionable response.
 - 1. Forms indicated in the Project Manual are adequate for this purpose, and must be used.
- D. Limit each request to a single proposed substitution item.
 - 1. Submit an electronic document, combining the request form with supporting data into single, bookmarked document.

3.02 SUBSTITUTION PROCEDURES DURING PROCUREMENT

- A. Submittal Time Restrictions:
 - 1. Owner will consider requests for substitutions only if submitted at least 10 days prior to the date for receipt of bids.
- B. Submittal Form (before award of contract):

1. Submit substitution requests by completing CSI/CSC Form 1.5C - Substitution Request. See this form for additional information and instructions. Use only this form; other forms of submission are unacceptable.

3.03 SUBSTITUTION PROCEDURES DURING CONSTRUCTION

- A. Submittal Form (after award of contract):
 1. Submit substitution requests by completing CSI/CSC Form 13.1A - Substitution Request (After Bidding/Negotiating). See this form for additional information and instructions. Use only this form; other forms of submission are unacceptable.
- B. Submit request for Substitution for Cause within 14 days of discovery of need for substitution, but not later than 14 days prior to time required for review and approval by COR, in order to stay on approved project schedule.
- C. Submit request for Substitution for Convenience within 14 days of discovery of its potential advantage to the project, but not later than 14 days prior to time required for review and approval by COR, in order to stay on approved project schedule.
 1. In addition to meeting general documentation requirements, document how the requested substitution benefits the Owner through cost savings, time savings, greater energy conservation, or in other specific ways.
 2. Document means of coordinating of substitution item with other portions of the work, including work by affected subcontractors.
 3. Bear the costs engendered by proposed substitution of:
 - a. Owner's compensation to the Design Team for any required redesign, time spent processing and evaluating the request.
- D. Substitutions will not be considered under one or more of the following circumstances:
 1. When they are indicated or implied on shop drawing or product data submittals, without having received prior approval.
 2. Without a separate written request.

3.04 RESOLUTION

- A. COR may request additional information and documentation prior to rendering a decision. Provide this data in an expeditious manner.
- B. COR will notify Contractor in writing of decision to accept or reject request.

3.05 ACCEPTANCE

- A. Accepted substitutions change the work of the Project. They will be documented and incorporated into work of the project by Change Order, Construction Change Directive, Architectural Supplementary Instructions, or similar instruments provided for in the Conditions of the Contract.

END OF SECTION

**SECTION 013000
ADMINISTRATIVE REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Web-based project software service.
- B. Preconstruction meeting.
- C. Site mobilization meeting.
- D. Progress meetings.
- E. Construction progress schedule.
- F. Contractor's daily reports.
- G. Progress photographs.
- H. Coordination drawings.
- I. Submittals for review, information, and project closeout.
- J. Number of copies of submittals.
- K. Requests for Interpretation (RFI) procedures.
- L. Submittal procedures.

1.02 RELATED REQUIREMENTS

- A. Section 016000 - Product Requirements: General product requirements.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 WEB-BASED PROJECT SOFTWARE SERVICE

- A. Web-Based Project Software Service: Provide, administer, and use web-based project software to host and manage project communication and documentation.
 - 1. Include, at minimum, the following features:
 - a. Project directory, including Owner, Contractor, subcontractors, COR, COR's consultants, and other entities involved in the project. Include names of contact persons and contact information for each entity.
 - b. Access control for each entity and for each workflow process to determine each entity's digital rights to create, modify, view, and print documents.
 - c. Workflow planning, allowing customization of workflow for each project entity.
 - d. Creation, logging, tracking, and notification for project communications.
 - e. Tracking of project communication statuses in real time, including timestamped response log.
 - f. Procedures for viewing PDFs or similar file formats, allowing markups by each entity. Provide security features to lock markups against changes once submitted.
 - g. Processing and tracking of payment applications.
 - h. Processing and tracking of contract modifications.
 - i. Creation and distribution of meeting minutes.
 - j. Document management for drawings, specifications, and coordination drawings, including revision control.
 - k. Management of construction progress photographs.
 - l. Mobile device compatibility.
 - m. Creation of data analytics reports.
 - n. Creation and export of editable logs for software functions. Provide Owner, COR, and COR's consultants with rights and ability to download logs when requested.
 - 2. Cost: Pay cost of service. Include the cost of the service in the contract sum.

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3. Provide up to 20 user licenses for use by Owner, COR, COR's consultants, and other entities involved in the project.
4. Comply with the software service's current published licensing agreements.
5. Training: Provide one-hour, web-based training session for users of software service. Further training is the responsibility of the user.
 - a. Representatives of Owner are scheduled and included in this training.
6. Project Closeout: COR determines when to terminate the software service for the project and is responsible for obtaining archive copies of files for Owner.

3.02 PRECONSTRUCTION MEETING

- A. COR will schedule a meeting after Notice of Award.
- B. Attendance Required:
 1. Owner.
 2. COR.
 3. Contractor.
 4. Owner's Consultants.
- C. Agenda:
 1. Execution of Owner-Contractor Agreement.
 2. Submission of executed bonds and insurance certificates.
 3. Distribution of Contract Documents.
 4. Submission of list of subcontractors, list of products, schedule of values, and progress schedule.
 5. Designation of personnel representing the parties to Contract and COR.
 6. Procedures and processing of field decisions, submittals, substitutions, applications for payments, proposal request, Change Orders, and Contract closeout procedures.
 7. Scheduling.
- D. Record minutes and distribute copies within two days after meeting to participants, with copies to COR, Owner, participants, and those affected by decisions made.

3.03 SITE MOBILIZATION MEETING

- A. Schedule meeting at the Project site prior to Contractor occupancy.
- B. Attendance Required:
 1. Contractor.
 2. Owner.
 3. COR.
 4. Special consultants.
 5. Contractor's superintendent.
 6. Major subcontractors.
- C. Agenda:
 1. Use of premises by Owner and Contractor.
 2. Owner's requirements.
 3. Survey and building layout.
 4. Security and housekeeping procedures.
 5. Schedules.
 6. Application for payment procedures.
 7. Procedures for testing.
 8. Procedures for maintaining record documents.
 9. Inspection and acceptance of equipment put into service during construction period.
- D. Record minutes and distribute copies within two days after meeting to participants, with two copies to COR, Owner, participants, and those affected by decisions made.

3.04 PROGRESS MEETINGS

- A. Schedule and administer meetings throughout progress of the work at maximum bi-monthly intervals.
- B. Make arrangements for meetings, prepare agenda with copies for participants, preside at meetings.
- C. Attendance Required:
 - 1. Contractor.
 - 2. Owner.
 - 3. COR.
 - 4. Special consultants.
 - 5. Contractor's superintendent.
- D. Agenda:
 - 1. Review minutes of previous meetings.
 - 2. Review of work progress.
 - 3. Field observations, problems, and decisions.
 - 4. Identification of problems that impede, or will impede, planned progress.
 - 5. Review of submittals schedule and status of submittals.
 - 6. Review of RFIs log and status of responses.
 - 7. Review of off-site fabrication and delivery schedules.
 - 8. Maintenance of progress schedule.
 - 9. Corrective measures to regain projected schedules.
 - 10. Planned progress during succeeding work period.
 - 11. Maintenance of quality and work standards.
 - 12. Effect of proposed changes on progress schedule and coordination.
 - 13. Other business relating to work.
- E. Record minutes and distribute copies within two days after meeting to participants, with copies to COR, Owner, participants, and those affected by decisions made.

3.05 CONSTRUCTION PROGRESS SCHEDULE

- A. Within 10 days after date of the Agreement, submit preliminary schedule defining planned operations for the first 60 days of work, with a general outline for remainder of work.
- B. If preliminary schedule requires revision after review, submit revised schedule within 10 days.
- C. Within 20 days after review of preliminary schedule, submit draft of proposed complete schedule for review.
 - 1. Include written certification that major contractors have reviewed and accepted proposed schedule.
- D. Within 10 days after joint review, submit complete schedule.
- E. Submit updated schedule with each Application for Payment.

3.06 DAILY CONSTRUCTION REPORTS

- A. Include only factual information. Do not include personal remarks or opinions regarding operations and/or personnel.
- B. Prepare a daily construction report recording the following information concerning events at Project site and project progress:
 - 1. Date.
 - 2. High and low temperatures, and general weather conditions.
 - 3. List of subcontractors at Project site.
 - 4. Approximate count of personnel at Project site.
 - 5. Major equipment at Project site.
 - 6. Material deliveries.
 - 7. Safety, environmental, or industrial relations incidents.

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8. Meetings and significant decisions.
9. Stoppages, delays, shortages, and losses. Include comparison between scheduled work activities (in Contractor's most recently updated and published schedule) and actual activities. Explain differences, if any. Note days or periods when no work was in progress and explain the reasons why.
10. Change Orders received and implemented.
11. Testing and/or inspections performed.
12. Signature of Contractor's authorized representative.

3.07 PROGRESS PHOTOGRAPHS

- A. Submit new photographs weekly, within 3 days after being taken.
- B. Photography Type: Digital; electronic files.
- C. Provide photographs of site and construction throughout progress of work produced by an experienced photographer, acceptable to COR.
- D. In addition to periodic, recurring views, take photographs of each of the following events:
 1. Completion of site clearing.
 2. Excavations in progress.
 3. Foundations in progress and upon completion.
 4. Enclosure of building, upon completion.
 5. Final completion, minimum of twenty (20) photos.
- E. Views:
 1. Provide photographs from four cardinal views at each specified time, until date of Substantial Completion.
 2. Consult with COR for instructions on views required.
 3. Provide factual presentation.
 4. Provide correct exposure and focus, high resolution and sharpness, maximum depth of field, and minimum distortion.
- F. Digital Photographs: 24 bit color, minimum resolution of 1600 by 1200 ("2 megapixel"), in JPG format; provide files unaltered by photo editing software.
 1. Delivery Medium: Via email in addition to two copies of project record thumb drive.
 2. File Naming: Include project identification, date and time of view, and view identification.

3.08 COORDINATION DRAWINGS

- A. Review drawings prior to submission to COR.

3.09 REQUESTS FOR INTERPRETATION (RFI)

- A. Definition: A request seeking one of the following:
 1. An interpretation, amplification, or clarification of some requirement of Contract Documents arising from inability to determine from them the exact material, process, or system to be installed; or when the elements of construction are required to occupy the same space (interference); or when an item of work is described differently at more than one place in Contract Documents.
 2. A resolution to an issue which has arisen due to field conditions and affects design intent.
- B. Whenever possible, request clarifications at the next appropriate project progress meeting, with response entered into meeting minutes, rendering unnecessary the issuance of a formal RFI.
- C. Preparation: Prepare an RFI immediately upon discovery of a need for interpretation of Contract Documents. Failure to submit a RFI in a timely manner is not a legitimate cause for claiming additional costs or delays in execution of the work.
 1. Prepare a separate RFI for each specific item.
 - a. Review, coordinate, and comment on requests originating with subcontractors and/or materials suppliers.
 - b. Do not forward requests which solely require internal coordination between subcontractors.

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2. Prepare in a format and with content acceptable to Owner.
3. Combine RFI and its attachments into a single electronic file. PDF format is preferred.
- D. Reason for the RFI: Prior to initiation of an RFI, carefully study all Contract Documents to confirm that information sufficient for their interpretation is definitely not included.
 1. Unacceptable Uses for RFIs: Do not use RFIs to request the following:
 - a. Approval of submittals (use procedures specified elsewhere in this section).
 - b. Approval of substitutions (see Section - 016000 - Product Requirements)
 2. Frivolous RFIs: Requests regarding information that is clearly indicated on, or reasonably inferable from, Contract Documents, with no additional input required to clarify the question. They will be returned without a response, with an explanatory notation.
- E. Content: Include identifiers necessary for tracking the status of each RFI, and information necessary to provide an actionable response.
 1. Official Project name and number, and any additional required identifiers established in Contract Documents.
 2. Owner's and Contractor's names.
 3. Discrete and consecutive RFI number, and descriptive subject/title.
 4. Issue date, and requested reply date.
 5. Reference to particular Contract Document(s) requiring additional information/interpretation. Identify pertinent drawing and detail number and/or specification section number, title, and paragraph(s).
 6. Annotations: Field dimensions and/or description of conditions which have engendered the request.
 7. Contractor's suggested resolution: A written and/or a graphic solution, to scale, is required in cases where clarification of coordination issues is involved, for example; routing, clearances, and/or specific locations of work shown diagrammatically in Contract Documents. If applicable, state the likely impact of the suggested resolution on Contract Time or the Contract Sum.
- F. Attachments: Include sketches, coordination drawings, descriptions, photos, submittals, and other information necessary to substantiate the reason for the request.
- G. RFI Log: Prepare and maintain a tabular log of RFIs for the duration of the project.
 1. Indicate current status of every RFI. Update log promptly and on a regular basis.
- H. Review Time: COR will respond and return RFIs to Contractor within ten calendar days of receipt. For the purpose of establishing the start of the mandated response period, RFIs received after 12:00 noon will be considered as having been received on the following regular working day.
 1. Response period may be shortened or lengthened for specific items, subject to mutual agreement, and recorded in a timely manner in progress meeting minutes.
- I. Responses: Content of answered RFIs will not constitute in any manner a directive or authorization to perform extra work or delay the project. If in Contractor's belief it is likely to lead to a change to Contract Sum or Contract Time, promptly issue a notice to this effect, and follow up with an appropriate Change Order request to Owner.
 1. Upon receipt of a response, promptly review and distribute it to all affected parties, and update the RFI Log.
 2. Notify COR within seven calendar days if an additional or corrected response is required by submitting an amended version of the original RFI, identified as specified above.

3.10 SUBMITTAL SCHEDULE

- A. Submit to COR for review a schedule for submittals in tabular format.
 1. Submit at the same time as the preliminary schedule specified in Section - 013216 - Construction Progress Schedule.
 2. Format schedule to allow tracking of status of submittals throughout duration of construction.

3. Arrange information to include scheduled date for initial submittal, specification number and title, submittal category (for review or for information), description of item of work covered, and role and name of subcontractor.
4. Account for time required for preparation, review, manufacturing, fabrication and delivery when establishing submittal delivery and review deadline dates.

3.11 SUBMITTALS FOR REVIEW

- A. When the following are specified in individual sections, submit them for review:
 1. Product data.
 2. Shop drawings.
 3. Samples for selection.
 4. Samples for verification.
- B. Submit to COR for review for the limited purpose of checking for compliance with information given and the design concept expressed in Contract Documents.
- C. Samples will be reviewed for aesthetic, color, or finish selection.
- D. After review, provide copies and distribute in accordance with SUBMITTAL PROCEDURES article below and for record documents purposes described in Section 017800 - Closeout Submittals.

3.12 SUBMITTALS FOR INFORMATION

- A. When the following are specified in individual sections, submit them for information:
 1. Design data.
 2. Certificates.
 3. Test reports.
 4. Inspection reports.
 5. Manufacturer's instructions.
 6. Manufacturer's field reports.
 7. Field engineering daily reports.
 8. Other types indicated.
- B. Submit for COR's knowledge as contract administrator or for Owner.

3.13 SUBMITTALS FOR PROJECT CLOSEOUT

- A. Submit Correction Punch List for Substantial Completion.
- B. Submit Final Correction Punch List for Substantial Completion.
- C. When the following are specified in individual sections, submit them at project closeout in compliance with requirements of Section 017800 - Closeout Submittals:
 1. Project record documents.
 2. Operation and maintenance data.
 3. Warranties.
 4. Bonds.
 5. Other types as indicated.
- D. Final Property Survey.
- E. Submit for Owner's benefit during and after project completion.

3.14 NUMBER OF COPIES OF SUBMITTALS

- A. Electronic Documents: Submit one electronic copy in PDF format; an electronically-marked up file will be returned. Create PDFs at native size and right-side up; illegible files will be rejected.
- B. Samples: Submit the number specified in individual specification sections; one of which will be retained by COR.
 1. After review, produce duplicates.
 2. Retained samples will not be returned to Contractor unless specifically so stated.

3.15 SUBMITTAL PROCEDURES

- A. General Requirements:
 - 1. Use a single transmittal for related items.
 - 2. Sequentially identify each item. For revised submittals use original number and a sequential numerical suffix.
 - 3. Identify: Project; Contractor; subcontractor or supplier; pertinent drawing and detail number; and specification section number and article/paragraph, as appropriate on each copy.
 - 4. Apply Contractor's stamp, signed or initialed certifying that review, approval, verification of products required, field dimensions, adjacent construction work, and coordination of information is in accordance with the requirements of the work and Contract Documents.
 - a. Submittals from sources other than the Contractor, or without Contractor's stamp will not be acknowledged, reviewed, or returned.
 - 5. Schedule submittals to expedite the Project, and coordinate submission of related items.
 - a. For each submittal for review, allow 15 days excluding delivery time to and from the Contractor.
 - b. For sequential reviews involving COR's consultants, Owner, or another affected party, allow an additional 7 days.
 - 6. Identify variations from Contract Documents and product or system limitations that may be detrimental to successful performance of the completed work.
 - 7. Provide space for Contractor and COR review stamps.
 - 8. When revised for resubmission, identify all changes made since previous submission.
 - 9. Distribute reviewed submittals. Instruct parties to promptly report inability to comply with requirements.
 - 10. Incomplete submittals will not be reviewed, unless they are partial submittals for distinct portion(s) of the work, and have received prior approval for their use.
- B. Product Data Procedures:
 - 1. Submit only information required by individual specification sections.
 - 2. Collect required information into a single submittal.
 - 3. Submit concurrently with related shop drawing submittal.
 - 4. Do not submit (Material) Safety Data Sheets for materials or products.
- C. Shop Drawing Procedures:
 - 1. Prepare accurate, drawn-to-scale, original shop drawing documentation by interpreting Contract Documents and coordinating related work.
 - 2. Do not reproduce Contract Documents to create shop drawings.
 - 3. Generic, non-project-specific information submitted as shop drawings do not meet the requirements for shop drawings.
- D. Samples Procedures:
 - 1. Transmit related items together as single package.
 - 2. Identify each item to allow review for applicability in relation to shop drawings showing installation locations.

3.16 SUBMITTAL REVIEW

- A. Submittals for Review: COR will review each submittal, and approve, or take other appropriate action.
- B. Submittals for Information: COR will acknowledge receipt and review. See below for actions to be taken.
- C. COR's actions will be reflected by marking each returned submittal using virtual stamp on electronic submittals.
- D. COR's and consultants' actions on items submitted for review:
 - 1. Authorizing purchasing, fabrication, delivery, and installation:
 - a. "Approved", or language with same legal meaning.

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- b. "Approved as Noted, Resubmission not required", or language with same legal meaning.
 - 1) At Contractor's option, submit corrected item, with review notations acknowledged and incorporated.
 - c. "Approved as Noted, Resubmit for Record", or language with same legal meaning.
- 2. Not Authorizing fabrication, delivery, and installation:
 - a. "Revise and Resubmit".
 - 1) Resubmit revised item, with review notations acknowledged and incorporated.
 - b. "Rejected".
 - 1) Submit item complying with requirements of Contract Documents.
- E. COR's and consultants' actions on items submitted for information:
 - 1. Items for which no action was taken:
 - a. "Received" - to notify the Contractor that the submittal has been received for record only.
 - 2. Items for which action was taken:
 - a. "Reviewed" - no further action is required from Contractor.

END OF SECTION

**SECTION 013216
CONSTRUCTION PROGRESS SCHEDULE**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Preliminary schedule.
- B. Construction progress schedule, bar chart type.

1.02 REFERENCE STANDARDS

- A. AGC (CPSM) - Construction Planning and Scheduling Manual; 2004.
- B. M-H (CPM) - CPM in Construction Management - Project Management with CPM; 2016, with Addendum (2021).

1.03 SUBMITTALS

- A. Within 10 days after date of Agreement, submit preliminary schedule.
- B. If preliminary schedule requires revision after review, submit revised schedule within 10 days.
- C. Within 20 days after review of preliminary schedule, submit draft of proposed complete schedule for review.
 - 1. Include written certification that major contractors have reviewed and accepted proposed schedule.
- D. Within 10 days after joint review, submit complete schedule.
- E. Submit updated schedule with each Application for Payment.
- F. Submit in PDF format.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 PRELIMINARY SCHEDULE

- A. Prepare preliminary schedule in the form of a horizontal bar chart.

3.02 CONTENT

- A. Show complete sequence of construction by activity, with dates for beginning and completion of each element of construction.
- B. Identify each item by specification section number.
- C. Show accumulated percentage of completion of each item, and total percentage of Work completed, as of the first day of each month.
- D. Provide legend for symbols and abbreviations used.

3.03 BAR CHARTS

- A. Include a separate bar for each major portion of Work or operation.
- B. Identify the first work day of each week.

3.04 NETWORK ANALYSIS

- A. Prepare network analysis diagrams and supporting mathematical analyses using the Critical Path Method.
- B. Illustrate order and interdependence of activities and sequence of work; how start of a given activity depends on completion of preceding activities, and how completion of the activity may restrain start of subsequent activities.
- C. Mathematical Analysis: Tabulate each activity of detailed network diagrams, using calendar dates, and identify for each activity:
 - 1. Preceding and following event numbers.
 - 2. Activity description.

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3. Estimated duration of activity, in maximum 15 day intervals.
 4. Earliest start date.
 5. Earliest finish date.
 6. Actual start date.
 7. Actual finish date.
 8. Latest start date.
 9. Latest finish date.
 10. Total and free float; float time shall accrue to Owner and to Owner's benefit.
 11. Monetary value of activity, keyed to Schedule of Values.
 12. Percentage of activity completed.
 13. Responsibility.
- D. Analysis Program: Capable of compiling monetary value of completed and partially completed activities, accepting revised completion dates, and recomputation of all dates and float.
- E. Required Reports: List activities in sorts or groups:
1. By preceding work item or event number from lowest to highest.
 2. By amount of float, then in order of early start.

3.05 REVIEW AND EVALUATION OF SCHEDULE

- A. Participate in joint review and evaluation of schedule with COR at each submittal.
- B. Evaluate project status to determine work behind schedule and work ahead of schedule.
- C. After review, revise as necessary as result of review, and resubmit within 10 days.

3.06 UPDATING SCHEDULE

- A. Maintain schedules to record actual start and finish dates of completed activities.
- B. Indicate progress of each activity to date of revision, with projected completion date of each activity.
- C. Annotate diagrams to graphically depict current status of Work.
- D. Identify activities modified since previous submittal, major changes in Work, and other identifiable changes.
- E. Indicate changes required to maintain Date of Substantial Completion.
- F. Submit reports required to support recommended changes.

END OF SECTION

**SECTION 014000
QUALITY REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Quality assurance.
- B. References and standards.
- C. Testing and inspection agencies and services.
- D. Contractor's construction-related professional design services.
- E. Contractor's design-related professional design services.
- F. Control of installation.
- G. Mock-ups.
- H. Tolerances.
- I. Defect Assessment.

1.02 RELATED REQUIREMENTS

- A. Section 014216 - Definitions.
- B. Section 016000 - Product Requirements: Requirements for material and product quality.

1.03 REFERENCE STANDARDS

- A. IAS AC89 - Accreditation Criteria for Testing Laboratories; 2021.

1.04 DEFINITIONS

- A. Contractor's Quality Control Plan: Contractor's management plan for executing the Contract for Construction.
- B. Contractor's Professional Design Services: Design of some aspect or portion of the project by party other than the design professional of record. Provide these services as part of the Contract for Construction.
 - 1. Design Services Types Required:
 - a. Construction-Related: Services Contractor needs to provide in order to carry out the Contractor's sole responsibilities for construction means, methods, techniques, sequences, and procedures.
 - b. Design-Related: Design services explicitly required to be performed by another design professional due to highly-technical and/or specialized nature of a portion of the project. Services primarily involve engineering analysis, calculations, and design, and are not intended to alter the aesthetic aspects of the design.
- C. Design Data: Design-related, signed and sealed drawings, calculations, specifications, certifications, shop drawings and other submittals provided by Contractor, and prepared directly by, or under direct supervision of, appropriately licensed design professional.

1.05 CONTRACTOR'S CONSTRUCTION-RELATED PROFESSIONAL DESIGN SERVICES

- A. Coordination: Contractor's professional design services are subject to requirements of project's Conditions for Construction Contract.
- B. Provide such engineering design services as may be necessary to plan and safely conduct certain construction operations, pertaining to, but not limited to the following:
 - 1. Temporary sheeting, shoring, or supports.
 - 2. Temporary bracing.
 - 3. Temporary stairs or steps required for construction access only.
 - 4. Temporary hoist(s) and rigging.

1.06 CONTRACTOR'S DESIGN-RELATED PROFESSIONAL DESIGN SERVICES

- A. Coordination: Contractor's professional design services are subject to requirements of project's Conditions for Construction Contract.
- B. Base design on performance and/or design criteria indicated in individual specification sections.

1.07 QUALITY ASSURANCE

- A. Testing Agency Qualifications:
 - 1. Prior to start of work, submit agency name, address, and telephone number, and names of full time registered Engineer and responsible officer.
 - 2. Qualification Statement: Provide documentation showing testing laboratory is accredited under IAS AC89.
- B. Designer Qualifications: Where professional engineering design services and design data submittals are specifically required of Contractor by Contract Documents, provide services of a Professional Engineer experienced in design of this type of work and licensed in the State of Washington.
- C. Contractor's Quality Control (CQC) Plan:
 - 1. Prior to start of work, submit a comprehensive plan describing how contract deliverables will be produced. Tailor CQC plan to specific requirements of the project. Include the following information:
 - a. Management Structure: Identify personnel responsible for quality. Include a chart showing lines of authority.
 - b. Acceptance of the plan is required prior to start of construction activities not including mobilization work. Owner's acceptance of the plan will be conditional and predicated on continuing satisfactory adherence to the plan. Owner reserves the right to require Contractor to make changes to the plan and operations, including removal of personnel, as necessary, to obtain specified quality of work results.
- D. Quality-Control Personnel Qualifications. Engage a person with requisite training and experience to implement and manage quality assurance (QA) and quality control (QC) for the project.

1.08 REFERENCES AND STANDARDS

- A. For products and workmanship specified by reference to a document or documents not included in the Project Manual, also referred to as reference standards, comply with requirements of the standard, except when more rigid requirements are specified or are required by applicable codes.
- B. Comply with reference standard of date of issue current on date of Contract Documents, except where a specific date is established by applicable code.
- C. Obtain copies of standards where required by product specification sections.
- D. Maintain copy at project site during submittals, planning, and progress of the specific work, until Substantial Completion.
- E. Should specified reference standards conflict with Contract Documents, request clarification from COR before proceeding.
- F. Neither the contractual relationships, duties, or responsibilities of the parties in Contract nor those of COR shall be altered from Contract Documents by mention or inference otherwise in any reference document.

1.09 TESTING AND INSPECTION AGENCIES AND SERVICES

- A. Contractor shall employ and pay for services of an independent testing agency to perform specified testing.

PART 3 EXECUTION

2.01 CONTROL OF INSTALLATION

- A. Monitor quality control over suppliers, manufacturers, products, services, site conditions, and workmanship, to produce work of specified quality.
- B. Comply with manufacturers' instructions, including each step in sequence.
- C. Should manufacturers' instructions conflict with Contract Documents, request clarification from COR before proceeding.
- D. Comply with specified standards as minimum quality for the work except where more stringent tolerances, codes, or specified requirements indicate higher standards or more precise workmanship.
- E. Have work performed by persons qualified to produce required and specified quality.
- F. Verify that field measurements are as indicated on shop drawings or as instructed by the manufacturer.
- G. Secure products in place with positive anchorage devices designed and sized to withstand stresses, vibration, physical distortion, and disfigurement.

2.02 MOCK-UPS

- A. Before installing portions of the Work where mock-ups are required, construct mock-ups in location and size indicated for each form of construction and finish required to comply with the following requirements, using materials indicated for the completed Work. The purpose of mock-up is to demonstrate the proposed range of aesthetic effects and workmanship.
- B. Tests shall be performed under provisions identified in this section and identified in the respective product specification sections.
- C. Assemble and erect specified items with specified attachment and anchorage devices, flashings, seals, and finishes.
- D. COR will use accepted mock-ups as a comparison standard for the remaining Work.
- E. Where mock-up has been accepted by COR and is specified in product specification sections to be removed, protect mock-up throughout construction, remove mock-up and clear area when directed to do so by COR.

2.03 TOLERANCES

- A. Monitor fabrication and installation tolerance control of products to produce acceptable Work. Do not permit tolerances to accumulate.
- B. Comply with manufacturers' tolerances. Should manufacturers' tolerances conflict with Contract Documents, request clarification from COR before proceeding.
- C. Adjust products to appropriate dimensions; position before securing products in place.

2.04 TESTING AND INSPECTION

- A. See individual specification sections for testing and inspection required.
- B. Testing Agency Duties:
 - 1. Provide qualified personnel at site. Cooperate with COR and Contractor in performance of services.
 - 2. Perform specified sampling and testing of products in accordance with specified standards.
 - 3. Ascertain compliance of materials and mixes with requirements of Contract Documents.
 - 4. Promptly notify COR and Contractor of observed irregularities or non-compliance of Work or products.
 - 5. Perform additional tests and inspections required by COR.
 - 6. Submit reports of all tests/inspections specified.
- C. Limits on Testing/Inspection Agency Authority:

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1. Agency may not release, revoke, alter, or enlarge on requirements of Contract Documents.
 2. Agency may not approve or accept any portion of the Work.
 3. Agency may not assume any duties of Contractor.
 4. Agency has no authority to stop the Work.
- D. Contractor Responsibilities:
1. Deliver to agency at designated location, adequate samples of materials proposed to be used that require testing, along with proposed mix designs.
 2. Cooperate with laboratory personnel, and provide access to the Work and to manufacturers' facilities.
 3. Provide incidental labor and facilities:
 - a. To provide access to Work to be tested/inspected.
 - b. To obtain and handle samples at the site or at source of Products to be tested/inspected.
 - c. To facilitate tests/inspections.
 - d. To provide storage and curing of test samples.
 4. Notify COR and laboratory 24 hours prior to expected time for operations requiring testing/inspection services.
 5. Employ services of an independent qualified testing laboratory and pay for additional samples, tests, and inspections required by Contractor beyond specified requirements.
 6. Arrange with Owner's agency and pay for additional samples, tests, and inspections required by Contractor beyond specified requirements.
- E. Re-testing required because of non-compliance with specified requirements shall be performed by the same agency on instructions by COR.
- F. Re-testing required because of non-compliance with specified requirements shall be paid for by Contractor.

2.05 DEFECT ASSESSMENT

- A. Replace Work or portions of the Work not complying with specified requirements.

END OF SECTION

SECTION 014216
DEFINITIONS

PART 1 GENERAL

1.01 SUMMARY

- A. This section supplements the definitions contained in the General Conditions.
- B. Other definitions are included in individual specification sections.

1.02 DEFINITIONS

- A. Furnish: To supply, deliver, unload, and inspect for damage.
- B. Install: To unpack, assemble, erect, apply, place, finish, cure, protect, clean, start up, and make ready for use.
- C. Product: Material, machinery, components, equipment, fixtures, and systems forming the work result. Not materials or equipment used for preparation, fabrication, conveying, or erection and not incorporated into the work result. Products may be new, never before used, or re-used materials or equipment.
- D. Project Manual: The book-sized volume that includes the procurement requirements (if any), the contracting requirements, and the specifications.
- E. Provide: To furnish and install.
- F. Supply: Same as Furnish.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

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**SECTION 015000
TEMPORARY FACILITIES AND CONTROLS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Temporary telecommunications services.
- B. Temporary Controls: Barriers and fencing.
- C. Vehicular access and parking.
- D. Waste removal facilities and services.

1.02 TELECOMMUNICATIONS SERVICES

- A. Provide, maintain, and pay for telecommunications services to field office at time of project mobilization.

1.03 BARRIERS

- A. Provide barriers to prevent unauthorized entry to construction areas, to prevent access to areas that could be hazardous to workers or the public, to allow for owner's use of site and to protect existing facilities and adjacent properties from damage from construction operations and demolition.

1.04 FENCING

- A. Construction: Contractor's option.

1.05 VEHICULAR ACCESS AND PARKING - SEE SECTION 015500

- A. Coordinate with COR to establish parking and related requirements at base of mountain.
- B. Coordinate access and haul routes with governing authorities and Owner.

1.06 WASTE REMOVAL

- A. Provide waste removal facilities and services as required to maintain the site in clean and orderly condition.
- B. Provide containers with lids. Remove trash from site periodically.
- C. If materials to be recycled or re-used on the project must be stored on-site, provide suitable non-combustible containers; locate containers holding flammable material outside the structure unless otherwise approved by the authorities having jurisdiction.

1.07 REMOVAL OF UTILITIES, FACILITIES, AND CONTROLS

- A. Remove temporary utilities, equipment, facilities, materials, prior to Date of Substantial Completion inspection.
- B. Clean and repair damage caused by installation or use of temporary work.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION - NOT USED

END OF SECTION

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**SECTION 015713
TEMPORARY EROSION AND SEDIMENT CONTROL**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Prevention of erosion due to construction activities.
- B. Restoration of areas eroded due to insufficient preventive measures.
- C. Compensation of Owner for fines levied by authorities having jurisdiction due to non-compliance by Contractor.

1.02 REFERENCE STANDARDS

- A. EPA (NPDES) - National Pollutant Discharge Elimination System (NPDES), Construction General Permit; Current Edition.
- B. FHWA FLP-94-005 - Best Management Practices for Erosion and Sediment Control; 1995.

1.03 PERFORMANCE REQUIREMENTS

- A. Comply with requirements of EPA (NPDES) for erosion and sedimentation control, as specified by the NPDES, for Phases I and II, and in compliance with requirements of Construction General Permit (CGP), whether the project is required by law to comply or not.
- B. Best Management Practices Standard: FHWA FLP-94-005.
- C. Develop and follow an Erosion and Sedimentation Prevention Plan and submit periodic inspection reports.
- D. Do not begin clearing, grading, or other work involving disturbance of ground surface cover until applicable permits have been obtained; furnish all documentation required to obtain applicable permits.
- E. Timing: Put preventive measures in place as soon as possible after disturbance of surface cover and before precipitation occurs.
- F. Erosion On Site: Minimize wind, water, and vehicular erosion of soil on project site due to construction activities for this project.
 - 1. Control movement of sediment and soil from temporary stockpiles of soil.
 - 2. Prevent development of ruts due to equipment and vehicular traffic.
 - 3. If erosion occurs due to non-compliance with these requirements, restore eroded areas at no cost to Owner.
- G. Maintenance: Maintain temporary preventive measures until permanent measures have been established.

1.04 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Erosion and Sedimentation Control Plan:
 - 1. Submit within 2 weeks after Notice to Proceed.
 - 2. Include:
 - a. Site plan identifying soils, existing erosion problems, and areas vulnerable to erosion due to topography, soils, or drainage.
 - b. Site plan showing grading; new improvements; and other temporary construction; and proposed preventive measures.
 - c. Schedule of temporary preventive measures, in relation to ground disturbing activities.
 - d. Other information required by law.
 - e. Format required by law is acceptable, provided any additional information specified is also included.
 - 3. Obtain the approval of the Plan by authorities having jurisdiction.
 - 4. Obtain the approval of the Plan by Owner.

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- C. Inspection Reports: Submit report of each inspection; identify each preventive measure, indicate condition, and specify maintenance or repair required and accomplished.

PART 2 PRODUCTS: NOT USED.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Examine site and identify existing features that contribute to erosion resistance; maintain such existing features to greatest extent possible.

3.02 PREPARATION

- A. Schedule work so that soil surfaces are left exposed for the minimum amount of time.

3.03 MAINTENANCE

- A. Inspect preventive measures weekly, within 24 hours after the end of any storm that produces 0.5 inches (13 mm) or more rainfall at the project site, and daily during prolonged rainfall.
- B. Repair deficiencies immediately.

3.04 CLEAN UP

- A. Remove temporary measures after permanent measures have been installed, unless permitted to remain by COR.
- B. Clean out temporary sediment control structures that are to remain as permanent measures.
- C. Where removal of temporary measures would leave exposed soil, shape surface to an acceptable grade and finish to match adjacent ground surfaces.

END OF SECTION

**SECTION 016000
PRODUCT REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. General product requirements.
- B. Sustainable design-related product requirements.
- C. Transportation, handling, storage and protection.
- D. Product option requirements.
- E. Substitution limitations.

1.02 RELATED REQUIREMENTS

- A. Section 012500 - Substitution Procedures: Substitutions made during procurement and/or construction phases.
- B. Section 014000 - Quality Requirements: Product quality monitoring.
- C. Section 017419 - Construction Waste Management and Disposal: Waste disposal requirements potentially affecting product selection, packaging and substitutions.

1.03 SUBMITTALS

- A. Product Data Submittals: Submit manufacturer's standard published data. Mark each copy to identify applicable products, models, options, and other data. Supplement manufacturers' standard data to provide information specific to this Project.
- B. Shop Drawing Submittals: Prepared specifically for this Project; indicate utility and electrical characteristics, utility connection requirements, and location of utility outlets for service for functional equipment and appliances.
- C. Sample Submittals: Illustrate functional and aesthetic characteristics of the product, with integral parts and attachment devices. Coordinate sample submittals for interfacing work.
 - 1. For selection from standard finishes, submit samples of the full range of the manufacturer's standard colors, textures, and patterns.

PART 2 PRODUCTS

2.01 NEW PRODUCTS

- A. Provide new products unless specifically required or permitted by Contract Documents.
- B. Use of products having any of the following characteristics is not permitted:
 - 1. Made of wood from newly cut old growth timber.
 - 2. Containing lead, cadmium, or asbestos.
- C. Where other criteria are met, Contractor shall give preference to products that:
 - 1. Are extracted, harvested, and/or manufactured closer to the location of the project.
 - 2. Have longer documented life span under normal use.
 - 3. Result in less construction waste. See Section 017419
 - 4. If made of wood, are made of sustainably harvested wood, wood chips, or wood fiber.

2.02 PRODUCT OPTIONS

- A. Products Specified by Reference Standards or by Description Only: Use any product meeting those standards or description.
- B. Products Specified by Naming One or More Manufacturers: Use a product of one of the manufacturers named and meeting specifications, no options or substitutions allowed.
- C. Products Specified by Naming One or More Manufacturers with a Provision for Substitutions: Submit a request for substitution for any manufacturer not named.
- D. Products Specified as Basis-of-Design: Submit substitution requests for unnamed products to COR for evaluation.

PART 3 EXECUTION

3.01 SUBSTITUTION LIMITATIONS

- A. See Section 012500 - Substitution Procedures.

3.02 TRANSPORTATION AND HANDLING

- A. Package products for shipment in manner to prevent damage; for equipment, package to avoid loss of factory calibration.
- B. If special precautions are required, attach instructions prominently and legibly on outside of packaging.
- C. Coordinate schedule of product delivery to designated prepared areas in order to minimize site storage time and potential damage to stored materials.
- D. Transport and handle products in accordance with manufacturer's instructions.
- E. Transport materials in covered trucks to prevent contamination of product and littering of surrounding areas.
- F. Promptly inspect shipments to ensure that products comply with requirements, quantities are correct, and products are undamaged.
- G. Provide equipment and personnel to handle products by methods to prevent soiling, disfigurement, or damage, and to minimize handling.
- H. Arrange for the return of packing materials, such as wood pallets, where economically feasible.

3.03 STORAGE AND PROTECTION

- A. Provide protection of stored materials and products against theft, casualty, or deterioration.
- B. Designate receiving/storage areas for incoming products so that they are delivered according to installation schedule and placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. See Section 017419.
- C. Store and protect products in accordance with manufacturers' instructions.
- D. Store with seals and labels intact and legible.
- E. Store sensitive products in weathertight, climate-controlled enclosures in an environment favorable to product.
- F. For exterior storage of fabricated products, place on sloped supports above ground.
- G. Protect products from damage or deterioration due to construction operations, weather, precipitation, humidity, temperature, sunlight and ultraviolet light, dirt, dust, and other contaminants.
- H. Comply with manufacturer's warranty conditions, if any.
- I. Cover products subject to deterioration with impervious sheet covering. Provide ventilation to prevent condensation and degradation of products.
- J. Prevent contact with material that may cause corrosion, discoloration, or staining.
- K. Provide equipment and personnel to store products by methods to prevent soiling, disfigurement, or damage.
- L. Arrange storage of products to permit access for inspection. Periodically inspect to verify products are undamaged and are maintained in acceptable condition.

END OF SECTION

**SECTION 017000
EXECUTION AND CLOSEOUT REQUIREMENTS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Examination, preparation, and general installation procedures.
- B. Surveying for laying out the work.
- C. Cleaning and protection.
- D. Starting of systems and equipment.
- E. Demonstration and instruction of Owner personnel.
- F. Closeout procedures, including Contractor's Correction Punch List, except payment procedures.

1.02 RELATED REQUIREMENTS

- A. Section 013000 - Administrative Requirements: Submittals procedures, Electronic document submittal service.
- B. Section 014000 - Quality Requirements: Testing and inspection procedures.
- C. Section 015713 - Temporary Erosion and Sediment Control: Additional erosion and sedimentation control requirements.
- D. Section 017419 - Construction Waste Management and Disposal: Additional procedures for trash/waste removal, recycling, salvage, and reuse.
- E. Section 024100 - Demolition: Demolition of whole structures and parts thereof; site utility demolition.

1.03 REFERENCE STANDARDS

- A. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations; 2022, with Errata (2021).

1.04 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Survey work: Submit name, address, and telephone number of Surveyor before starting survey work.
 - 1. On request, submit documentation verifying accuracy of survey work.
 - 2. Submit a copy of site drawing signed by the Land Surveyor, that the elevations and locations of the work are in compliance with Contract Documents.
 - 3. Submit surveys and survey logs for the project record.
- C. Demolition Plan: Submit demolition plan as specified by OSHA and local authorities.
 - 1. Indicate extent of demolition, removal sequence, bracing and shoring, and location and construction of barricades and fences. Include design drawings and calculations for bracing and shoring.
 - 2. Identify demolition firm and submit qualifications.
 - 3. Include a summary of safety procedures.

1.05 QUALIFICATIONS

- A. For surveying work, employ a land surveyor registered in the State of Washington and acceptable to COR. Submit evidence of surveyor's Errors and Omissions insurance coverage in the form of an Insurance Certificate. Employ only individual(s) trained and experienced in collecting and recording accurate data relevant to ongoing construction activities,
- B. For design of temporary shoring and bracing, employ a Professional Engineer experienced in design of this type of work and licensed in the State of Washington.

1.06 PROJECT CONDITIONS

- A. Use of explosives is not permitted.
- B. Grade site to drain. Maintain excavations free of water.
- C. Protect site from puddling or running water. Provide water barriers as required to protect site from soil erosion.
- D. Dust Control: Execute work by methods to minimize raising dust from construction operations. Provide positive means to prevent air-borne dust from dispersing into atmosphere and over adjacent property.
- E. Erosion and Sediment Control: Plan and execute work by methods to control surface drainage from cuts and fills, from borrow and waste disposal areas. Prevent erosion and sedimentation.
- F. Noise Control: Provide methods, means, and facilities to minimize noise produced by construction operations.

1.07 COORDINATION

- A. Coordinate scheduling, submittals, and work of the various sections of the Project Manual to ensure efficient and orderly sequence of installation of interdependent construction elements, with provisions for accommodating items installed later.
- B. Notify affected utility companies and comply with their requirements.
- C. Verify that utility requirements and characteristics of new operating equipment are compatible with building utilities. Coordinate work of various sections having interdependent responsibilities for installing, connecting to, and placing in service, such equipment.
- D. Coordinate space requirements, supports, and installation of mechanical and electrical work that are indicated diagrammatically on drawings. Follow routing indicated for pipes, ducts, and conduit, as closely as practicable; place runs parallel with lines of building. Utilize spaces efficiently to maximize accessibility for other installations, for maintenance, and for repairs.
- E. Coordinate completion and clean-up of work of separate sections.
- F. After Owner occupancy of premises, coordinate access to site for correction of defective work and work not in accordance with Contract Documents, to minimize disruption of Owner's activities.

PART 2 PRODUCTS: NOT USED

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that existing site conditions and substrate surfaces are acceptable for subsequent work. Start of work means acceptance of existing conditions.
- B. Examine and verify specific conditions described in individual specification sections.
- C. Take field measurements before confirming product orders or beginning fabrication, to minimize waste due to over-ordering or misfabrication.
- D. Verify that utility services are available, of the correct characteristics, and in the correct locations.

3.02 PREPARATION

- A. Clean substrate surfaces prior to applying next material or substance.
- B. Seal cracks or openings of substrate prior to applying next material or substance.
- C. Apply manufacturer required or recommended substrate primer, sealer, or conditioner prior to applying any new material or substance in contact or bond.

3.03 LAYING OUT THE WORK

- A. Verify locations of survey control points prior to starting work.
- B. Promptly notify COR of any discrepancies discovered.

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- C. Protect survey control points prior to starting site work; preserve permanent reference points during construction.
- D. Promptly report to COR the loss or destruction of any reference point or relocation required because of changes in grades or other reasons.
- E. Replace dislocated survey control points based on original survey control. Make no changes without prior written notice to COR.
- F. Utilize recognized engineering survey practices.
- G. Establish elevations, lines and levels. Locate and lay out by instrumentation and similar appropriate means:
 - 1. Site improvements including pavements; stakes for grading, fill and topsoil placement; utility locations, slopes, and invert elevations.
 - 2. Grid or axis for structures.
 - 3. Building foundation, column locations, ground floor elevations.
- H. Periodically verify layouts by same means.
- I. Maintain a complete and accurate log of control and survey work as it progresses.

3.04 GENERAL INSTALLATION REQUIREMENTS

- A. Install products as specified in individual sections, in accordance with manufacturer's instructions and recommendations, and so as to avoid waste due to necessity for replacement.
- B. Make vertical elements plumb and horizontal elements level, unless otherwise indicated.
- C. Install equipment and fittings plumb and level, neatly aligned with adjacent vertical and horizontal lines, unless otherwise indicated.
- D. Make consistent texture on surfaces, with seamless transitions, unless otherwise indicated.
- E. Make neat transitions between different surfaces, maintaining texture and appearance.

3.05 PROGRESS CLEANING

- A. Maintain areas free of waste materials, debris, and rubbish. Maintain site in a clean and orderly condition.
- B. Remove debris and rubbish from pipe chases, plenums, attics, crawl spaces, and other closed or remote spaces, prior to enclosing the space.
- C. Broom and vacuum clean interior areas prior to start of surface finishing, and continue cleaning to eliminate dust.
- D. Collect and remove waste materials, debris, and trash/rubbish from site periodically and dispose off-site; do not burn or bury.

3.06 PROTECTION OF INSTALLED WORK

- A. Protect installed work from damage by construction operations.
- B. Provide special protection where specified in individual specification sections.
- C. Provide temporary and removable protection for installed products. Control activity in immediate work area to prevent damage.
- D. Provide protective coverings at walls, projections, jambs, sills, and soffits of openings.
- E. Protect finished floors, stairs, and other surfaces from traffic, dirt, wear, damage, or movement of heavy objects, by protecting with durable sheet materials.
- F. Prohibit traffic or storage upon waterproofed or roofed surfaces. If traffic or activity is necessary, obtain recommendations for protection from waterproofing or roofing material manufacturer.
- G. Remove protective coverings when no longer needed; reuse or recycle coverings if possible.

3.07 SYSTEM STARTUP

- A. Coordinate schedule for start-up of various equipment and systems.

- B. Notify COR and Owner seven days prior to start-up of each item.
- C. Verify that each piece of equipment or system has been checked for proper lubrication, drive rotation, belt tension, control sequence, and for conditions that may cause damage.
- D. Verify tests, meter readings, and specified electrical characteristics agree with those required by the equipment or system manufacturer.
- E. Verify that wiring and support components for equipment are complete and tested.
- F. Execute start-up under supervision of applicable Contractor personnel and manufacturer's representative in accordance with manufacturers' instructions.
- G. Submit a written report that equipment or system has been properly installed and is functioning correctly.

3.08 DEMONSTRATION AND INSTRUCTION

- A. Demonstrate start-up, operation, control, adjustment, trouble-shooting, servicing, maintenance, and shutdown of each item of equipment at scheduled time, at equipment location.
- B. For equipment or systems requiring seasonal operation, perform demonstration for other season within six months.
- C. Provide a qualified person who is knowledgeable about the Project to perform demonstration and instruction of Owner's personnel.
- D. Utilize operation and maintenance manuals as basis for instruction. Review contents of manual with Owner's personnel in detail to explain all aspects of operation and maintenance.
- E. Prepare and insert additional data in operations and maintenance manuals when need for additional data becomes apparent during instruction.

3.09 ADJUSTING

- A. Adjust operating products and equipment to ensure smooth and unhindered operation.

3.10 FINAL CLEANING

- A. Use cleaning materials that are nonhazardous.
- B. Clean interior and exterior glass, surfaces exposed to view; remove temporary labels, stains and foreign substances, polish transparent and glossy surfaces, vacuum carpeted and soft surfaces.
- C. Remove all labels that are not permanent. Do not paint or otherwise cover fire test labels or nameplates on mechanical and electrical equipment.
- D. Clean equipment and fixtures to a sanitary condition with cleaning materials appropriate to the surface and material being cleaned.
- E. Clean site; rake clean landscaped surfaces.
- F. Remove waste, surplus materials, trash/rubbish, and construction facilities from the site; dispose of in legal manner; do not burn or bury.

3.11 CLOSEOUT PROCEDURES

- A. Make submittals that are required by governing or other authorities.
- B. Accompany Project Coordinator on preliminary inspection to determine items to be listed for completion or correction in the Contractor's Correction Punch List for Contractor's Notice of Substantial Completion.
- C. Notify COR when work is considered ready for COR's Substantial Completion inspection.
- D. Submit written certification containing Contractor's Correction Punch List, that Contract Documents have been reviewed, work has been inspected, and that work is complete in accordance with Contract Documents and ready for COR's Substantial Completion inspection.

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- E. Conduct Substantial Completion inspection and create Final Correction Punch List containing COR's and Contractor's comprehensive list of items identified to be completed or corrected and submit to COR.
- F. Correct items of work listed in Final Correction Punch List and comply with requirements for access to Owner-occupied areas.
- G. Notify COR when work is considered finally complete and ready for COR's Substantial Completion final inspection.
- H. Complete items of work determined by COR listed in executed Certificate of Substantial Completion.

END OF SECTION

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SECTION 017419
CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

PART 1 GENERAL

1.01 WASTE MANAGEMENT REQUIREMENTS

- A. Owner requires that this project generate the least amount of trash and waste possible.
- B. Employ processes that ensure the generation of as little waste as possible due to error, poor planning, breakage, mishandling, contamination, or other factors.
- C. Minimize trash/waste disposal in landfills; reuse, salvage, or recycle as much waste as economically feasible.
- D. Methods of trash/waste disposal that are not acceptable are:
 - 1. Burning on the project site.
 - 2. Burying on the project site.
 - 3. Dumping or burying on other property, public or private.
 - 4. Other illegal dumping or burying.
- E. Regulatory Requirements: Contractor is responsible for knowing and complying with regulatory requirements, including but not limited to Federal, state and local requirements, pertaining to legal disposal of all construction and demolition waste materials.

1.02 RELATED REQUIREMENTS

- A. Section 013000 - Administrative Requirements: Additional requirements for project meetings, reports, submittal procedures, and project documentation.
- B. Section 015000 - Temporary Facilities and Controls: Additional requirements related to trash/waste collection and removal facilities and services.
- C. Section 016000 - Product Requirements: Waste prevention requirements related to product substitutions.
- D. Section 016000 - Product Requirements: Waste prevention requirements related to delivery, storage, and handling.
- E. Section 017000 - Execution and Closeout Requirements: Trash/waste prevention procedures related to demolition, cutting and patching, installation, protection, and cleaning.

1.03 DEFINITIONS

- A. Clean: Untreated and unpainted; not contaminated with oils, solvents, caulk, or the like.
- B. Construction and Demolition Waste: Solid wastes typically including building materials, packaging, trash, debris, and rubble resulting from construction, remodeling, repair and demolition operations.
- C. Hazardous: Exhibiting the characteristics of hazardous substances, i.e., ignitibility, corrosivity, toxicity or reactivity.
- D. Nonhazardous: Exhibiting none of the characteristics of hazardous substances, i.e., ignitibility, corrosivity, toxicity, or reactivity.
- E. Nontoxic: Neither immediately poisonous to humans nor poisonous after a long period of exposure.
- F. Recyclable: The ability of a product or material to be recovered at the end of its life cycle and remanufactured into a new product for reuse by others.
- G. Recycle: To remove a waste material from the project site to another site for remanufacture into a new product for reuse by others.
- H. Recycling: The process of sorting, cleansing, treating and reconstituting solid waste and other discarded materials for the purpose of using the altered form. Recycling does not include burning, incinerating, or thermally destroying waste.
- I. Return: To give back reusable items or unused products to vendors for credit.

- J. Reuse: To reuse a construction waste material in some manner on the project site.
- K. Salvage: To remove a waste material from the project site to another site for resale or reuse by others.
- L. Sediment: Soil and other debris that has been eroded and transported by storm or well production run-off water.
- M. Source Separation: The act of keeping different types of waste materials separate beginning from the first time they become waste.
- N. Toxic: Poisonous to humans either immediately or after a long period of exposure.
- O. Trash: Any product or material unable to be reused, returned, recycled, or salvaged.
- P. Waste: Extra material or material that has reached the end of its useful life in its intended use. Waste includes salvageable, returnable, recyclable, and reusable material.

PART 3 EXECUTION

2.01 WASTE MANAGEMENT PLAN IMPLEMENTATION

- A. Manager: Designate an on-site person or persons responsible for instructing workers and overseeing and documenting results of the Waste Management Plan.
- B. Communication: Distribute copies of the Waste Management Plan to job site foreman, each subcontractor, Owner, and COR.
- C. Instruction: Provide on-site instruction of appropriate separation, handling, and recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the project.
- D. Meetings: Discuss trash/waste management goals and issues at project meetings.
 - 1. Prebid meeting.
 - 2. Preconstruction meeting.
 - 3. Regular job-site meetings.
- E. Facilities: Provide specific facilities for separation and storage of materials for recycling, salvage, reuse, return, and trash disposal, for use by all contractors and installers.
 - 1. Provide containers as required.
 - 2. Keep recycling and trash/waste bin areas neat and clean and clearly marked in order to avoid contamination of materials.
- F. Hazardous Wastes: Separate, store, and dispose of hazardous wastes according to applicable regulations.
- G. Recycling: Separate, store, protect, and handle at the site identified recyclable waste products in order to prevent contamination of materials and to maximize recyclability of identified materials. Arrange for timely pickups from the site or deliveries to recycling facility in order to prevent contamination of recyclable materials.
- H. Reuse of Materials On-Site: Set aside, sort, and protect separated products in preparation for reuse.
- I. Salvage: Set aside, sort, and protect products to be salvaged for reuse off-site.

END OF SECTION

**SECTION 017800
CLOSEOUT SUBMITTALS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Project record documents.
- B. Operation and maintenance data.
- C. Warranties and bonds.

1.02 RELATED REQUIREMENTS

- A. Section 013000 - Administrative Requirements: Submittals procedures, shop drawings, product data, and samples.
- B. Section 017000 - Execution and Closeout Requirements: Contract closeout procedures.
- C. Individual Product Sections: Specific requirements for operation and maintenance data.
- D. Individual Product Sections: Warranties required for specific products or Work.

1.03 SUBMITTALS

- A. Project Record Documents: Submit documents to COR with claim for final Application for Payment.
- B. Operation and Maintenance Data:
 - 1. For equipment, or component parts of equipment put into service during construction and operated by Owner, submit completed documents within ten days after acceptance.
 - 2. Submit one copy of completed documents 15 days prior to final inspection. This copy will be reviewed and returned after final inspection, with COR comments. Revise content of all document sets as required prior to final submission.
 - 3. Submit two sets of revised final documents in final form within 10 days after final inspection.
- C. Warranties and Bonds:
 - 1. For equipment or component parts of equipment put into service during construction with Owner's permission, submit documents within 10 days after acceptance.
 - 2. Make other submittals within 10 days after Date of Substantial Completion, prior to final Application for Payment.
 - 3. For items of Work for which acceptance is delayed beyond Date of Substantial Completion, submit within 10 days after acceptance, listing the date of acceptance as the beginning of the warranty period.

PART 2 PRODUCTS - NOT USED

PART 3 EXECUTION

3.01 PROJECT RECORD DOCUMENTS

- A. Maintain on site one set of the following record documents; record actual revisions to the Work:
 - 1. Drawings.
 - 2. Specifications.
 - 3. Addenda.
 - 4. Change Orders and other modifications to the Contract.
 - 5. Reviewed shop drawings, product data, and samples.
 - 6. Manufacturer's instruction for assembly, installation, and adjusting.
- B. Ensure entries are complete and accurate, enabling future reference by Owner.
- C. Store record documents separate from documents used for construction.
- D. Record information concurrent with construction progress.

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- E. Specifications: Legibly mark and record at each product section description of actual products installed, including the following:
 - 1. Manufacturer's name and product model and number.
 - 2. Product substitutions or alternates utilized.
 - 3. Changes made by Addenda and modifications.
- F. Record Drawings and Shop Drawings: Legibly mark each item to record actual construction including:
 - 1. Measured depths of foundations in relation to finish first floor datum.
 - 2. Field changes of dimension and detail.
 - 3. Details not on original Contract drawings.

3.02 OPERATION AND MAINTENANCE DATA

- A. Source Data: For each product or system, list names, addresses and telephone numbers of Subcontractors and suppliers, including local source of supplies and replacement parts.
- B. Product Data: Mark each sheet to clearly identify specific products and component parts, and data applicable to installation. Delete inapplicable information.
- C. Drawings: Supplement product data to illustrate relations of component parts of equipment and systems, to show control and flow diagrams. Do not use Project Record Documents as maintenance drawings.
- D. Typed Text: As required to supplement product data. Provide logical sequence of instructions for each procedure, incorporating manufacturer's instructions.

3.03 OPERATION AND MAINTENANCE DATA FOR MATERIALS AND FINISHES

- A. For Each Product, Applied Material, and Finish:
 - 1. Product data, with catalog number, size, composition, and color and texture designations.
 - 2. Information for re-ordering custom manufactured products.
- B. Instructions for Care and Maintenance: Manufacturer's recommendations for cleaning agents and methods, precautions against detrimental cleaning agents and methods, and recommended schedule for cleaning and maintenance.
- C. Moisture protection and weather-exposed products: Include product data listing applicable reference standards, chemical composition, and details of installation. Provide recommendations for inspections, maintenance, and repair.
- D. Additional information as specified in individual product specification sections.
- E. Where additional instructions are required, beyond the manufacturer's standard printed instructions, have instructions prepared by personnel experienced in the operation and maintenance of the specific products.

3.04 OPERATION AND MAINTENANCE DATA FOR EQUIPMENT AND SYSTEMS

- A. For Each Item of Equipment and Each System:
 - 1. Description of unit or system, and component parts.
 - 2. Identify function, normal operating characteristics, and limiting conditions.
 - 3. Include performance curves, with engineering data and tests.
 - 4. Complete nomenclature and model number of replaceable parts.
- B. Where additional instructions are required, beyond the manufacturer's standard printed instructions, have instructions prepared by personnel experienced in the operation and maintenance of the specific products.
- C. Maintenance Requirements: Include routine procedures and guide for preventative maintenance and trouble shooting; disassembly, repair, and reassembly instructions; and alignment, adjusting, balancing, and checking instructions.
 - 1. Include Carbon Monoxide Monitoring Protocol.
- D. Include manufacturer's printed operation and maintenance instructions.

- E. Provide original manufacturer's parts list, illustrations, assembly drawings, and diagrams required for maintenance.

3.05 ASSEMBLY OF OPERATION AND MAINTENANCE MANUALS

- A. Assemble operation and maintenance data into durable manuals for Owner's personnel use, with data arranged in the same sequence as, and identified by, the specification sections.
- B. Where systems involve more than one specification section, provide separate tabbed divider for each system.
- C. Binders: Commercial quality, 8-1/2 by 11 inch (216 by 280 mm) three D side ring binders with durable plastic covers; 2 inch (50 mm) maximum ring size. When multiple binders are used, correlate data into related consistent groupings.
- D. Cover: Identify each binder with typed or printed title OPERATION AND MAINTENANCE INSTRUCTIONS; identify title of Project; identify subject matter of contents.
- E. Project Directory: Title and address of Project; names, addresses, and telephone numbers of COR, Consultants, Contractor and subcontractors, with names of responsible parties.
- F. Tables of Contents: List every item separated by a divider, using the same identification as on the divider tab; where multiple volumes are required, include all volumes Tables of Contents in each volume, with the current volume clearly identified.
- G. Dividers: Provide tabbed dividers for each separate product and system; identify the contents on the divider tab; immediately following the divider tab include a description of product and major component parts of equipment.
- H. Text: Manufacturer's printed data, or typewritten data on 20 pound paper.
- I. Drawings: Provide with reinforced punched binder tab. Bind in with text; fold larger drawings to size of text pages.

3.06 WARRANTIES AND BONDS

- A. Obtain warranties and bonds, executed in duplicate by responsible Subcontractors, suppliers, and manufacturers, within 10 days after completion of the applicable item of work. Except for items put into use with Owner's permission, leave date of beginning of time of warranty until Date of Substantial completion is determined.
- B. Verify that documents are in proper form, contain full information, and are notarized.
- C. Co-execute submittals when required.
- D. Retain warranties and bonds until time specified for submittal.

END OF SECTION

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**SECTION 024100
DEMOLITION**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Building demolition .

1.02 RELATED REQUIREMENTS

- A. Section 011000 - Summary: Limitations on Contractor's use of site and premises.
- B. Section 015000 - Temporary Facilities and Controls: Site fences, security, protective barriers, and waste removal.
- C. Section 015713 - Temporary Erosion and Sediment Control.
- D. Section 016000 - Product Requirements: Handling and storage of items removed for salvage and relocation.
- E. Section 017000 - Execution and Closeout Requirements: Project conditions; protection of bench marks, survey control points, and existing construction to remain; reinstallation of removed products; temporary bracing and shoring.

1.03 DEFINITIONS

- A. Demolish: Dismantle, raze, destroy, or wreck any building or structure or any part thereof.
- B. Remove: Detach or dismantle items from existing construction and dispose of them off site, unless items are indicated to be salvaged or reinstalled.
- C. Existing to Remain: Designation for existing items that are not to be removed and that are not otherwise indicated to be salvaged or reinstalled.

1.04 REFERENCE STANDARDS

- A. 29 CFR 1926 - Safety and Health Regulations for Construction; Current Edition.
- B. NFPA 241 - Standard for Safeguarding Construction, Alteration, and Demolition Operations; 2022, with Errata (2021).

1.05 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Site Plan: Indicate:
 - 1. Areas for temporary construction and field offices.
 - 2. Areas for temporary and permanent placement of removed materials.
- C. Demolition Plan: Submit demolition plan as required by OSHA and local AHJs.
 - 1. Indicate extent of demolition, removal sequencing, bracing and shoring, and location and construction of barricades and fences.
 - 2. Summary of safety procedures.

PART 2 PRODUCTS

2.01 MATERIALS

PART 3 EXECUTION

3.01 DEMOLITION

- A. Remove the entire building indicated on the drawings.
- B. Remove other items indicated, for salvage and relocation.

3.02 GENERAL PROCEDURES AND PROJECT CONDITIONS

- A. Comply with requirements in Section 017000.
- B. Comply with applicable codes and regulations for demolition operations and safety of adjacent structures and the public.

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1. Obtain required permits.
 2. Comply with applicable requirements of NFPA 241.
 3. Take precautions to prevent catastrophic or uncontrolled collapse of structures to be removed; do not allow worker or public access within range of potential collapse of unstable structures.
 4. Provide, erect, and maintain temporary barriers and security devices.
 5. Use physical barriers to prevent access to areas that could be hazardous to workers or the public.
 6. Conduct operations to minimize effects on and interference with adjacent structures and occupants.
 7. Do not close or obstruct roadways or sidewalks without permits from authority having jurisdiction.
 8. Conduct operations to minimize obstruction of public and private entrances and exits. Do not obstruct required exits at any time. Protect persons using entrances and exits from removal operations.
 9. Obtain written permission from owners of adjacent properties when demolition equipment will traverse, infringe upon, or limit access to their property.
- C. Do not begin removal until receipt of notification to proceed from Owner.
- D. Protect existing structures and other elements to remain in place and not removed.
1. Provide bracing and shoring.
 2. Prevent movement or settlement of adjacent structures.
 3. Stop work immediately if adjacent structures appear to be in danger.
- E. Minimize production of dust due to demolition operations. Do not use water if that will result in ice, flooding, sedimentation of public waterways or storm sewers, or other pollution.
- F. Hazardous Materials:
1. Hazardous Materials: Comply with 29 CFR 1926 and state and local regulations.

3.03 DEBRIS AND WASTE REMOVAL

- A. Remove debris, junk, and trash from site.
- B. Leave site in clean condition, ready for subsequent work.
- C. Clean up spillage and wind-blown debris from public and private lands.

END OF SECTION

SECTION 044313
STONE MASONRY VENEER

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Stone Gabion Cladding at walls and roof.
- B. Metal angles, anchors and accessories for anchored gabion baskets.

1.02 RELATED REQUIREMENTS

- A. Section 05 12 00 - Structural Steel Framing: material requirements for steel angles, anchorage devices and other components as part of gabion basket assemblies.

1.03 REFERENCE STANDARDS

- A. ASTM C1528/C1528M - Standard Guide for Selection of Dimension Stone; 2020.

1.04 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Product Data: Provide data on stone units, gabion baskets ,and reinforcement.
- C. Shop Drawings: Indicate component details, bearing, anchorage, loading, welds, and type and location of fasteners, and accessories or items required of related work.

1.05 MOCK-UP

- A. Locate as indicated on drawings.
- B. Mock-up may remain as part of the Work.

PART 2 PRODUCTS

2.01 STONE

- A. Stone, General: See recommendations in ASTM C1528/C1528M.
- B. Stone to be sourced from quarry within the park. COR will provide information on available quarry.

2.02 ACCESSORIES - ANCHORED VENEER

- A. Gabion Baskets: Gabions shall be of a single unit construction. The base, ends, sides, and lid shall be either welded into a single unit or shall be connected in such a manner that strength and flexibility at the connection are at least equal to that of the wire mesh. The gabions shall be fabricated in such a manner that they can be assembled at the construction site with spiral binders or high tensile steel hog rings and pre-formed stiffeners into rectangular baskets of the specified size. Gabions shall be divided into cells of equal length, not more than 1 cubic yard by diaphragms made of the same wire mesh as used for the gabion body. Each gabion shall be fabricated with the necessary diaphragms secured in proper position on the base in such a manner that no additional tying at the base will be necessary.
- B. Wire: 6 Gage, Welded Wire Fabric, ASTM A1064,A370.
- C. Steel Angles and Supports: provide steel angles, supports and other items as indicated on drawings. Confirm to material requirements in Section 05 12 00.
- D. Configuration: Wire mesh at 2" x 2" square grid.
- E. Construction: Gabions shall first be assembled individually as empty units. Each gabion shall be manufactured with the necessary panels, perimeter angle (structural backbone) as indicated, properly spaced and secured, so they can be hung into position at the construction site with no additional tying of the rotation joint.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that support work and site conditions are ready to receive work of this section.

3.02 PREPARATION - GABION VENEER

- A. Establish lines, levels, and coursing. Protect from disturbance.
- B. Shop fabricated gabion baskets with integral steel angle frames and attachment devices.
- C. Install attachment devices plumb and true.
- D. At contractor's option, gabions may be prefilled with stone at base of mountain and hoisted to site for installation on building, or gabions may be installed on building prior to filling baskets with stone.

3.03 INSTALLATION - GABION VENEER

- A. Size stone to fill Gabion Baskets fully of a stone diameter that exceeds the wire fabric opening size.
- B. Arrange Gabion Baskets consistent joint width.

3.04 TOLERANCES

- A. Maximum Variation From Unit to Adjacent Unit: 1/8 inch (3.2 mm).
- B. Maximum Variation from Plane of Wall: 1/4 inch in 10 feet (6 mm in 3 m) and 1/2 inch in 20 feet (13 mm in 6 m) or more.
- C. Maximum Variation from Plumb: 1/4 inch (6 mm) per story non-cumulative.
- D. Maximum Variation of Joint Thickness: 1/8 inch in 3 feet (3 mm in 1 m).

3.05 CLEANING

- A. Remove excess mortar as work progresses, and upon completion of work.
- B. Use non-metallic tools in cleaning operations.

END OF SECTION

SECTION 05 12 00
STRUCTURAL STEEL FRAMING

PART 1 GENERAL

3.01 REFERENCE STANDARDS

- A. AISC (MAN) - Steel Construction Manual; 2023, with Errata (2024).
- B. AISC 303 - Code of Standard Practice for Steel Buildings and Bridges; 2022.
- C. ASTM A36/A36M - Standard Specification for Carbon Structural Steel; 2019.
- D. ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless; 2022.
- E. ASTM A123/A123M - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products; 2017.
- F. ASTM A500/A500M - Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes; 2023.
- G. ASTM A563/A563M - Standard Specification for Carbon and Alloy Steel Nuts (Inch and Metric); 2021a.
- H. ASTM A992/A992M - Standard Specification for Structural Steel Shapes; 2022.
- I. ASTM A1011/A1011M - Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength; 2023.
- J. ASTM E94/E94M - Standard Guide for Radiographic Examination Using Industrial Radiographic Film; 2017.
- K. ASTM E164 - Standard Practice for Contact Ultrasonic Testing of Weldments; 2019.
- L. ASTM E165/E165M - Standard Practice for Liquid Penetrant Testing for General Industry; 2023.
- M. ASTM E709 - Standard Guide for Magnetic Particle Testing; 2021.
- N. ASTM F436/F436M - Standard Specification for Hardened Steel Washers Inch and Metric Dimensions; 2019.
- O. ASTM F959/F959M - Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series; 2017a.
- P. ASTM F1554 - Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength; 2020.
- Q. ASTM F3125/F3125M - Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength; 2023.
- R. AWS A2.4 - Standard Symbols for Welding, Brazing, and Nondestructive Examination; 2020.
- S. AWS B2.1/B2.1M - Specification for Welding Procedure and Performance Qualification; 2021, with Errata (2023).
- T. AWS D1.1/D1.1M - Structural Welding Code - Steel; 2020, with Errata (2023).
- U. IAS AC172 - Accreditation Criteria for Fabricator Inspection Programs for Structural Steel AC172; 2019.
- V. RCSC (HSBOLT) - Specification for Structural Joints Using High-Strength Bolts; Research Council on Structural Connections; 2020.

3.02 SUBMITTALS

- A. See Section 01 30 00 - Administrative Requirements, for submittal procedures.
- B. Shop Drawings:
 - 1. Indicate profiles, sizes, spacing, locations of structural members, openings, attachments, and fasteners.
 - 2. Connections not detailed.
 - 3. Indicate cambers and loads.
 - 4. Indicate welded connections with AWS A2.4 welding symbols. Indicate net weld lengths.
- C. Manufacturer's Mill Certificate: Certify that products meet or exceed specified requirements.
- D. Mill Test Reports: Indicate structural strength, destructive test analysis and non-destructive test analysis.
- E. Fabricator Test Reports: Comply with ASTM A1011/A1011M.
- F. Welders' Qualification Statement: Welders' certificates in accordance with AWS B2.1/B2.1M and dated no more than 12 months before start of scheduled welding work.
- G. Designer's Qualification Statement.
- H. Fabricator's Qualification Statement.
- I. Fabricator's Qualification Statement: Provide documentation showing steel fabricator is accredited under IAS AC172.

3.03 QUALITY ASSURANCE

- A. Fabricate structural steel members in accordance with AISC (MAN) "Steel Construction Manual."
- B. Welder Qualifications: Welding processes and welding operators qualified in accordance with AWS D1.1/D1.1M and no more than 12 months before start of scheduled welding work.
- C. Fabricator Qualifications: A qualified steel fabricator that is accredited by the International Accreditation Service (IAS) Fabricator Inspection Program for Structural Steel in accordance with IAS AC172.
- D. Erector: Company specializing in performing the work of this section with minimum 5 years of documented experience.
- E. Design connections not detailed on drawings under direct supervision of a Professional Structural Engineer experienced in design of this work and licensed in the State in which the Project is located.

PART 2 PRODUCTS

4.01 MATERIALS

- A. Steel Angles and Channels: ASTM A36/A36M.
- B. Steel Plate: ASTM A572/A572M, Grade 50
- C. Steel W Shapes and Tees: ASTM A992/A992M.
- D. Rolled Steel Structural Shapes: ASTM A992/A992M.
- E. Steel Plates and Bars: ASTM A572/A572M, Grade 50 (345) high-strength, columbium-vanadium steel.
- F. Cold-Formed Structural Tubing: ASTM A500/A500M, Grade C.
- G. Pipe: ASTM A53/A53M, Grade B, Finish black.
- H. High-Strength Structural Bolts, Nuts, and Washers: ASTM F3125/F3125M, Type 1, with matching compatible ASTM A563/A563M nuts and ASTM F436/F436M washers.
- I. Tension Control Bolts: Twist-off type; ASTM F3125/F3125M.
- J. Unheaded Anchor Rods: ASTM F1554, Grade 36, plain, with matching ASTM A563/A563M nuts and ASTM F436/F436M Type 1 washers.

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- K. Headed Anchor Rods: ASTM F1554 Grade 36, plain.
- L. Load Indicator Washers: Provide washers complying with ASTM F959/F959M at connections requiring high-strength bolts.
- M. Welding Materials: AWS D1.1/D1.1M; type required for materials being welded.
- N. Grout: ASTM C1107/C1107M; Non-shrink; premixed compound consisting of non-metallic aggregate, cement, water reducing and plasticizing agents.
 - 1. Minimum Compressive Strength at 48 Hours: 2,000 pounds per square inch.
 - 2. Minimum Compressive Strength at 28 Days: 7,000 pounds per square inch.
- O. Shop and Touch-Up Primer: Fabricator's standard, complying with VOC limitations of authorities having jurisdiction.
- P. Touch-Up Primer for Galvanized Surfaces: Fabricator's standard, complying with VOC limitations of authorities having jurisdiction.

4.02 FABRICATION

- A. Shop fabricate to greatest extent possible.
- B. Fabricate connections for bolt, nut, and washer connectors.

4.03 FINISH

- A. Shop prime structural steel members. Do not prime surfaces that will be fireproofed, field welded, in contact with concrete, or high strength bolted.
- B. Galvanize exterior structural steel members exposed to weather or where indicated in the drawings to comply with ASTM A123/A123M. Provide minimum 1.7 oz/sq ft galvanized coating.

PART 3 EXECUTION

5.01 ERECTION

- A. Erect structural steel in compliance with AISC 303.
- B. Allow for erection loads and provide sufficient temporary bracing to maintain structure in safe condition, plumb, and in true alignment until completion of erection and installation of permanent bracing.
- C. Do not field cut or alter structural members without approval of Engineer.
- D. After erection, prime welds, abrasions, and surfaces not shop primed, except surfaces to be in contact with concrete.

5.02 FIELD QUALITY CONTROL

- A. An independent testing agency will perform field quality control tests, as specified in Section 01 40 00 - Quality Requirements.
- B. High-Strength Bolts: Provide testing and verification of field-bolted connections in accordance with RCSC (HSBOLT) "Specification for Structural Joints Using High-Strength Bolts," testing at least 10% percent of bolts at each connection.
- C. Welded Connections: Visually inspect all field-welded connections and test at least 5 percent of welds using one of the following:
 - 1. Radiographic testing performed in accordance with ASTM E94/E94M.
 - 2. Ultrasonic testing performed in accordance with ASTM E164.
 - 3. Liquid penetrant inspection performed in accordance with ASTM E165/E165M.
 - 4. Magnetic particle inspection performed in accordance with ASTM E709.

END OF SECTION

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**SECTION 054000
COLD-FORMED METAL FRAMING**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Slotted channel framing.

1.02 DEFINITIONS

- A. General: See AISI S240 for definitions of terms used in this section.

1.03 REFERENCE STANDARDS

- A. AISI S240 - North American Standard for Cold-Formed Steel Structural Framing; 2015, with Errata (2020).
- B. ASTM A1003/A1003M - Standard Specification for Steel Sheet, Carbon, Metallic- and Nonmetallic-Coated for Cold-Formed Framing Members; 2015.
- C. ASTM C1007 - Standard Specification for Installation of Load Bearing (Transverse and Axial) Steel Studs and Related Accessories; 2020 (Reapproved 2024).

1.04 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Provide data on cold-formed steel structural members; include material descriptions and base steel thickness.
- C. Product Data: Provide manufacturer's data on factory-made connectors and mechanical fasteners, showing compliance with requirements.

PART 2 PRODUCTS

2.01 MATERIALS

- A. Steel Sheet: ASTM A1003/A1003M, subject to the ductility limitations indicated in AISI S240.

2.02 SLOTTED CHANNEL FRAMING

- A. Framing shall be a strut type metal framing system (Strut System)
- B. Strut System shall be used to support electrical equipment, solar panels and other devices.
- C. Hot-Dipped Galvanized per ASTM A123 or A153
 - 1. Zinc coated after all manufacturing operations are complete
 - 2. Zinc coating thickness shall be G65 (2.6 mils = 1.50 oz./ sq. ft. surface area)

2.03 CONNECTIONS

- A. Performance Requirements: Provide connections in compliance with requirements of AISI S240.
- B. Fixed Connections: Provide nonmovement devices for tie-down to gabion structures and other components.
- C. Bridging Connections: Provide mechanical load-transferring devices that accommodate wind load torsion and weak axis buckling induced by axial compression loads. Provide bridging connectors where indicated on the drawings.

PART 3 EXECUTION

3.01 INSTALLATION - GENERAL

- A. Install structural members and connections in compliance with ASTM C1007.

END OF SECTION

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**SECTION 055000
METAL FABRICATIONS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Shop fabricated steel items.

1.02 RELATED REQUIREMENTS

- A. Section 055100 - Metal Stairs.

1.03 REFERENCE STANDARDS

- A. ASTM A36/A36M - Standard Specification for Carbon Structural Steel; 2019.
- B. ASTM A283/A283M - Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates; 2024.
- C. ASTM A307 - Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength; 2021.
- D. ASTM F3125/F3125M - Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi, 144 ksi, and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength; 2025.
- E. AWS D1.1/D1.1M - Structural Welding Code - Steel; 2025.

1.04 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Shop Drawings: Indicate profiles, sizes, connection attachments, reinforcing, anchorage, size and type of fasteners, and accessories. Include erection drawings, elevations, and details where applicable.

PART 2 PRODUCTS

2.01 MATERIALS - STEEL

- A. Steel Sections: ASTM A36/A36M.
- B. Plates: ASTM A283/A283M.
- C. Mechanical Fasteners: Same material as or compatible with materials being fastened; type consistent with design and specified quality level.
- D. Bolts, Nuts, and Washers: ASTM A307, Grade A, plain.
- E. Bolts, Nuts, and Washers: ASTM F3125/F3125M, Type 1, plain.
- F. Welding Materials: AWS D1.1/D1.1M; type required for materials being welded.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install items plumb and level, accurately fitted, free from distortion or defects.
- B. Provide for erection loads, and for sufficient temporary bracing to maintain true alignment until completion of erection and installation of permanent attachments.
- C. Obtain approval prior to site cutting or making adjustments not scheduled.

END OF SECTION

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**SECTION 055100
METAL STAIRS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Stairs and landings with grating treads.
- B. Structural steel stair framing and supports.
- C. Handrails and guards.
- D. Metal grating.

1.02 RELATED REQUIREMENTS

- A. Section 051200 - Structural Steel Framing: Structural members that stairs, railings and landings are attached to.

1.03 REFERENCE STANDARDS

- A. ASCE 7 - Minimum Design Loads and Associated Criteria for Buildings and Other Structures; Most Recent Edition Cited by Referring Code or Reference Standard.
- B. ASTM A36/A36M - Standard Specification for Carbon Structural Steel; 2019.
- C. ASTM A53/A53M - Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless; 2024.
- D. ASTM A123/A123M - Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products; 2024.
- E. ASTM A153/A153M - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware; 2023.
- F. ASTM A500/A500M - Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes; 2023.
- G. ASTM A501/A501M - Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing; 2021.
- H. ASTM F3125/F3125M - Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi, 144 ksi, and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength; 2025.
- I. AWS A2.4 - Standard Symbols for Welding, Brazing, and Nondestructive Examination; 2020.
- J. AWS B2.1/B2.1M - Specification for Welding Procedure and Performance Qualification; 2021, with Errata (2023).
- K. AWS D1.1/D1.1M - Structural Welding Code - Steel; 2025.
- L. NAAMM AMP 510 - Metal Stairs Manual; 1992.
- M. NAAMM MBG 531 - Metal Bar Grating Manual; 2024.
- N. NAAMM MBG 532 - Heavy Duty Metal Bar Grating Manual; 2024.
- O. SSPC-Paint 20 - Zinc-Rich Coating (Type I - Inorganic, and Type II - Organic); 2019.

1.04 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Shop Drawings: Indicate profiles, sizes, connection attachments, reinforcing, anchorage, size and type of fasteners, and accessories.
 - 1. Indicate welded connections using standard AWS A2.4 welding symbols. Indicate net weld lengths.
 - 2. Include the design engineer's seal and signature on each sheet of shop drawings.

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- C. Design Data, Seismic Performance: Submit documentation that stairs meet performance requirements specified.
- D. Welders' Qualification Statement: Welders' certificates in accordance with AWS B2.1/B2.1M and dated no more than 12 months before start of scheduled welding work.
- E. Designer's Qualification Statement.

1.05 QUALITY ASSURANCE

- A. Structural Designer Qualifications: Professional Structural Engineer experienced in design of this work and licensed in the State of Washington, or personnel under direct supervision of such an engineer.
- B. Welder Qualifications: Welding processes and welding operators qualified in accordance with AWS D1.1/D1.1M and dated no more than 12 months before start of scheduled welding work.
- C. Fabricator Qualifications:
 - 1. A company specializing in manufacturing products specified in this section, with not less than five years of documented experience.

PART 2 PRODUCTS

2.01 METAL STAIRS - GENERAL

- A. Metal Stairs: Provide stairs of the design specified, complete with landing platforms, vertical and horizontal supports, railings, and guards, fabricated accurately for anchorage to each other and to building structure.
 - 1. Regulatory Requirements: Provide stairs and railings that comply with most stringent requirements of local, state, and federal regulations; where requirements of Contract Documents exceed those of regulations, comply with Contract Documents.
 - 2. Structural Design: Provide complete stair and railing assemblies that comply with the following:
 - a. Stair Capacity: Uniform live load of 100 lb/sq ft (4.7 kPa) and a concentrated load of 300 lb (and a concentrated load of 14.4 kg) with deflection of stringer or landing framing not to exceed 1/360 of span.
 - b. Railing Assemblies: Comply with applicable local code.
 - c. Seismic Performance: Stairs designed to withstand the effects of earthquake motions determined according to ASCE 7.
 - 1) Component Importance Factor: 1.5.
 - 2) No permanent inelastic deformation occurs under movements equal 2.5 percent interstory drift, minimum.
 - 3. Dimensions: As indicated on drawings.
 - 4. Shop assemble components; disassemble into largest practical sections suitable for transport and access to site.
 - 5. No sharp or rough areas on exposed travel surfaces and surfaces accessible to touch.
 - 6. Separate dissimilar metals using paint or permanent tape.
- B. Metal Jointing and Finish Quality Levels:
 - 1. Architectural: All joints as inconspicuous as possible, whether welded or mechanical.
 - a. Welded Joints: Continuously welded and ground smooth and flush.
 - b. Mechanical Joints: Butted tight, flush, and hairline; concealed fastenings only.
 - c. Exposed Edges and Corners: Eased to small uniform radius.
 - d. Metal Surfaces to be Painted: Sanded or ground smooth, suitable for highest quality gloss finish.
- C. Fasteners: Same material or compatible with materials being fastened; type consistent with design and specified quality level.
- D. Anchors and Related Components: Same material and finish as item to be anchored, except where specifically indicated otherwise; provide all anchors and fasteners required.

2.02 METAL STAIRS WITH GRATING TREADS

- A. Jointing and Finish Quality Level: Architectural, as defined above.
- B. Risers: Closed.
- C. Treads: Steel bar grating.
 - 1. Grating Type: Welded.
 - 2. Bearing Bar Depth: 1 inch (25 mm), minimum.
 - 3. Top Surface: Standard.
 - 4. Nosing: Checkered plate.
 - 5. Nosing Width: 1-1/4 inch (32 mm), minimum.
 - 6. Anchorage to Stringers: End plates welded to grating, bolted to stringers.
- D. Stringers: Rolled steel channels.
 - 1. Stringer Depth: 10 inches (250 mm).
 - 2. End Closure: Sheet steel, 14 gauge, 0.075 inch (1.9 mm) minimum; welded across ends.
- E. Railings: Steel pipe railings.
- F. Finish: Galvanized after fabrication.

2.03 HANDRAILS AND GUARDS

- A. Guards:
 - 1. Top Rails: Round pipe or tube rails unless otherwise indicated.
 - a. Outside Diameter: 1-1/4 inch (32 mm), minimum, to 1-1/2 inches (38 mm), maximum.
 - 2. Infill at Pipe Railings: Pipe or tube rails sloped parallel to stair.
 - a. Outside Diameter: 1-1/4 inch (31 mm).
 - b. Material: Steel pipe or tube, round.
 - c. Jointing: Welded and ground smooth and flush.
 - 3. End and Intermediate Posts: Same material and size as top rails.
 - a. Horizontal Spacing: As indicated on drawings.
 - b. Mounting: Welded to top surface of stringer.

2.04 MATERIALS

- A. Steel Sections: ASTM A36/A36M.
- B. Steel Tubing: ASTM A500/A500M or ASTM A501/A501M structural tubing, round and shapes as indicated.
- C. Pipe: ASTM A53/A53M Grade B Schedule 40, black finish.
- D. Gratings: Bar gratings that comply with NAAMM MBG 531 or NAAMM MBG 532, whichever applies based on bar sizes.

2.05 ACCESSORIES

- A. Steel Bolts, Nuts, and Washers: ASTM F3125/F3125M, Type 1, and galvanized to ASTM A153/A153M where connecting galvanized components.
- B. Welding Materials: AWS D1.1/D1.1M; type required for materials being welded.
- C. Touch-Up Primer for Galvanized Surfaces: SSPC-Paint 20 Type I - Inorganic, and comply with VOC limitations of authorities having jurisdiction.

2.06 SHOP FINISHING

- A. Clean surfaces of rust, scale, grease, and foreign matter prior to finishing.
- B. Galvanizing: Hot-dip galvanize to minimum requirements of ASTM A123/A123M.
 - 1. Touch up abraded areas after fabrication using specified touch-up primer for galvanized surfaces.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install components plumb and level, accurately fitted, free from distortion or defects.
- B. Allow for erection loads, and for sufficient temporary bracing to maintain true alignment until completion of erection and installation of permanent attachments.
- C. Provide welded field joints where specifically indicated on drawings. Perform field welding in accordance with AWS D1.1/D1.1M.
- D. Other field joints may be either welded or bolted provided the result complies with the limitations specified for jointing quality levels.
- E. Obtain approval prior to site cutting or creating adjustments not scheduled.
- F. After erection, prime welds, abrasions, and surfaces not shop primed or galvanized, except surfaces to be in contact with concrete.

3.02 TOLERANCES

- A. Maximum Variation From Plumb: 1/4 inch (6 mm) per story, non-cumulative.
- B. Maximum Offset From True Alignment: 1/4 inch (6 mm).

END OF SECTION

**SECTION 061000
ROUGH CARPENTRY**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Nonstructural dimension lumber framing.
- B. Sheathing.
- C. Plywood flooring.
- D. Plywood snow shutter.

1.02 RELATED REQUIREMENTS

- A. Section 051200 - Structural Steel Framing: Prefabricated beams and columns for support of wood framing.

1.03 REFERENCE STANDARDS

- A. PS 1 - Structural Plywood; 2023.
- B. PS 2 - Performance Standard for Wood Structural Panels; 2018.
- C. PS 20 - American Softwood Lumber Standard; 2025.

PART 2 PRODUCTS

2.01 GENERAL REQUIREMENTS

- A. Dimension Lumber: Comply with PS 20 and requirements of specified grading agencies.
 - 1. Species: Douglas Fir-Larch, unless otherwise indicated.
 - 2. If no species is specified, provide species graded by the agency specified; if no grading agency is specified, provide lumber graded by grading agency meeting the specified requirements.
 - 3. Grading Agency: Grading agency whose rules are approved by the Board of Review, American Lumber Standard Committee at www.alsc.org, and who provides grading service for the species and grade specified; provide lumber stamped with grade mark unless otherwise indicated.

2.02 CONSTRUCTION PANELS

- A. Roof and Wall Sheathing: PS 2 type, rated Structural I Sheathing.
 - 1. Bond Classification: Exterior.
 - 2. Span Rating: 60.
 - 3. Performance Category: 1/2 PERF CAT.
- B. Plywood flooring
 - 1. Grade: A1 Hardwood/B
 - 2. Thickness: as indicated on drawings.
 - 3. Edges: Square.
- C. Plywood window snow shutter
 - 1. Grade: A/C
 - 2. Thickness: as indicated on drawings.
 - 3. Finish: Painted

PART 3 EXECUTION

3.01 INSTALLATION - GENERAL

- A. Select material sizes to minimize waste.
- B. Reuse scrap to the greatest extent possible; clearly separate scrap for use on site as accessory components, including: shims, bracing, and blocking.

3.02 INSTALLATION OF CONSTRUCTION PANELS

- A. Roof Sheathing: Secure panels with long dimension perpendicular to framing members, with ends staggered and over firm bearing.
 - 1. Nail panels to framing; staples are not permitted.
- B. Wall Sheathing: Secure with long dimension perpendicular to wall studs, with ends over firm bearing and staggered, using nails, screws, or staples.
- C. Flooring Plywood: Secure in layout per Finish Floor Plan. Utilize countersunk screws. Predrill. 24" OC min. each direction.

3.03 TOLERANCES

- A. Framing Members: 1/4 inch (6 mm) from true position, maximum.
- B. Surface Flatness of Floor: 1/8 inch in 10 feet (1 mm/m) maximum, and 1/4 inch in 30 feet (7 mm in 10 m) maximum.
- C. Variation from Plane, Other than Floors: 1/4 inch in 10 feet (2 mm/m) maximum, and 1/4 inch in 30 feet (7 mm in 10 m) maximum.

3.04 CLEANING

- A. Waste Disposal: See Section 017419 - Construction Waste Management and Disposal.
 - 1. Comply with applicable regulations.
 - 2. Do not burn scrap on project site.
 - 3. Do not burn scraps that have been pressure treated.
 - 4. Do not send materials treated with pentachlorophenol, CCA, or ACA to co-generation facilities or "waste-to-energy" facilities.
- B. Do not leave wood, shavings, sawdust, etc. on the ground or buried in fill.
- C. Prevent sawdust and wood shavings from entering the storm drainage system.

END OF SECTION

SECTION 06 15 29 - NAIL-LAMINATED TIMBER

PART 1 – GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

A. Section includes nail-laminated timber (NLT) floor and roof decking, prefabricated in panels.

B. Related Sections:

- Section 01 81 13 “Sustainable Design Requirements” for LEED requirements.
- Section 05 12 00 “Structural Steel Framing” for custom-fabricated steel connection brackets.
- Section 06 10 00 “Rough Carpentry” for dimension lumber framing.
- Section 06 16 00 “Sheathing” for floor and roof sheathing.
- Section 09 91 00 “Painting” for sealing and finishing requirements.

C. References:

- ALSC, American Lumber Standard Committee Board of Review.
- ALSC DOC PS20-15 American Softwood Lumber Standard.
- APA, The Engineered Wood Association.
- ASTM A153/A153M-16a Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
- ASTM E488/E488M-15 Standard Test Methods for Strength of Anchors in Concrete Elements.
- ASTM F1667-15 Standard Specification for Driven Fasteners: Nails, Spikes, and Staples.
- ICC-ES ESR-1539 Power-Driven Staples and Nails.
- NDS 2024 National Design Specification for Wood Construction.
- 2024 International Building Code.
- For Projects overseen by a Construction Manager or Design-Build Contractor in lieu of a General Contractor, references to “Contractor” shall apply to the relevant Subcontractor(s).

1.3 ACTION SUBMITTALS

A. Product Data: For each type of factory-fabricated product. Submit proposed sealer for review and approval.

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B. Shop Drawings: Connections and details, joint patterns, material specifications, and finishes, including an erection layout.

C. Provide a letter outlining steps to be taken during construction to ensure adequate weather protection of wood structures.

D. Sustainable Design Submittals: Per Section 01 81 13.1.4 INFORMATIONAL SUBMITTALS

A. In lieu of grade stamping lumber exposed to view, submit manufacturer's certificate certifying that products meet or exceed specified requirements.

B. The fabricator and erector shall submit a QA/QC log of items such as but not limited to:

- Environmental conditions at all stages, such as during fabrication, storage, transportation, erection, and ideally until building is completely finished.
- Actual length, thickness, and width of the NLT panels. Length, width, thickness, and diagonal measurement are to be noted on the top surface of the panels.
- Site deliveries, including verified load manifests with notes of damaged or missing materials and elements.
- Material and element install with sign off for QC on hardware/fastener installation.
- Equipment used, such as but not limited to torque drills (with torque clutch) for screw installation through steel plates etc.
- Any changes or modifications. The inclusion of representative pictures within the log is required.

1.5 QUALITY ASSURANCE

A. Perform Work in accordance with NDS and the following agencies:

- Lumber Grading Agency: Certified by ALSC.
- Plywood Grading Agency: Certified by APA.

B. Build mockups to demonstrate aesthetic effects and set quality standards for materials and execution.

- Build one complete NLT panel with a minimum width of 48 inches.
- Mockup must illustrate typical wood appearance, coating, and finish.
- Keep mockup available to view as the standard of Work for remaining fabrication.
- Approved mockups may become part of the completed Work if undisturbed at the time of Substantial Completion.

C. Build NLT panels in a shop environment for quality control. Shop fit panels during fabrication. Review with Consultant prior to proceeding further.

1.6 DELIVERY, STORAGE, AND HANDLING

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A. Store all materials and assembled NLT panels under cover with proper drainage. Take particular care to protect exposed end grain. Protect from staining and damage at all times during fabrication, transportation, and installation.

B. Take all necessary precautions to keep NLT dry during and after installation, including temporary sloping tarps and UV protection.

PART 2 – PRODUCTS

2.1 DIMENSION LUMBER

A. Lumber Grading Rules: DOC PS20 and applicable rules of grading agencies indicated. If no grading agency is indicated, comply with the applicable rules of any rules-writing agency certified by the ALSC Board of Review. Grade lumber by an agency certified by the ALSC Board of Review to inspect and grade lumber under the rules indicated.

B. Do not grade stamp lumber exposed to view. Deliver to site with certificates as to species, grades, stress grades, seasoning, moisture content, and other evidence as required to show compliance with the Specifications.

C. Dress lumber, S4S, unless otherwise indicated.

D. Maximum Moisture Content: 19% unless noted otherwise.

E. Wood Members: SPF #1/#2 unless noted otherwise on the Drawings.

F. Finger-Jointed Lumber: Certified Exterior Joints, Heat-Resistant Adhesive (HRA), in accordance with ALSC.2.2 CONNECTORSA. Provide fasteners of size and type indicated that comply with requirements specified in this article for material and manufacture. Provide fasteners with hot-dip zinc coating complying with ASTM A153 or of Type 304 stainless steel.

B. Nails, Brads, and Staples: ASTM F1667.

C. Power-Driven Fasteners: ICC-ES ESR-1539.

D. Screws, Tight-Fit Pins and Bolts, Through-Bolts, Glued-In Rods, and Specialty Connectors: As specified on the Drawings.

E. Expansion Anchors: Anchor bolt and sleeve assembly with capability to sustain, without failure, a load equal to six times the load imposed when installed in unit masonry assemblies and equal to four times the load imposed when installed in concrete as determined by testing per ASTM E488 conducted by a qualified independent testing and inspecting agency.

F. Metal Straps and Ties: Galvanized Simpson Strong-Tie connectors or approved equal where required.

G. Structural Steel Connectors: As specified in Section 05 12 00. All steel and connectors shall be hot dip galvanized unless noted otherwise. Fabricate steel hardware and connections with joints neatly fitted, welded, and ground smooth. Test fit in shop.

2.3 MISCELLANEOUS MATERIALS

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A. Moisture Barrier: Light gauge metal, asphalt-impregnated building paper, 1/4-inch-thick closed-cell foam gasket material, saturated felt roll roofing, or 6 mil polyethylene.

B. Wood Sealer: As specified in Section 09 91 00. Sealer shall be compatible with indicated finish. End sealer shall be effective in retarding the transmission of moisture at cross-grain cuts.

C. Specialty and/or proprietary products shown on the Drawings have been selected and specified based on the manufacturer's representation. The Consultant shall not become guarantor of the product. Install specialty products in strict conformance with the manufacturer's recommendations. Contractor is responsible for proper workmanship during installation. 2.4 PREFABRICATIONA. Hand select members to ensure straightness and architectural-quality appearance.

- No wane, knot holes, grade stamps, or stains are permitted to be visible in the completed structure.
- Where pine beetle kill wood is specified, hand select all members to ensure beetle staining is visible. Ensure staining is spatially distributed throughout NLT panels; avoid clusters of stained boards.
- Assume a minimum of 30% - 40% lumber rejection rate to achieve acceptable appearance with #2 grade material. Higher grade material (e.g. J-grade or MSR lumber) will reduce the rejection rate and may be substituted for #2 grade material at Contractor's option.

B. Place soffits of timbers so the least number of checks and knots will be visible in the completed structure.

C. Arrange laminations in staggered pattern or aligned joint pattern as indicated on the Drawings.

- For staggered pattern, stagger and nail together as indicated on the Drawings.
- For aligned joint pattern, place laminations with joints centered over support members below. No joints are to be visible from below. Nail together as indicated on the Drawings.

D. Use common steel wire nails unless noted otherwise. Make tight connections between members. Install fasteners without splitting wood. Drive nails snug but do not countersink nail heads unless noted otherwise.

E. Substitution of common nails with power-driven nails of the same length and diameter is acceptable. Substitution of power-driven nails of smaller diameter is permitted only with the Consultant's approval.

F. Clearly mark top surface of NLT panels for identification during erection.

G. Apply a saturation coat of end sealer to ends of members at time of cutting and treat field-cut holes similarly.

PART 3 – EXECUTION

3.1 EXAMINATION

A. Examine substrates and conditions, with Installer present, for compliance with requirements for installation tolerances and other conditions affecting performance of NLT. Proceed with installation only

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after unsatisfactory conditions have been corrected.3.2 PREPARATIONA. Coordinate layout and installation of NLT panels with other construction that penetrates panels or is supported by them, including mechanical and electrical work.

B. Protect NLT from weather. If, despite protection, NLT becomes wet, apply end sealer to cut ends and allow to dry before installation.

3.3 INSTALLATION

A. Set NLT true and straight to correct layout, levels, and locations. Provide temporary bracing as required.

B. Securely attach NLT to supports with fasteners as indicated. Use concealed fasteners where possible.

C. Install continuous NLT panels without end joints where possible. Where end joints are necessary, stagger joints in adjacent rows.

D. Install plywood or OSB sheathing over NLT where required for diaphragm action or additional strength, fastened with nails or screws as specified.

E. Comply with manufacturer's written instructions for installation.

F. Do not install damaged components; replace as necessary.

3.4 FIELD QUALITY CONTROL

A. Testing Agency: Owner will engage a qualified testing agency to perform tests and inspections.

B. Tests: As determined by testing agency from within specified test methods.

3.5 PROTECTION

A. Protect installed products from damage during construction.

B. Protect NLT from weather exposure until building enclosure is complete.

END OF SECTION 06 15 29

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SECTION 06 16 00
STRUCTURAL SHEATHING

PART 1 – GENERAL

1.1 SUMMARY

A. Section Includes:

1. Structural sheathing for walls, roofs, and floors, including plywood and oriented strand board (OSB).
2. Accessories including fasteners, edge supports, and blocking.

B. Related Requirements:

1. Section 06 10 00 - Rough Carpentry for general wood framing and blocking.
2. Section 06 17 00 - Shop-Fabricated Structural Wood for coordination with other engineered wood products.
3. Section 07 21 00 - Thermal Insulation for coordination with insulating sheathing where applicable.

1.2 REFERENCES

A. Reference Standards:

1. APA - The Engineered Wood Association: PRP-108 - Performance Standards and Policies for Structural-Use Panels.
2. ASTM International: ASTM D3043 - Standard Test Methods for Structural Panels in Flexure.
3. National Institute of Standards and Technology (NIST): PS 1 - Structural Plywood; PS 2 - Performance Standard for Wood Structural Panels.
4. National Design Specification (NDS) for Wood Construction, published by the American Wood Council.
5. International Code Council (ICC): International Building Code (IBC) and International Residential Code (IRC).

1.3 SUBMITTALS

A. Product Data: Submit manufacturer's technical data sheets, including panel grades, span ratings, sizes, and installation instructions for structural sheathing.

B. Shop Drawings: Submit details showing sheathing layout, panel orientations, joint treatments, and fastening patterns.

C. Certificates: Submit certification that structural sheathing complies with specified standards, including APA trademarks or equivalent.

D. Warranty Documentation: Submit manufacturer's warranty information.

1.4 QUALITY ASSURANCE

A. Manufacturer Qualifications: Manufacturer with products certified by APA or equivalent agency, complying with PS 1 or PS 2.

B. Installer Qualifications: Experienced in installing wood structural panels similar to those specified.

C. Source Limitations: Obtain structural sheathing from a single manufacturer to ensure compatibility.

D. Identification: Each panel shall bear the stamp of a qualified inspection and testing agency, indicating grade, bond classification, span rating, and performance category.

1.5 DELIVERY, STORAGE, AND HANDLING

A. Delivery: Deliver materials in manufacturer's original packaging, protected from weather.

B. Storage: Store panels flat, off the ground, supported at intervals to prevent warping, and protected from moisture and direct sunlight.

C. Handling: Handle with care to avoid damage; do not drop or drag panels.

1.6 WARRANTY

A. Manufacturer's Warranty: Provide a limited warranty against manufacturing defects, typically 25 years or as specified by manufacturer.

PART 2 – PRODUCTS

2.1 MANUFACTURERS

A. Acceptable Manufacturers:

1. LP Building Solutions (LP OSB Sheathing).
2. APA - The Engineered Wood Association member mills for plywood and OSB.
3. Or approved equal, subject to compliance with requirements.

2.2 MATERIALS

A. Structural Sheathing:

1. Description: Wood structural panels complying with PS 1 for plywood or PS 2 for OSB and other mat-formed panels, suitable for subfloor, wall, roof, diaphragm, and shear wall applications.
2. Types: a. Plywood: Veneer plies arranged in cross-aligned layers, bonded with waterproof adhesive. b. Oriented Strand Board (OSB): Thin rectangular wood strands arranged in oriented layers, bonded with waterproof adhesive.
3. Bond Classification: Exposure 1 for protected applications; Exterior where exposed to weather.
4. Moisture Content: Less than 16 percent at time of shipment.
5. Treatment (Where Required): Preservative-treated complying with AWPA standards for specified use category.
6. Grades: APA Rated Sheathing or Structural I Sheathing as indicated on Drawings.

B. Sizes:

1. Nominal Dimensions: 4 feet by 8 feet standard; other sizes as required.
2. Performance Categories (Thickness): as indicated.
3. Edges: Square edges standard

C. Span Ratings and Design Values: Panels shall meet performance criteria for concentrated loads, uniform loads, wall racking, and fastener holding per PS 2.

Adjust per NDS for conditions.

Performance Category (in.)	Span Rating	Typical Applications
3/8	24/0	Roof sheathing over 24" spans, no floor rating
7/16	24/16	Roof over 24", subfloor over 16"
15/32	32/16	Roof over 32", subfloor over 16"
19/32	40/20	Roof over 40", subfloor over 20"
23/32	48/24	Roof over 48", subfloor over 24"

D. Accessories:

1. Fasteners: Corrosion-resistant nails, screws, or staples complying with manufacturer's recommendations and code requirements.
2. Edge Supports: H-clips or blocking for roof sheathing where spans exceed panel ratings.

2.3 FABRICATION

A. Fabricate panels to sizes indicated; provide factory-applied edge sealer where cut.

PART 3 – EXECUTION

3.1 EXAMINATION

A. Verify framing and substrate conditions are suitable for sheathing installation, including proper spacing and alignment.

B. Coordinate with related trades for penetrations and openings.

3.2 INSTALLATION

A. Install structural sheathing in accordance with manufacturer's instructions, APA Engineered Wood Construction Guide, and approved shop drawings.

B. Orientation: Install panels with strength axis perpendicular to supports unless otherwise indicated.

C. Spacing: Provide 1/8-inch gaps at panel edges and ends for expansion.

D. Fastening: Use specified fasteners at recommended spacing (e.g., 6 inches at edges, 12 inches in field for walls); stagger joints.

E. Roof Sheathing: Install with clips or blocking at unsupported edges for spans over 24 inches.

F. Cutting: Field-cut as necessary; apply sealer to cuts.

3.3 ADJUSTING

A. Adjust installed panels for proper alignment and secure fastening.

3.4 PROTECTION

A. Protect installed sheathing from weather and damage until completion of Work.

END OF SECTION

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SECTION 06 17 23
PARALLEL STRAND LUMBER

PART 1 – GENERAL

1.1 SUMMARY

A. Section Includes:

1. Parallel strand lumber (PSL) for structural framing, beams, headers, columns, and other applications as indicated on Drawings.
2. Accessories including connectors, fasteners, and hardware compatible with PSL.

1.2 REFERENCES

A. Reference Standards:

1. American Wood Protection Association (AWPA): U1 - Use Category System; T1 - Processing and Treatment Standard.
2. ASTM International: ASTM D5456 - Standard Specification for Evaluation of Structural Composite Lumber Products.
3. ICC Evaluation Service: ESR-1387 - Weyerhaeuser Engineered Wood Products.
4. National Design Specification (NDS) for Wood Construction, published by the American Wood Council.
5. PS 20 - American Softwood Lumber Standard, published by the U.S. Department of Commerce.

1.3 SUBMITTALS

- A. Product Data: Submit manufacturer's technical data sheets, including material properties, design values, sizes, and installation instructions for PSL.
- B. Shop Drawings: Submit detailed drawings showing PSL member sizes, locations, connections, and bracing.
- C. Certificates: Submit certification that PSL complies with specified standards and grades. D. Warranty Documentation: Submit manufacturer's warranty information.

1.4 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Manufacturer with a minimum of 10 years' experience producing PSL, with products evaluated by ICC-ES or equivalent.
- B. Installer Qualifications: Experienced in installing engineered wood products similar to those specified.
- C. Source Limitations: Obtain PSL from a single manufacturer to ensure compatibility.
- D. Identification: Each PSL member shall be stamped with grade, mill identification, and compliance marks.

1.5 DELIVERY, STORAGE, AND HANDLING

- A. Delivery: Deliver materials in manufacturer's original packaging, protected from weather.
- B. Storage: Store PSL flat, off the ground, supported at intervals to prevent warping, and protected from moisture and direct sunlight. C. Handling: Handle with care to avoid damage; do not drop or drag members.

1.6 WARRANTY

- A. Manufacturer's Warranty: Provide a 30-year limited warranty against manufacturing defects, termites, and fungal decay for treated PSL products.

PART 2 – PRODUCTS

2.1 MANUFACTURERS

- A. Acceptable Manufacturers:
1. Weyerhaeuser (Trus Joist Parallam PSL).
 2. Or approved equal, subject to compliance with requirements.

2.2 MATERIALS

- A. Parallel Strand Lumber (PSL):
1. Description: Composite lumber made from wood strands oriented parallel to the member length, bonded with exterior-type structural adhesive.
 2. Wood Species: Primarily Douglas fir, southern pine, or equivalent, as specified by manufacturer.
 3. Moisture Content: Kiln-dried to 19 percent maximum at time of manufacture.
 4. Grades: As indicated on Drawings, or minimum 2.0E grade unless otherwise noted.
- B. Sizes:
1. Beams and Headers: Widths and depths as indicated.
 2. Columns: Cross-sections width and depth as indicated.
 3. Lengths: Up to 66 feet, as required.
- C. Accessories:
1. Fasteners: Corrosion-resistant where exposed to weather or treated wood; comply with manufacturer's recommendations for type and size.
 2. Connectors: Simpson Strong-Tie or equivalent, designed for PSL loads.

2.3 FABRICATION

- A. Fabricate PSL members to sizes and profiles indicated, with factory-applied end sealer where cut.
- B. Provide holes and notches only as permitted by manufacturer; reinforce as required.

PART 3 – EXECUTION

3.1 EXAMINATION

- A. Verify substrate conditions are suitable for PSL installation, including level supports and proper alignment.
- B. Coordinate with related trades for embedded items and openings.

3.2 INSTALLATION

- A. Install PSL in accordance with manufacturer's instructions and approved shop drawings.
- B. Bearing: Provide minimum bearing lengths of 1-1/2 inches at ends and 3-1/2 inches at intermediate supports; ensure full-width bearing.
- C. Connections: Use specified fasteners and connectors; adjust design values for wet conditions (e.g., reduce lateral connection values by 0.7 for moisture $\leq 28\%$, 0.4 for $>28\%$).
- D. Bracing: Provide temporary and permanent bracing to prevent buckling or twisting.
- E. Cutting and Drilling: Field-cut only as necessary; apply field sealer to cuts and holes.
- F. Treated PSL: Use in exterior or ground-contact applications as specified; avoid contact with aluminum.

3.3 ADJUSTING

- A. Adjust installed members for proper fit and alignment before final connections.

3.4 PROTECTION

- A. Protect installed PSL from weather, construction activities, and damage until completion of Work.

END OF SECTION

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**SECTION 062000
FINISH CARPENTRY**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Finish carpentry items.
- B. Wood door frames and wood door slabs.
- C. Hardware and attachment accessories.

1.02 REFERENCE STANDARDS

- A. AWI/AWMAC/WI (AWS) - Architectural Woodwork Standards, 2nd Edition; 2014, with Errata (2016).
- B. AWMAC/WI (NAAWS) - North American Architectural Woodwork Standards; 2021, with Errata.
- C. BHMA A156.9 - Cabinet Hardware; 2020.
- D. PS 1 - Structural Plywood; 2023.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Shop Drawings: Indicate materials, component profiles, fastening methods, jointing details, and accessories.
 - 1. Provide information as required by AWI/AWMAC/WI (AWS) or AWMAC/WI (NAAWS).

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Protect from moisture damage.
- B. Handle materials and products to prevent damage to edges, ends, or surfaces.

PART 2 PRODUCTS

2.01 FINISH CARPENTRY ITEMS

- A. Quality Standard: Custom Grade, in accordance with AWI/AWMAC/WI (AWS) or AWMAC/WI (NAAWS), unless noted otherwise.
- B. Interior Woodwork Items:
 - 1. Door Frames and Door Leafs: Douglas fir, clear finish.
 - 2. Miscellaneous Trim: Douglas fir, clear finish.
 - 3. Bunk Trim Pieces: Douglas fir and/or hem-fir, clear finish.
 - 4. Ladders: Douglas fir, clear finish.
 - 5. Shelves: Douglas fir, clear finish.

2.02 SHEET MATERIALS

- A. Softwood Plywood: Face species as indicated, plain sawn, veneer core; PS 1 Grade A-B, glue type as recommended for application.

2.03 FASTENINGS

- A. Fasteners: Of size and type to suit application; standard finish in concealed locations and standard finish in exposed locations.
- B. Fasteners for Exterior Applications: Stainless steel; length required to penetrate wood substrate 1-1/2 inch (38 mm) minimum.

2.04 HARDWARE

- A. Hardware: Comply with BHMA A156.9.
- B. Flush Lifting Handle: 3" x 2-1/4" flush lifting handle with hinged lifting loop. Basis of Design: Perko Flush Lifting Handle model #255711.

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- C. Ceiling Mounting Hook: Heavy duty ceiling mounting hook, 2" round padeye type, stainless steel, cast grade 15-5PH high-tensile stainless steel. Secured with 4 @ 1/4" x 3" lag screws. Basis of Design: Ronstan Heavy Duty Padeyes model # RF2429-06.
- D. Stainless Steel Wool: 4 3/8" stainless steel wool in reels, #2, medium coarse.

2.05 FABRICATION

- A. Shop assemble work for delivery to site, permitting passage through building openings.
- B. When necessary to cut and fit on site, provide materials with ample allowance for cutting. Provide trim for scribing and site cutting.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install custom fabrications in accordance with AWI/AWMAC/WI (AWS) or AWMAC/WI (NAAWS) requirements for grade indicated.
- B. Set and secure materials and components in place, plumb and level.
- C. Carefully scribe work abutting other components, with maximum gaps of 1/32 inch (0.79 mm). Do not use additional overlay trim to conceal larger gaps.

END OF SECTION

SECTION 072700 AIR BARRIERS

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Air barriers (also referred to as weather barrier on drawings).

1.02 DEFINITIONS

- A. Air Barrier: Airtight barrier made of material that is virtually air impermeable but water vapor permeable, both to amount as specified, with sealed seams and sealed joints to adjacent surfaces.

1.03 REFERENCE STANDARDS

- A. ASTM D1970/D1970M - Standard Specification for Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection; 2021.
- B. ASTM E84 - Standard Test Method for Surface Burning Characteristics of Building Materials; 2024.
- C. ASTM E96/E96M - Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials; 2024a.
- D. ASTM E2178 - Standard Test Method for Determining Air Leakage Rate and Calculation of Air Permeance of Building Materials; 2021a.

PART 2 PRODUCTS

2.01 AIR BARRIER MATERIALS (AIR IMPERMEABLE AND WATER VAPOR PERMEABLE)

- A. Air Barrier Sheet, Self-Adhered:
 - 1. Air Permeance: 0.004 cfm/sq ft (0.02 L/(s sq m)), maximum, when tested in accordance with ASTM E2178.
 - 2. Water Vapor Permeance: 41 perms (2345 ng/(Pa s sq m)), minimum, when tested in accordance with ASTM E96/E96M using Procedure A - Desiccant Method, at 73.4 degrees F (23 degrees C).
 - 3. Ultraviolet (UV) and Weathering Resistance: Approved by manufacturer for up to 90 days of weather exposure.
 - 4. Surface Burning Characteristics: Flame spread index of 25 or less, smoke developed index of 450 or less, Class A when tested in accordance with ASTM E84.
 - 5. Seam and Perimeter Tape: As recommended by sheet manufacturer.
 - 6. Products:
 - a. VaproShield, LLC; RevealShield SA - Self-Adhered: www.vaproshield.com/#sle.
 - b. Grace Construction Products; Perm-A-Barrier VPS 30: <https://gcpat.com/en/products/perm-a-barrier-air-barrier-system..>
 - c. Substitutions: See Section 016000 - Product Requirements.

2.02 ACCESSORIES

- A. Sealants, Tapes, and Accessories for Sealing Air Barrier and Adjacent Substrates: As indicated or in compliance with air barrier manufacturer's installation instructions.
- B. Flexible Flashing: Self-adhesive sheet flashing complying with ASTM D1970/D1970M, except slip resistance requirement waived if not installed on roof.
 - 1. Width: 4 inches (100 mm) and as indicated on drawings.
 - 2. Ultraviolet (UV) and Weathering Resistance: Approved by manufacturer for up to 30 days of weather exposure.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that surfaces and conditions are ready for work of this section.

- B. Where existing conditions are responsibility of another installer, notify COR of unsatisfactory conditions.
- C. Do not proceed with this work until unsatisfactory conditions have been corrected.

3.02 PREPARATION

- A. Remove projections, protruding fasteners, and loose or foreign matter that might interfere with proper installation.
- B. Clean and prime substrate surfaces to receive adhesives and sealants in accordance with manufacturer's installation instructions.

3.03 INSTALLATION

- A. Install materials in accordance with manufacturer's installation instructions.
- B. Air Barriers: Install continuous airtight barrier over surfaces indicated, with sealed seams and with sealed joints to adjacent surfaces.
- C. Apply sealants and adhesives within recommended temperature range in accordance with manufacturer's installation instructions.
- D. Self-Adhered Sheets:
 - 1. Prepare substrate in accordance with sheet manufacturer's installation instructions; fill and tape joints in substrate and between dissimilar materials.
 - 2. Lap sheets shingle fashion to shed water and seal laps airtight.
 - 3. Once sheets are in place, press firmly into substrate with resilient hand roller; ensure that laps are firmly adhered with no gaps or fishmouths.
 - 4. Use same material, or other material approved by sheet manufacturer, to seal to adjacent substrates, and as flashing.
 - 5. At wide joints, provide extra flexible membrane allowing joint movement.
- E. Openings and Penetrations in Exterior Air Barriers:
 - 1. Install flashing over sills, covering entire sill frame member, extending at least 5 inches (125 mm) onto air barrier and at least 6 inches (150 mm) up jambs; mechanically fasten stretched edges.
 - 2. At openings with frames having nailing flanges, seal head and jamb flanges using a continuous bead of sealant compressed by flange and cover flanges with sealing tape at least 4 inches (100 mm) wide; do not seal sill flange.
 - 3. At openings with nonflanged frames, seal air barrier to each side of framing at opening using flashing at least 9 inches (230 mm) wide, and covering entire depth of framing.
 - 4. At head of openings, install flashing under air barrier extending at least 2 inches (50 mm) beyond face of jambs; seal air barrier to flashing.
 - 5. At interior face of openings, seal gap between window/door frame and rough framing, using joint sealant over backer rod.
 - 6. Service and Other Penetrations: Form flashing around penetrating item and seal to air barrier surface.

END OF SECTION

**SECTION 075300
ELASTOMERIC MEMBRANE ROOFING**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Elastomeric roofing membrane application.
- B. Roofing walkway pads.

1.02 REFERENCE STANDARDS

- A. ASCE 7 - Minimum Design Loads and Associated Criteria for Buildings and Other Structures; Most Recent Edition Cited by Referring Code or Reference Standard.
- B. ASTM D4637/D4637M - Standard Specification for EPDM Sheet Used in Single-Ply Roof Membrane; 2015, with Editorial Revision (2022).

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Provide data indicating membrane materials, flashing materials, surfacing, and fasteners.
- C. Shop Drawings: Indicate joint or termination detail conditions, conditions of interface with other materials, and paver layout.
- D. Manufacturer's Installation Instructions: Indicate membrane seaming precautions and perimeter conditions requiring special attention.

1.04 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing the products specified in this section with minimum three years of documented experience.
- B. Installer Qualifications: Company specializing in performing the work of this section with minimum three years documented experience.

1.05 FIELD CONDITIONS

- A. Do not apply roofing membrane during unsuitable weather.
- B. Do not apply roofing membrane to damp or frozen deck surface or when precipitation is expected or occurring.
- C. Do not expose materials vulnerable to water or sun damage in quantities greater than can be weatherproofed the same day.
- D. Schedule applications so that no partially completed sections of roof are left exposed at end of workday.

1.06 WARRANTY

- A. See Section 017800 - Closeout Submittals for additional warranty requirements.
- B. Provide twenty year manufacturer's material and labor warranty to cover failure to prevent penetration of water.

PART 2 PRODUCTS

2.01 ROOFING

- A. Elastomeric Membrane Roofing: One ply membrane, fully adhered.
- B. Primer, Roof Coating: Water-based primer with high-tack finish that promotes adhesion for elastomeric roof coatings.

2.02 ROOFING MEMBRANE AND ASSOCIATED MATERIALS

- A. Membrane: Ethylene-propylene-diene-monomer (EPDM); internally reinforced with fabric or scrim; complying with minimum properties of ASTM D4637/D4637M.
 - 1. Thickness: 75 mil, 0.075 inch (1.9 mm), minimum.

- 2. Color: Black.
- B. Seaming Materials: As recommended by membrane manufacturer.
- C. Flexible Flashing Material: Same material as membrane.

2.03 ACCESSORIES

- A. Membrane Adhesive: As recommended by membrane manufacturer.
- B. Surface Conditioner for Adhesives: Compatible with membrane and adhesives.
- C. Surface Primer Over Roof Substrate for Adhesion of Membrane: Low VOC acrylic primer for application over roofing substrates.
 - 1. Color: White.
 - 2. Roof Membrane Substrate: Includes wood as indicated on drawings.
- D. Walkway Pads: Heavy duty type with dimples (round or square). Minimum base material thickness of 160 mils; overall height with dimples approximately 3/8".
 - 1. Composition: Roofing membrane manufacturer's standard.

PART 3 EXECUTION

3.01 PREPARATION - WOOD DECK

- A. Verify flatness and tightness of joints in wood decking; fill knot holes with latex filler.
- B. Confirm dry deck by moisture meter with 12 percent moisture maximum.

3.02 INSTALLATION - MEMBRANE

- A. Roll out membrane, free from wrinkles or tears. Place sheet into place without stretching.
- B. Shingle joints on sloped substrate in direction of drainage.
- C. Fully Adhered Application: Apply adhesive to substrate at rate recommended by membrane manufacturer for project conditions. Fully embed membrane in adhesive except in areas directly over or within 3 inches (76 mm) of expansion joints. Fully adhere one roll before proceeding to adjacent rolls.
- D. Overlap edges and ends and seal seams by contact adhesive, minimum 3 inches (76 mm). Seal permanently waterproof. Apply uniform bead of sealant to joint edge.
- E. Around roof penetrations, seal flanges and flashings with flexible flashing.

3.03 PROTECTION

- A. Protect installed roofing and flashings from construction operations.
- B. Where traffic must continue over finished roof membrane, protect surfaces using durable materials.

END OF SECTION

SECTION 076200
SHEET METAL FLASHING AND TRIM

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Fabricated sheet metal items, including flashings and counterflashings.
- B. Sealants for joints within sheet metal fabrications.

1.02 REFERENCE STANDARDS

- A. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2023.
- B. ASTM C920 - Standard Specification for Elastomeric Joint Sealants; 2018 (Reapproved 2024).
- C. SMACNA (ASMM) - Architectural Sheet Metal Manual; 2012.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Shop Drawings: Indicate material profile, jointing pattern, jointing details, fastening methods, flashings, terminations, and installation details.

1.04 QUALITY ASSURANCE

- A. Perform work in accordance with SMACNA (ASMM) and CDA A4050 requirements and standard details, except as otherwise indicated.

PART 2 PRODUCTS

2.01 SHEET MATERIALS

- A. Galvanized Steel: ASTM A653/A653M, with G90/Z275 zinc coating; minimum 24-gauge, 0.0239-inch (0.61 mm) thick base metal.

2.02 FABRICATION

- A. Form sections true to shape, accurate in size, square, and free from distortion or defects.
- B. Form pieces in longest possible lengths.
- C. Hem exposed edges on underside 1/2 inch (13 mm); miter and seam corners.
- D. Form material with flat lock seams, except where otherwise indicated; at moving joints, use sealed lapped, bayonet-type or interlocking hooked seams.
- E. Fabricate corners from one piece with minimum 18-inch (450 mm) long legs; seam for rigidity, seal with sealant.
- F. Fabricate flashings to allow toe to extend 2 inches (50 mm) over roofing gravel. Return and brake edges.

2.03 FLASHING

- A. Flashing Panels for Exterior Wall Penetrations: Premanufactured components and accessories as required to preserve integrity of building envelope; suitable for acade materials to be installed.

2.04 ACCESSORIES

- A. Fasteners: Galvanized steel, with soft neoprene washers.
- B. Primer Type: Zinc chromate.
- C. Concealed Sealants: Non-curing butyl sealant.
- D. Exposed Sealants: ASTM C920; elastomeric sealant, with minimum movement capability as recommended by manufacturer for substrates to be sealed; color to match adjacent material.

PART 3 EXECUTION

3.01 PREPARATION

- A. Install starter and edge strips, and cleats before starting installation.
- B. Back paint concealed metal surfaces with protective backing paint to a minimum dry film thickness of 15 mil, 0.015 inch (0.38 mm).

3.02 INSTALLATION

- A. Comply with drawing details.
- B. Secure flashings in place using concealed fasteners, and use exposed fasteners only where permitted.
- C. Apply plastic cement compound between metal flashings and felt flashings.
- D. Fit flashings tight in place; make corners square, surfaces true and straight in planes, and lines accurate to profiles.

END OF SECTION

SECTION 079200 JOINT SEALANTS

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Nonsag gunnable joint sealants.

1.02 REFERENCE STANDARDS

- A. ASTM C920 - Standard Specification for Elastomeric Joint Sealants; 2018 (Reapproved 2024).
- B. ASTM C1193 - Standard Guide for Use of Joint Sealants; 2025.
- C. ASTM C1248 - Standard Test Method for Staining of Porous Substrate by Joint Sealants; 2022.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Submit manufacturer's technical datasheets for each product to be used; include the following:
 - 1. Physical characteristics, including movement capability, VOC content, hardness, cure time, and color availability.
 - 2. List of backing materials approved for use with the specific product.
 - 3. Substrates that product is known to satisfactorily adhere to and with which it is compatible.
 - 4. Substrates the product should not be used on.

PART 2 PRODUCTS

2.01 JOINT SEALANTS - GENERAL

2.02 NONSAG JOINT SEALANTS

- A. Nonstaining Silicone Sealant: ASTM C920, Grade NS, Uses M, A, G, and O; not expected to withstand continuous water immersion or traffic.
 - 1. Movement Capability: Plus and minus 35 percent, minimum.
 - 2. Nonstaining to Porous Stone: Nonstaining to light-colored natural stone when tested in accordance with ASTM C1248.
 - 3. Dirt Pick-Up: Reduced dirt pick-up compared to other silicone sealants.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that joints are ready to receive work.
- B. Verify that backing materials are compatible with sealants.

3.02 PREPARATION

- A. Remove loose materials and foreign matter that could impair adhesion of sealant.
- B. Clean joints, and prime as necessary, in accordance with manufacturer's instructions.
- C. Perform preparation in accordance with manufacturer's instructions and ASTM C1193.
- D. Mask elements and surfaces adjacent to joints from damage and disfigurement due to sealant work; be aware that sealant drips and smears may not be completely removable.

3.03 INSTALLATION

- A. Install this work in accordance with sealant manufacturer's requirements for preparation of surfaces and material installation instructions.
- B. Provide joint sealant installations complying with ASTM C1193.
- C. Install bond breaker backing tape where backer rod cannot be used.
- D. Install sealant free of air pockets, foreign embedded matter, ridges, and sags, and without getting sealant on adjacent surfaces.

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- E. Do not install sealant when ambient temperature is outside manufacturer's recommended temperature range, or will be outside that range during the entire curing period, unless manufacturer's approval is obtained and instructions are followed.
- F. Nonsag Sealants: Tool surface concave, unless otherwise indicated; remove masking tape immediately after tooling sealant surface.

3.04 FIELD QUALITY CONTROL

- A. See Section 014000 - Quality Requirements for additional requirements.
- B. Perform field quality control inspection/testing as specified in PART 1 under QUALITY ASSURANCE article.
- C. Remove and replace failed portions of sealants using same materials and procedures as indicated for original installation.

END OF SECTION

**SECTION 085113
ALUMINUM WINDOWS**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Extruded aluminum windows with fixed sash and operating sash.
- B. Factory glazing.
- C. Operating hardware.
- D. Insect screens.

1.02 REFERENCE STANDARDS

- A. AAMA/WDMA/CSA 101/I.S.2/A440 - North American Fenestration Standard/Specification for Windows, Doors, and Skylights; 2022, with Errata (2023).
- B. AAMA CW-10 - Care and Handling of Architectural Aluminum from Shop to Site; 2015.
- C. AAMA 611 - Specification for Anodized Architectural Aluminum; 2024.
- D. AAMA 1503 - Voluntary Test Method for Thermal Transmittance and Condensation Resistance of Windows, Doors and Glazed Wall Sections; 2009.
- E. ASTM E283/E283M - Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Skylights, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen; 2019.
- F. ASTM E331 - Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference; 2000 (Reapproved 2023).
- G. ASTM E1996 - Standard Specification for Performance of Exterior Windows, Curtain Walls, Doors, and Impact Protective Systems Impacted by Windborne Debris in Hurricanes; 2023.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Include component dimensions, information on glass and glazing, internal drainage details, and descriptions of hardware and accessories.
- C. Shop Drawings: Indicate opening dimensions, elevations of different types, framed opening tolerances, anchorage locations, and installation requirements.
- D. Grade Substantiation: Prior to submitting shop drawings or starting fabrication, submit one of the following showing compliance with specified grade:
 - 1. Evidence of AAMA Certification.
 - 2. Evidence of WDMA Certification.
 - 3. Evidence of CSA Certification.
 - 4. Test report(s) by independent testing agency itemizing compliance and acceptable to authorities having jurisdiction.
- E. Test Reports: Prior to submitting shop drawings or starting fabrication, submit test report(s) by independent testing agency showing compliance with performance requirements in excess of those prescribed by specified grade.

1.04 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with minimum three years of documented experience.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Comply with requirements of AAMA CW-10.
- B. Protect finished surfaces with wrapping paper or strippable coating during installation. Do not use adhesive papers or sprayed coatings that bond to substrate when exposed to sunlight or weather.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Basis of Design: Wasau Invent 4250i-HP-XLT Hurricane Windows.
- B. Other Acceptable - Aluminum Window Manufacturers:
 - 1. Kawneer Optiq AA 4325 Series Windows.

2.02 ALUMINUM WINDOWS

- A. Aluminum Windows: Extruded aluminum frame and sash, factory fabricated, factory finished, with operating hardware, related flashings, and anchorage and attachment devices.
 - 1. Frame Depth: 4 inch (102 mm).
 - 2. Fabrication: Joints and corners flush, hairline, and weatherproof, accurately fitted and secured; prepared to receive anchors; fasteners and attachments concealed from view; reinforced as required for operating hardware and imposed loads.
 - 3. Perimeter Clearance: Minimize space between framing members and adjacent construction while allowing expected movement.
 - 4. Movement: Accommodate movement between window and perimeter framing and deflection of lintel, without damage to components or deterioration of seals.
 - 5. System Internal Drainage: Drain to the exterior by means of a weep drainage network any water entering joints, condensation occurring in glazing channel, and migrating moisture occurring within system.
- B. Outswinging Awning Type:
 - 1. Construction: Thermally broken.
 - 2. Provide screens.
 - 3. Glazing: Single; clear; transparent.
 - 4. Exterior Finish: Class I color anodized.
 - 5. Interior Finish: Class I color anodized.

2.03 PERFORMANCE REQUIREMENTS

- A. Grade: AAMA/WDMA/CSA 101/I.S.2/A440 requirements for specific window type:
 - 1. Performance Class (PC): R.
- B. Wind-Borne-Debris Resistance: Identical full-size glazed assembly without auxiliary protection, tested by independent agency in accordance with ASTM E1996 for Wind Zone 4 - Additional Protection for Large and Small Missile impact and pressure cycling at design wind pressure.
- C. Water Leakage: No uncontrolled leakage on interior face when tested in accordance with ASTM E331 at differential pressure of 12.11 psf (580 Pa).
- D. Air Leakage: 0.1 cfm/sq ft (0.5 L/sec sq m) maximum leakage per unit area of outside window frame dimension when tested at 1.57 psf (75 Pa) pressure difference in accordance with ASTM E283/E283M.
- E. Condensation Resistance Factor of Frame: 50, measured in accordance with AAMA 1503.

2.04 COMPONENTS

- A. Frames: Aluminum; thermally broken with interior portion of frame insulated from exterior portion; flush glass stops of snap-on type.
- B. Glazing: 1" thick insulated glass units with laminated safety glass at exterior lite.
- C. Insect Screens: Extruded aluminum frame with mitered and reinforced corners; screen mesh taut and secure to frame; secured to window with adjustable hardware allowing screen removal without use of tools.
 - 1. Hardware: Spring loaded steel pins; four per screen unit.
 - 2. Screen Mesh: Vinyl-coated fiberglass, window manufacturer's standard mesh.
 - 3. Frame Finish: Same as frame and sash.
- D. Operable Sash Weatherstripping: Wool pile; permanently resilient, profiled to achieve effective weather seal.

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- E. Fasteners: Stainless steel.
- F. Sealant for Setting Sills and Sill Flashing: Non-curing butyl type.

2.05 HARDWARE

- A. Sash lock: Lever handle with cam lock.

2.06 FINISHES

- A. Class I Natural Anodized Finish: AAMA 611 AA-M12C22A41, clear anodic coating not less than 0.7 mil (0.018 mm) thick.

PART 3 EXECUTION

3.01 PRIME WINDOW INSTALLATION

- A. Install windows in accordance with manufacturer's instructions.
- B. Attach window frame and shims to perimeter opening to accommodate construction tolerances and other irregularities.
- C. Align window plumb and level, free of warp or twist. Maintain dimensional tolerances and alignment with adjacent work.
- D. Install sill and sill end angles.
- E. Provide thermal isolation where components penetrate or disrupt building insulation. Pack fibrous insulation in shim spaces at perimeter of assembly to maintain continuity of thermal barrier.
- F. Install operating hardware not pre-installed by manufacturer.

3.02 ADJUSTING

- A. Adjust hardware for smooth operation and secure weathertight closure.

3.03 CLEANING

- A. See Section 017419 - Construction Waste Management and Disposal for additional requirements.
- B. Remove protective material from factory finished aluminum surfaces.
- C. Wash surfaces by method recommended and acceptable to window manufacturer; rinse and wipe surfaces clean.

END OF SECTION

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**SECTION 087100
DOOR HARDWARE**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Hardware for wood doors.
- B. Weatherstripping and gasketing.

1.02 REFERENCE STANDARDS

- A. BHMA A156.1 - Standard for Butts and Hinges; 2021.
- B. BHMA A156.13 - Mortise Locks & Latches Series 1000; 2022.
- C. BHMA A156.16 - Standard for Auxiliary Hardware; 2023.
- D. BHMA A156.18 - Standard for Materials and Finishes; 2020.
- E. BHMA A156.22 - Standard for Gasketing; 2021.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Manufacturer's catalog literature for each type of hardware, marked to clearly show products to be furnished for this project, and includes construction details, material descriptions, finishes, and dimensions and profiles of individual components.
- C. Shop Drawings - Door Hardware Schedule: Submit detailed listing that includes each item of hardware to be installed on each door. Use door numbering scheme as included in Contract Documents.
 - 1. Prepared by or under supervision of Architectural Hardware Consultant (AHC).
 - 2. Provide complete description for each door listed.

1.04 QUALITY ASSURANCE

- A. Manufacturer Qualifications: Company specializing in manufacturing products specified in this section with minimum three years of documented experience.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Package hardware items individually; label and identify each package with door opening code to match door hardware schedule.

1.06 WARRANTY

- A. See Section 017800 - Closeout Submittals for additional warranty requirements.
- B. Manufacturer's Warranty: Provide warranty against defects in material and workmanship for period indicated. Complete forms in Owner's name and register with manufacturer.
 - 1. Locksets and Cylinders: Three years, minimum.
 - 2. Other Hardware: Three years, minimum., minimum.

PART 2 PRODUCTS

2.01 DESIGN AND PERFORMANCE CRITERIA

- A. Provide specified door hardware as required to make doors fully functional, compliant with applicable codes, and secure to extent indicated.
- B. Provide individual items of single type, of same model, and by same manufacturer.
- C. Provide door hardware products that comply with the following requirements:
 - 1. Applicable provisions of federal, state, and local codes.
- D. Fasteners:
 - 1. Provide fasteners of proper type, size, quantity, and finish that comply with commercially recognized standards for proposed applications.
 - a. Aluminum fasteners are not permitted.

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- b. Provide phillips flat-head screws with heads finished to match door surface hardware unless otherwise indicated.

2.02 HINGES

- A. Manufacturers:
 1. McKinney; an Assa Abloy Group company: www.assaabloydss.com/#sle.
 2. Hager Companies: www.hagerco.com/#sle.
 3. BEST, dormakaba Group: www.bestaccess.com/#sle.
- B. Hinges: Comply with BHMA A156.1, Grade 1.
 1. Provide hinges on every swinging door.
 2. Provide five-knuckle full mortise butt hinges unless otherwise indicated.
 3. Provide following quantity of butt hinges for each door:
 - a. Doors up to 24 inches (0.6 m) High: Two hinges.
 - b. Doors From 24 inches (0.6 m) High up to 60 inches (1.5 m) High: Three hinges.
 - c. Doors 60 inches (1.5 m) High up to 84 inches (2.14 m) High: Four hinges.

2.03 SURFACE BOLTS

- A. Manufacturers:
 1. Adams Rite, an Assa Abloy Group company: www.assaabloydss.com/#sle.
 2. Hager Companies: www.hagerco.com/#sle.
 3. Ives, an Allegion brand; Basis of Design: SB360 Heavy Duty Surface Bolt: www.allegion.com/us/#sle.
 4. Pamex, Inc; Flush Bolts: www.pamexinc.com/#sle.
 5. Substitutions: See Section 016000 - Product Requirements.
- B. Surface Bolts: ANSI/BHMA A156.16
 1. Throw: 1-1/4"
 2. Grade: Heavy Duty
 3. Finish: US2C/604 Clear Zinc
 4. Length: 12"
 5. Bar: 3/4" wide x 3/4" thick

2.04 MORTISE LOCKS

- A. Manufacturers:
 1. BEST, dormakaba Group; 45H Series: www.bestaccess.com/#sle.
 2. Hager Companies: www.hagerco.com/#sle.
 3. Schlage, an Allegion brand: www.allegion.com/us/#sle.
- B. Mortise Locks: Comply with BHMA A156.13, Grade 1, Security, 1000 Series.
 1. Latchbolt Throw: 3/4 inch (19 mm), minimum.
 2. Backset: 2-3/4 inch (70 mm) unless otherwise indicated.
 3. Strikes: Provide manufacturer's standard strike for each latchset or lockset with strike box and curved lip extending to protect frame in compliance with indicated requirements.
 - a. Extra-Long-Lip Strikes: Provide for locks used on frames with applied wood casing trim.
 - b. Finish: To match lock or latch.

2.05 WALL STOPS

- A. Wall Stops: Comply with BHMA A156.16, Grade 1 and Resilient Material Retention Test as described in this standard.
 1. Provide wall stops to prevent damage to wall surface upon opening door.
 2. Type: Bumper, concave, wall stop.
 3. Material: Stainless steel housing with rubber insert.

2.06 WEATHERSTRIPPING AND GASKETING

- A. Weatherstripping and Gasketing: Comply with BHMA A156.22.

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1. Head and Jamb Type: Adjustable.
2. Door Sweep Type: Encased in retainer.
3. Material: Aluminum, with brush weatherstripping.
4. Provide weatherstripping on each exterior door at head, jambs, and meeting stiles of door pairs, unless otherwise indicated.

2.07 SILENCERS

- A. Silencers: Provide at equal locations on door frame to mute sound of door's impact upon closing.
 1. Single Door: Provide three on strike jamb of frame.
 2. Material: Rubber, gray color.

2.08 FINISHES

- A. Finishes: Provide door hardware of same finish, unless otherwise indicated.
 1. Finish: 630; satin stainless steel, with stainless steel 300 series base material (former US equivalent US32D); BHMA A156.18.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install hardware in accordance with manufacturer's instructions and applicable codes.
- B. Use templates provided by hardware item manufacturer.
- C. Door Hardware Mounting Heights: Distance from finished floor to center line of hardware item. As indicated in following list; unless noted otherwise in Door Hardware Schedule or on drawings.
- D. Set exterior door thresholds with full-width bead of elastomeric sealant at each point of contact with floor providing a continuous weather seal; anchor thresholds with stainless steel countersunk screws.

3.02 ADJUSTING

- A. Adjust work under provisions of Section 017000 - Execution and Closeout Requirements.
- B. Adjust hardware for smooth operation.
- C. Adjust gasketing for complete, continuous seal; replace if unable to make complete seal.

3.03 CLEANING

- A. Clean finished hardware in accordance with manufacturer's written instructions after final adjustments have been made.

3.04 PROTECTION

- A. Protect finished Work under provisions of Section 017000 - Execution and Closeout Requirements.
- B. Do not permit adjacent work to damage hardware or finish.

END OF SECTION

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SECTION 089100
LOUVERS

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Louvers, frames, and accessories.
 - 1. Operable Louvers
 - 2. Steel Floor Register with Operable Dampers
 - 3. Steel Grate Vent Cap

1.02 REFERENCE STANDARDS

- A. AAMA 2605 - Voluntary Specification, Performance Requirements and Test Procedures for Superior Performing Organic Coatings on Aluminum Extrusions and Panels (with Coil Coating Appendix); 2022.
- B. AMCA 500-L - Laboratory Methods of Testing Louvers for Rating; 2023.
- C. AMCA 511 - Certified Ratings Program Product Rating Manual for Air Control Devices; 2021, with Editorial Revision (2022).
- D. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2023.
- E. FEMA P-361 - Safe Rooms for Tornadoes and Hurricanes: Guidance for Community and Residential Safe Rooms; 2024.
- F. ICC 500 - ICC/NSSA Standard for the Design and Construction of Storm Shelters; 2020.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Provide data describing design characteristics, maximum recommended air velocity, design free area, materials and finishes.
- C. Shop Drawings: Indicate louver layout plan and elevations, opening and clearance dimensions, and tolerances; head, jamb and sill details; blade configuration, screens, blank-off areas required, and frames.
- D. Test Reports: Independent agency reports showing compliance with specified performance criteria.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Louvers:
 - 1. Airolite Company, LLC: www.airolite.com/#sle.
 - 2. Ruskin Company: www.ruskin.com/#sle.
 - 3. Greenheck; www.greenheck.com.

2.02 LOUVERS

- A. Louvers: Factory fabricated and assembled, complete with frame, mullions, and accessories; AMCA Certified in accordance with AMCA 511.
 - 1. High-Velocity Wind Load Resistance: Design to comply with applicable requirements of ICC 500 and FEMA P-361, including resistance to horizontal debris impact.
 - 2. Intake Louvers: Design to allow maximum of 0.01 oz/sq ft (3.1 g/sq m) water penetration at calculated intake design velocity based on design air flow and actual free area, when tested in accordance with AMCA 500-L.
 - 3. Drainable Blades: Continuous rain stop at front or rear of blade aligned with vertical gutter recessed into both jambs of frame.
 - 4. Screens: Provide insect screens at all louvers and grilles.

- B. Stationary Louvers, Type : "STEEL GRATE VENT CAP": Vertical blade, formed galvanized sheet or extruded aluminum construction.
 - 1. Free Area: 50 percent, minimum.
 - 2. Blades: Straight.
 - 3. Frame: Depth as indicated on drawings, channel profile; corner joints .
 - 4. Steel Thickness, Galvanized: Frame 16 gauge, 0.0598 inch (1.52 mm) minimum base metal; blades 16 gauge, 0.0598 inch (1.52 mm) minimum base metal.
 - 5. Aluminum Finish: Class I natural anodized; finish welded units after fabrication.
 - 6. Steel Finish: High performance organic coating, finished after fabrication.
- C. Operable Louvers: Combination horizontal fixed blade and operable damper unit, formed galvanized steel sheet construction.
 - 1. Operation: Manual operation with extended operation bar, min 48 inches long, allowing operation from a height of 60" above finish floor.
 - 2. Movable Blades: Drainable, pivoted at, with vinyl, rubber, or polyethylene blade edge and jamb seals; rattle-free linkage.
 - 3. Fixed Blades: Straight.
 - 4. Frame: 6 inches deep (150 mm deep), channel profile; corner joints mitered and , with continuous recessed caulking channel each side.
 - 5. Steel Thickness, Galvanized: Frame 16 gauge, 0.0598 inch (1.52 mm) minimum base metal; blades 16 gauge, 0.0598 inch (1.52 mm) minimum base metal.
 - 6. Steel Finish: Superior performing organic coatings, finished after fabrication.
 - 7. Basis of Design: Greenheck EAC-601.
- D. Steel Floor Register with Operable Dampers
 - 1. Commercial grade heavysteel floor register with fully adjustable damper.
 - 2. Size: As indicated on drawings.
 - 3. Manufacturers:
 - a. Continental Industries F20 Heavy Duty Floor Register:
<https://www.continentalindustries.com/F20>.
 - b. Hart Cooley #210 Heavy Duty Steel Floor Register:
<https://www.hartandcooley.com/product/210-heavy-duty-steel-floor-register>.
 - c. Other Acceptable Manufacturers: Dayus or Truair.

2.03 MATERIALS

- A. Steel Sheet: Hot-dipped galvanized steel sheet, ASTM A653/A653M, with G90/Z275 coating.

2.04 FINISHES

- A. Superior Performing Organic Coatings System: Manufacturer's standard multi-coat superior performing organic coatings system complying with AAMA 2605, including at least 70 percent polyvinylidene fluoride (PVDF) resin, and at least 80 percent of aluminum extrusion and panels surfaces having minimum total dry film thickness (DFT) of 1.2 mils, 0.0012 inch (0.030 mm).

2.05 ACCESSORIES

- A. Screens: Frame of same material as louver, with reinforced corners; removable, screw attached; installed on inside face of louver frame.
- B. Flashings: Of same material as louver frame, formed to required shape, single length in one piece per location.
- C. Sealant for Setting Sills and Sill Flashing: Non-curing butyl type.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that prepared openings and flashings are ready to receive this work and opening dimensions are as indicated on shop drawings.

3.02 INSTALLATION

- A. Install louver assembly in accordance with manufacturer's instructions.
- B. Install louvers level and plumb.
- C. Install flashings and align louver assembly to ensure moisture shed from flashings and diversion of moisture to exterior.
- D. Secure louver frames in openings with concealed fasteners.

3.03 ADJUSTING

- A. Adjust operable louvers for freedom of movement of control mechanism. Lubricate operating joints.

3.04 CLEANING

- A. Strip protective finish coverings.
- B. Clean surfaces and components.

END OF SECTION

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**SECTION 101423
PANEL SIGNAGE**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Panel signage.

1.02 REFERENCE STANDARDS

- A. ADA Standards - 2010 ADA Standards for Accessible Design; 2010.
- B. ICC A117.1 - Accessible and Usable Buildings and Facilities; 2017.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Package signs as required to prevent damage before installation.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Panel Signage:
 - 1. Seton: www.seton.com
 - 2. Big City Manufacturing: www.metaltags.com.
 - 3. Laser Engraved Memories: www.laserengravedmemories.com.

2.02 REGULATORY REQUIREMENTS

- A. Accessibility Requirements: Comply with ADA Standards and ICC A117.1 and applicable building codes, unless otherwise indicated; in the event of conflicting requirements, comply with the most restrictive requirements.

2.03 PANEL SIGNAGE

- A. Engraved Aluminum Panel Signage:
 - 1. Application: Informational signs.
 - 2. Description: Flat signs with engraved panel media, tactile characters.
 - 3. Sign Size: As indicated on drawings.
 - 4. Total Thickness: 1/8 inch (3 mm).
 - 5. Sign Edges: Squared.
 - 6. Color and Font, unless otherwise indicated:
 - a. Character Font: Helvetica, Arial, or other sans serif font.
 - b. Character Case: Upper case only.
 - c. Background Color: National Park Service Green.
 - d. Character Color: Brushed aluminum color.
 - 7. Material: Engraved aluminum plate.
 - 8. Profile: Flat panel without frame.
 - 9. Tactile Letters: Raised 1/32 inch minimum.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Install with horizontal edges level.
- C. Locate panel signs and mount at heights indicated on drawings and in accordance with ADA Standards and ICC A117.1.

END OF SECTION

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**SECTION 104400
FIRE PROTECTION SPECIALTIES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Fire extinguishers.

1.02 REFERENCE STANDARDS

- A. FM (AG) - FM Approval Guide; Current Edition.
- B. NFPA 10 - Standard for Portable Fire Extinguishers; 2022.
- C. UL (DIR) - Online Certifications Directory; Current Edition.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Product Data: Provide extinguisher operational features, extinguisher ratings and classifications, color and finish, anchorage details, and installation instructions.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Fire Extinguishers:
 - 1. Ansul, a Tyco Business: www.ansul.com/#sle.
 - 2. Fire-End & Croker Corporation: www.croker.com/#sle.
 - 3. Kidde, a unit of United Technologies Corp: www.kidde.com/#sle.
 - 4. Oval Brand Fire Products: www.ovalfireproducts.com/#sle.

2.02 FIRE EXTINGUISHERS

- A. Fire Extinguishers - General: Comply with product requirements of NFPA 10 and applicable codes, whichever is more stringent.
 - 1. Provide extinguishers labeled by UL (DIR) or FM (AG) for purpose specified and as indicated.
- B. Multipurpose Dry Chemical Type Fire Extinguishers: Carbon steel tank, with pressure gauge.
 - 1. Class: A:B:C type.
 - 2. Size: 5 pound (2.27 kg).
 - 3. Temperature range: Minus 40 degrees F (Minus 40 degrees C) to 120 degrees F (49 degrees C).

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install in accordance with manufacturer's instructions.

END OF SECTION

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SECTION 105617
WALL MOUNTED STANDARDS AND SHELVING

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Steel shelf standards, brackets, and accessories.
- B. Shelves.
- C. See drawings for locations and configurations.

1.02 SUBMITTALS

- A. See Section 013000 - Administrative Requirements for submittal procedures.
- B. Product Data: Manufacturer's data sheets on each product to be used.

PART 2 PRODUCTS

2.01 COMPONENTS

- A. Steel Shelf Standards, Brackets, and Accessories:
 - 1. Extra-Duty Shelf Standards and Brackets: Double-slotted channel standards for brackets adjustable in 1 inch (25 mm) increments along entire length of standard, drilled and countersunk for screws.
 - a. Load Capacity: Recommended by manufacturer for loading of 300 to 680 pounds (135 to 310 kg) per pair of standards.
 - b. Finish: Electroplated, chrome-look.
 - c. Brackets: Double tab type, locking into slots; size to suit shelves; same finish as standards.
 - d. Bracket Quantity: Provide one bracket for each 12 inches (305 mm) of standard length.
- B. Shelving:
 - 1. Wood Shelves: Solid board shelves, with edges sanded and eased.
 - a. Species: Douglas Fir or Hem-fir.
 - b. Thickness: 3/4 inch (19 mm), nominal.
 - c. Finish: Polyurethane varnish.
- C. Fasteners: Screws as recommended by manufacturer for intended application or as otherwise required by project conditions. Finish of exposed to view fasteners to match finish of standards and other components.

PART 3 EXECUTION

3.01 PREPARATION

- A. Clean surfaces thoroughly prior to installation.
- B. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

3.02 INSTALLATION

- A. Install in accordance with manufacturer's instructions.
- B. Mount standards or brackets to solid backing capable of supporting intended loads.
- C. Install brackets, shelving, and accessories.

END OF SECTION

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**SECTION 114000
FOODSERVICE EQUIPMENT**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Stainless Steel Table.
- B. Stainless Steel Wall Paneling.

1.02 REFERENCE STANDARDS

- A. ASTM A666/A666M - Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar; 2024.

1.03 SUBMITTALS

- A. See Section 013000 - Administrative Requirements, for submittal procedures.
- B. Product Data: Provide data on appliances; indicate configuration, sizes, materials, finishes, locations, and utility service connection locations, service characteristics, and wiring diagrams.
- C. Manufacturer's Installation Instructions: Indicate special procedures, perimeter conditions requiring special attention.

PART 2 PRODUCTS

2.01 STAINLESS STEEL WALL PANELING

- A. Materials
 - 1. Stainless steel sheet: ASTM A240, Type 304, No. 4 brushed finish unless otherwise indicated.
 - 2. Thickness: 18-gauge (1.2 mm) typical; heavier gauges where noted.
 - 3. Trim: Stainless steel angles, channels, or edge moldings to match panel material and finish.
- B. Fabrication
 - 1. Panels factory-formed to sizes indicated.
 - 2. Edges hemmed or turned for stiffness and safety.
 - 3. Provide back stiffeners or reinforcing where required to maintain flatness.
- C. Accessories
 - 1. Fasteners: Stainless steel screws, concealed where possible.

2.02 STAINLESS STEEL TABLE

- A. Basis of Design: Eagle Group Worktable with Backsplash andn Stainless Steel Base with Undershelf: www.eaglegrp.com.
- B. Other acceptable manufacturers:
 - 1. RDM Industrial.
 - 2. Hally Stainless Steel.
- C. Materials
 - 1. Top: 16-gauge Type 304 stainless steel, flat, with 1-1/2 in. (38 mm) turned-down edges and 5/8 in. (16 mm) marine edge at rear.
 - 2. Undershelf: 18-gauge Type 430 stainless steel, adjustable.
 - 3. Legs: 1-5/8 in. (41 mm) diameter stainless steel tubing, with adjustable stainless steel bullet feet.
- D. Construction
 - 1. Welded and polished joints, NSF-certified.
 - 2. Sound deadening under work surface.
 - 3. Provide gussets at leg connections.
- E. Dimensions
 - 1. Standard table: 24 in. (609 mm) deep x 66 in. (1676 mm) long x 36 in. (914 mm) high.

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2. Variations as indicated on drawings.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify ventilation outlets, service connections, and supports are correct and in required location.
- B. Verify that electric power is available and of the correct characteristics.

3.02 INSTALLATION

- A. Install items in accordance with manufacturers' instructions.
- B. Insulate to prevent electrolysis between dissimilar metals.

3.03 CLEANING

- A. Remove masking or protective covering from stainless steel and other finished surfaces.
- B. Wash and clean equipment.

3.04 PROTECTION

- A. Protect finished work from damage.

END OF SECTION

SECTION 260000 ELECTRICAL GENERAL REQUIREMENTS

PART 1 GENERAL

1.01 SUMMARY

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 1 Specifications Sections, apply to this Section.
- B. This section describes specific requirements, products, and methods of execution, which are typical throughout the electrical work of this project. Additional requirements for the specific systems may modify these requirements.
- C. This Section applies to all Division 26 and is part of all other Division 26 Sections.
- D. Index of Electrical Specifications:
 - 1. 260000 - Electrical General Requirements
 - 2. 260526 - Grounding and Bonding for Electrical Systems
 - 3. 260529 - Hangers and Supports for Electrical Systems
 - 4. 260533 - Raceway and Boxes for Electrical Systems
 - 5. 264100 - Facility Lightning Protection

1.02 REFERENCES

- A. Codes: Perform work in strict accordance with applicable national, state and local codes; including, but not limited to the latest legally enacted editions of the following specifically noted requirements:
 - 1. NFPA 70, National Electrical Code - NEC.
 - 2. ANSI-C2, National Electrical Safety Code - NESC.
 - 3. International Building Code - IBC.
 - 4. International Fire Code - IFC.
 - 5. Underwriters Laboratory (UL) or approved equal.
- B. Standards: Reference to the following standards infers that installation, equipment and material shall be within the limits for which it was designed, tested and approved, in conformance with the current publications and standards of the following organizations:
 - 1. American National Standards Institute - ANSI.
 - 2. American Society for Testing and Materials - ASTM.
 - 3. Institute of Electrical and Electronics Engineers - IEEE.
 - 4. Insulated Cable Engineers Association - ICEA.
 - 5. National Electrical Manufacturers' Association - NEMA.
 - 6. National Fire Protection Association - NFPA.

1.03 DEFINITIONS

- A. "Accessible" means arranged so that an appropriately dressed man, 6 feet-2 inches tall, weighing 250 pounds, may approach the area in question with the tools and products necessary for the work intended and may then position himself to properly and safely perform the task to be accomplished, without disassembly or damage to the surrounding installation.
- B. "Authority Having Jurisdiction" is the individual official, board, department, or agency established and authorized by the political subdivision created by law to administer and enforce the provisions of the Code as adopted or amended.
- C. "As Specified" denotes a product, system, or installation that:
 - 1. Includes all of the salient characteristics identified in the Drawings and Specifications;
 - 2. Meets all of the requirements of the "Basis of Design"; and
 - 3. Is produced by a manufacturer listed as acceptable on the Drawings or in the Specifications.
- D. "Basis of Design" refers to products around which the design was prepared. Some or all of the particular characteristics of Basis of Design products may be critical to the fit or performance of

the completed installation. Such characteristics are often subtle. Where substitutions are made to products that are the Basis of Design, the Contractor is alerted that nominally acceptable substitutions may produce undesirable side effects such as switchboards that no longer fit the space due to increased product dimensions. The Contractor is responsible for resolving all impacts of substitutions. Approval of a substitution request does not relieve the Contractor of complying with the design intent and all Codes.

- E. "Contracting Agency" is the Owner as defined in the General Conditions of the Contract.
- F. "Demolish" means to permanently remove a component, equipment, or system and its appurtenances with no intent for reuse and to properly dispose of it.
- G. "Furnish" means to purchase material as shown and specified, and cart the material to an approved location at the site or elsewhere as noted or agreed to be installed by supporting crafts.
- H. "Install" means to set in place and connect, ready for use and in complete and properly operating finished condition, material that has been furnished.
- I. "Product" is a generic term that includes materials, equipment, fixtures and any physical item used on the project.
- J. "Provide" means furnish all products, labor, subcontracts, and appurtenances required and install to a complete and properly operating, finished condition.
- K. "Remove" means to remove a component, equipment, or system and its appurtenances and either store it for re-installation, reuse, or turn it over to the Contracting Agency.
- L. "Rough-in and Connect" means provide an appropriate system connection such as conduit with junction boxes, wiring, switches, disconnects, etc., and wiring connections. Equipment furnished is received, uncrated, assembled, and set in place under the Division in which it is specified.
- M. "Serviceable" means arranged so that the component or product in question may be properly removed, and replaced without disassembly, destruction or damage to the surrounding installation. "Serviceable" components shall be "accessible".
- N. "Shop Drawings" are dimensioned working construction drawings drawn to scale to show an entire area of work in sufficient detail to demonstrate service and maintenance clearances and complete coordination of all trades.
- O. "Substitution" is a product, system or installation that is not by a listed manufacturer or does not conform to all salient characteristics identified in the Contract Documents, but which the Contractor warrants meets all specific requirements listed in the Contract Documents.
- P. "System Drawing" is a diagrammatic engineered drawing that shows the interconnection and relationship between products to demonstrate how the products interact to accomplish the function intended. Examples of system drawings include control and instrumentation diagrams, and wiring diagrams. Some drawings, such as dimensioned and complete Fire Suppression Drawings may be both System Drawings and Shop Drawings.

1.04 PERFORMANCE REQUIREMENTS

- A. Provide labor, products and services required for the complete installation, checkout and startup of electrical systems shown and specified. Where the work of several crafts is involved, coordinate related work to provide each system in complete and in proper operating order.
- B. Lay out the work in advance and avoid conflict with other work in progress. Physical dimensions shall be determined from existing conditions. Verify locations for junction boxes; disconnect switches, stub-ups, etc., for connection to equipment furnished by others, or in other Divisions of this Work.
- C. Cooperate with others involved in the project, with due regard to their work, to promote rapid completion of the entire project.

- D. Coordinate installation of conduit, junction boxes, system components, and other products to provide proper service areas and access for items requiring periodic maintenance inspection or replacement.
- E. Reference to a specific manufacturer's product (even as "Basis of Design") does not necessarily establish acceptability of that product without regard to compliance with all other provisions of these specifications.
- F. Local Conditions: The Contractor shall thoroughly familiarize himself with the work as well as the local conditions under which the work is to be performed. Schedule work with regard to seasons, weather, climatic conditions and other local conditions which may affect the progress and quality of the work.

1.05 SUBMITTALS

- A. Refer to Division 1 for general submittal requirements.
- B. In addition to Division 1 general submittal requirements, prepare Divisions Division 26 submittals in accordance with the following, and supplemental requirements listed within the specific specification section.
- C. The Contracting Agency's obligation to review submittals and to return them in a timely manner is conditioned upon the prior review and approval of the submittals by the Contractor as required by the Construction Contract.
- D. Submittal review is for general design and arrangement only and does not relieve the Contractor from any of the requirements of the Project Manual.
 - 1. Submittals will not be checked for quantity, dimension, fit, or for proper technical design of manufactured equipment.
 - 2. Provision of a complete and satisfactory working installation is the responsibility of the Contractor.
- E. Furnish suppliers with the applicable sections of the Project Manual and review and verify that the suppliers' submittals clearly represent products which comply with the Project Manual.
- F. Coordination
 - 1. Prior to submittal submission to the Contracting Agency for approval, coordinate with applicable subcontractors for conflicts, clearances, physical space allocation and routing, discrepancies, dimensional errors, omissions, contradictions, departures from the Contract requirements, correct electrical/mechanical services and connections, and provisions for commissioning.
 - 2. Review, revise, correct, and appropriately annotate submittals prior to submission for approval.
- G. Product Submittals
 - 1. Provide electronic submittals in PDF format.
 - 2. Reduce PDF file size by removing data and file creation elements not needed for final file presentation. Coordinate maximum file size with the Contracting Agency if necessary.
 - 3. Create PDF documents without security, to be searchable, and to allow for copy and paste functions.
 - 4. For scanned documents, run the optical character recognition (OCR) function to ensure the document is searchable and can be copied and pasted.
- H. Product Data
 - 1. General
 - a. Prepare electrical product submittals in accordance with Division 1, the general requirements of this section and the additional specific requirements of each individual specification section.
 - b. Submittals not provided as specified shall be rejected without review. This procedure is designed to accelerate and improve the accuracy of the technical review process, as well as simplify the future preparation of the IO&Ms.

- c. Product data for each specification section shall be submitted as a complete package, except as noted in this section.
- 2. Submittal Organization
 - a. Organize product submittal information in the same order as the products are specified within each specification section.
 - b. Provide a separate electronic file folder under each section number and name (e.g., 260529 –Hangers and Supports for Electrical Systems).
 - c. Within each section file, organize product cutsheet information in the same order as the products are specified in Part 2 of each applicable specification section.
 - d. Provide electronic bookmarks within each section for the first page of each separate product submittal. Provide product paragraph number/title for each bookmark.
 - e. Use PDF editing tools to clearly indicate the salient characteristics of each product and neatly cross out non-applicable information.
 - f. If a particular specified product is being omitted from the product submittal or will not be used for the project, provide a single page file identifying the product and annotated with a brief reason why the product is not being submitted (e.g., “NOT USED,” “NO SUBMITTAL REQUIRED,” “TO BE SUBMITTED BY (PROVIDE DATE),” etc.) This will inform the submittal reviewer that the product was not unintentionally omitted.
- 3. Product Information
 - a. Indicate manufacturer's name and address, and local supplier's name, address, phone number.
 - b. Indicate each product as “Basis of Design,” “Specified Equal” or “Proposed Substitution.”
 - c. Identify catalog designation and/or model number.
 - d. Provide manufacturer's product literature. Neatly annotate to indicate specified salient features, appurtenances and performance criteria for each product specified to demonstrate compliance with the Project Manual to include scheduled information, drawing information and specified information.
 - e. Indicate product deviations from the Project Manual and mark out non-applicable items on generic “cut sheets.”
 - f. Provide basic manufacturer's installation instructions.
- 4. Product Substitutions
 - a. Clearly indicate both in the section table of contents and on the individual product submittal information each proposed substitution, deviation or change from the product as described in the Project Manual.
 - b. Submittal approval does not include substitutions, deviations, or changes from the requirements of the Project Manual unless they are specifically itemized and approved. The term “No Exceptions Taken” will not apply to substitutions, deviations or changes not clearly identified.
 - c. Provision of a satisfactory working installation of equal quality to the system as described in the Project Manual shall be the responsibility of the Contractor.
 - d. Correct unapproved deviations from the Project Manual discovered in the field as directed by and at no additional cost to the Contracting Agency.
 - e. Cost of any design modifications because of proposed product substitutions shall be borne by the Contractor.
- I. Test and Evaluation Reports
 - 1. Review the submittal requirements for Test and Evaluation Reports for each Division 26 specification section.
 - 2. Submit copies of reports as specified. Also include these reports within the Installation, Operations and Maintenance (IO&M) Manuals as determined by the Contracting Agency.
- J. Installation, Operations and Maintenance (IO&M) Manuals
 - 1. Review the submittal requirements for IO&M manuals for each Division 26 specification section.

2. Begin the preparation of the IO&M manuals with a complete and fully approved set of product data submittals organized, annotated and with the product information as indicated within the "Product Data" submittals article above and in each Division 26 section.
3. Next, augment each individual product submittal with the written installation, operations, and maintenance information for each approved product. This type of information is not applicable (or available) for bulk commodity or simplistic products such as copper pipe, basic pipe hangers or equipment tags, etc.
4. Annotate the installation, operations, and maintenance information to indicate applicable information for the specific equipment model(s) installed.
5. Maintenance information shall include:
 - a. Preventive maintenance requirements for each product, including the recommended frequency of performing each preventive maintenance task.
 - b. Instructions for troubleshooting, minor repair and adjustments required for preventive maintenance routines, limited to repairs and adjustments that may be performed without special tools or test equipment and that require no extensive special training or skills.
 - c. Information of a maintenance nature covering warranty items, etc., that have not been discussed in the manufacturers' literature.
 - d. Information on the spare and replacement parts for each product and system. Properly identify each part by part number and manufacturer.
 - e. Recommended spare parts list.
6. Organize the IO&M manual information files by specification section (not by sub-contractor) with bookmarks for each section. Provide bookmarks for each section (and subsection as applicable) within the IO&M. Label bookmarks in a similar format as used for product submittals.
7. Provide a cover page for the IO&M manual volume, to include:
 - a. The Contracting Agency Name.
 - b. Project Name.
 - c. "Electrical Installation, Operations and Maintenance Manual."
8. Provide a table of contents which lists the Divisions specification sections and the Part 2 products in the same order as the applicable specification section.
9. Submit copies of Operation and Maintenance Manuals in electronic PDF format.

1.06 QUALITY ASSURANCE

- A. Qualifications: Perform the Work using qualified workmen that are experienced and usually employed in the trade.
- B. Product Testing and Certification:
 1. Nationally Recognized Testing Laboratory (NRTL) Labeling: Electrical equipment and conductors shall be "Approved," "Certified," "Identified," or "Listed" and "Labeled" to establish that the electrical equipment is safe, free of electrical shock and fire hazard, and suitable for the purpose for which it is intended to be used. The manufacturer shall have the specific authorization of one of the Occupational Safety and Health Administration (OSHA) approved Nationally Recognized Testing Laboratories (NRTLs) in accordance with the applicable national standards to label the equipment as suitable.
 2. Further details on the specific NRTLs, as well as the product standards that they are specifically recognized to evaluate equipment in accordance with, can be found on the OSHA Web site: <http://www.osha.gov/dts/otpca/nrtl/>
- C. Drawings and Specifications:
 1. The Drawings and specifications are complementary. Do not scale the Drawings. Locations of devices, fixtures, and equipment are approximate unless dimensioned.
 2. The Drawings are partly diagrammatic and do not show precise routing of conduits or exact location of all products, and may not show in minute detail all features of the installation; however, provide all systems complete and in proper operating order.

3. Drawing symbols used for basic materials, equipment and methods are commonly used by the industry. Special items are identified by a supplementary list of graphical illustrations, or called for on the Drawings or in the specifications.

1.07 SPARE PARTS

- A. Furnish spare parts for systems and equipment as listed in applicable sections of Division 26.
- B. Clearly label each part with name, manufacturers part number, system and/or equipment where used and location.
- C. Deliver parts to location and person designated by the Contracting Agency, in durable storage boxes.
- D. Group cartons containing smaller items by system or application and deliver in an appropriate number of storage boxes.

1.08 WARRANTY

- A. Warranty work shall be promptly coordinated and performed at the Contractor's sole expense. Workmanship, labor and materials (without limitation) in this Division shall be warranted for the longer of the following:
 1. As called for in the General Conditions of the Contract.
 2. For a minimum period of one year from the date of final acceptance.
 3. For the extended warranty period specified in a specific Section under this Division.
- B. Where a specific product carries a longer warranty as a standard offering of its manufacturer, extended warranty coverage beyond these requirements shall be retained by the Owner. The Owner will have recourse back to the manufacturer only in these cases, when the warranty as specified in A above has expired.

PART 2 PRODUCTS

2.01 MATERIALS AND EQUIPMENT FURNISHED IN DIVISION 26

- A. Materials furnished and installed in permanent construction shall be new, full-weight, standard in every way, and in first class condition.
- B. Materials shall conform to the standards of an organization acceptable to the Authority Having Jurisdiction and concerned with product evaluation that maintains periodic inspection of labeled equipment or materials and by whose labeling the manufacturer indicates compliance with appropriate standards or performance in a specified manner. Only materials designed for the purpose employed shall be used.
- C. Materials shall be identical with apparatus or equipment that has been in successful operation for at least two years. Materials of similar class or service shall be of one manufacturer.
- D. Capacities, sizes, and dimensions given are minimums unless otherwise indicated. Systems, materials and equipment proposed for use on this project shall be subject to review for adequacy and compliance with Contract Documents.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Skilled craftsmen shall install materials and equipment. The norms for execution of the work shall be in conformity with NEC Chapter 3 and the National Electrical Contractors' Association "National Electrical Installation Standards", which herewith is made part of these specifications.
- B. Repair surfaces and furnish all required material and labor to maintain fireproof, airtight and waterproof characteristics of the construction.
- C. Installation of equipment shall be in accordance with manufacturers' instructions.
- D. Coordinate exact device elevation with architectural elements including casework, equipment, etc.

3.02 CUTTING & PATCHING

- A. Obtain written permission of the Contracting Agency before cutting or piercing structural members.
- B. Floor penetrations shall be in accordance with Section 260529 - Hangers and Supports.
- C. Wall penetrations through exterior walls are not permitted.

3.03 PROTECTIVE FINISHES

- A. Take care not to scratch or deface factory finish of electrical apparatus and devices. Repaint all marred or scratched surfaces.
- B. Provide hot dip galvanized components for ferrous materials exposed to the weather.

3.04 SEPARATION OF SYSTEMS

- A. Conductors and equipment of different voltage levels, frequency, current characteristics (AC & DC) or functions (normal vs. emergency, etc.) shall not share the same raceways or enclosures unless specifically shown on the Drawings or approved by the Contracting Agency, or inherently necessary for correct system function (i.e., at transfer switches, transformers, etc.)

3.05 PROTECTION OF MATERIAL AND EQUIPMENT

- A. The Contractor shall be responsible for materials and equipment to be installed under this Contract. The Contractor shall make good at his own cost any damage which said materials or equipment may sustain from any cause before final acceptance.
- B. Cover and protect electrical equipment during construction from dust, dirt, debris, overspray, or other construction contaminates.

3.06 CLEANING AND REPAIR

- A. Throughout the work, the Contractor shall keep the work area reasonably neat and orderly by frequent periodic cleanups.
- B. Prior to substantial completion, clean equipment and systems used during construction.
- C. Repair surfaces damaged or impacted by the work. Restore to original condition or better. Retexture surfaces to match surrounding surfaces. Repaint affected surfaces, with extent of paint to include adjacent surfaces to next wall or other clean break to avoid mismatched finish.
- D. As independent parts of the installation are completed, they may be tested and utilized during construction.

END OF SECTION

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SECTION 260526
GROUNDING AND BONDING FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.01 SUMMARY

- A. This section includes general requirements, products and methods of execution relating to the furnishing and installation of a complete grounding system as required for this project.
- B. Related Sections:
 - 1. Section 264100 - Facility Lightning Protection

1.02 REFERENCES

- A. The publications listed below form a part of this specification. The publications are referred to in the text by basic designation only, latest edition.

NUMBER	TITLE
ANSI/IEEE C2	National Electrical Safety Code
ANSI/NFPA 70	National Electrical Code
ANSI/TIA/-606-C	Administration Standard for Commercial Telecommunications Infrastructure
ANSI/TIA-607-E	Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
IEEE C62.41	Recommended Practice on Surge Voltages in Low-Voltage Surge Protective Devices
IEEE C62.42	Guide for the Application of Gas Tube Arrester Low-Voltage Surge Protective Devices
IEEE Draft P1250 (D4)	Guide on Service to Equipment Sensitive to Momentary Voltage Disturbances
IEEE Std 1100	Recommended Practice for Powering and Grounding Sensitive Electronic Equipment
IEEE Std 142	Recommended Practice for Grounding of Industrial and Commercial Power Systems
IEEE STD 81	Recommended Guide for Measuring Ground Resistance and Potential Gradients in the Earth
NFPA 70	National Electric Code (NEC) - Codebook and Handbook
REA PE-33	(1985) Shield Bonding Connectors
UL 1449 Edition 3	Surge Protective Devices (SPDs)
UL 467 Edition 6	Grounding and Bonding Equipment
UL 497 Edition 5	Protectors for Paired Conductors for Communication Circuits
UL 497A Edition 1	Secondary Protectors for Communication Circuits
UL 497B Edition 1	Protectors for Data Communication and Fire Alarm Circuits

1.03 SUBMITTALS

- A. Provide submittals for products in accordance with Section 260000 - Electrical General Requirements and Division 1. Include copies of catalog cuts, data sheets and other descriptive information for all specified materials.

1.04 MINIMUM REQUIREMENTS

- A. The minimum requirements for the system shall conform to Article 250 of the NEC.
- B. The system shall comply with the requirements of the delegated design as detailed in section specification section 264100 Facility Lightning Protection.

1.05 SPECIAL REQUIREMENTS

- A. Unless specified elsewhere, the ohmic values for grounds and grounding systems from system to earth shall be as follows:
 - 1. For grounding metal enclosures and frames for electrical and electronically operated equipment -- 5 ohms maximum.
 - 2. For grounding systems to which electrical utilization equipment and appliances are connected -- 5 ohms maximum.
 - 3. For grounding secondary distribution systems, neutrals, noncurrent carrying metal parts associated with distribution systems, and enclosures of electrical equipment not normally within reach of other than authorized and qualified electrical operating and maintenance personnel -- 10 ohms maximum.

PART 2 PRODUCTS

2.01 EQUIPMENT

- A. Grounding conductors, ground rods, and equipment required for ground systems shall be listed for the purpose intended and approved by a Nationally Recognized Testing Laboratory (NRTL), and be in accordance with U.L. 467 and as follows:
 - 1. Ground rods shall be 3/4 inch by 10 foot copper bonded steel. Erico Eritech or approved equal.
 - 2. Ground busbar shall be minimum 1/4" x 4" x 20" copper mounted on fiberglass insulators. Burndy CB14420S or approved equal.
 - 3. Grounding conductors shall be copper unless otherwise required by lightning protection system design.

2.02 CONNECTIONS

- A. Joints in grounding conductors and mats below grade shall be made with exothermic welding process or hydraulically crimped fittings listed for direct burial. Terminations above grade shall be made with solderless lugs, securely bolted in place.
- B. Clamps, lugs, connectors, bonding bushings, and other such grounding and bonding items shall be:
 - 1. Labeled or listed for the purpose.
 - 2. Shall be made (both body and hardware) of hot dip galvanized steel, bronze, or other corrosion resistant alloy (except bushing throats shall be plastic).
 - 3. Shall be the products of O-Z/Gedney, T & B, Raco, or accepted equals.
 - 4. In outdoor, damp, or corrosive environments, metals for these items shall be copper (with or without tin-plating), bronze, or other corrosion resistant alloys only; O-Z/Gedney or accepted equal.

PART 3 EXECUTION

3.01 ELECTRICAL GROUND

- A. Create an equipotential plane for the grounding system for this project near the photovoltaic system inverter by connecting the following to the ground bus:
 - 1. Structural Steel.
 - 2. Grounds specified to be installed.
 - 3. Grounds required to be installed by the Facility Lightning Protection system design.
 - 4. Other items or equipment called for on the Drawings.
- B. Current carrying capacity of the grounding and bonding conductors shall be in conformity with Table 250.66 of the NEC.

- C. Measure resistance to earth of service grounds by the fall of potential method per IEEE STD 81 "Recommended Guide for Measuring Ground Resistance and Potential Gradients in the Earth". Record and submit test readings. Grounding system resistance to ground shall not exceed 5 ohms. Make any modifications or additions to the grounding electrode system necessary for compliance without additional cost. Final tests shall ensure that this requirement is met.

3.02 EQUIPMENT GROUND

- A. The raceway system shall be bonded in conformity with NEC requirements to provide a continuous ground path. Where required by Code or Ordinance or where called for on the plans an additional grounding conductor shall be provided, sized in conformity with Table 250.122 of the NEC, unless larger size is noted.
- B. Provide separate grounding conductor securely bonded and effectively grounded to the enclosures at both ends of all non-metallic raceways and all flexible conduit.

3.03 CONCEALED CONNECTIONS

- A. Permanent grounding connections, where permitted by the NEC to be concealed, shall not be so concealed until inspected and accepted by the Contracting Agency. Failure to comply with this requirement shall make the Contractor liable for all expenses incurred in the process of re-exposing the connections for inspection, and subsequent repair and patching of the concealing construction, including the work of other trades. The Contractor shall schedule inspection of such connections at least one work week in advance of concealment, and shall not be entitled to any additional compensation or time extension for delays caused by inability of the Contracting Agency's representative to be available at the desired time.

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SECTION 260529
HANGERS AND SUPPORTS FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.01 SUMMARY

- A. This Section includes:
 - 1. General hanger and support requirements for electrical equipment, conduit and cable trays not required to be vibration and/or seismically controlled.
 - 2. Penetrations, sleeves and seals.
- B. Related Sections:
 - 1. 260000 - Electrical General Requirements
 - 2. 260533 - Raceways and Boxes for Electrical Systems

1.02 REFERENCES

- A. NFPA 70: National Electrical Code (NEC) latest legally enacted edition.

1.03 DESCRIPTION

- A. Provide general hanger, support and seismic support requirements for electrical equipment, conduit and cable trays and other nonstructural equipment in accordance with the manufacture's written installation instructions and NFPA 70.

1.04 SUBMITTALS

- A. See Section 260000 - General Electrical Requirements for general submittal requirements
- B. Product Data:
 - 1. Provide manufacturers catalog data for each product specified. Indicate channel gauge and maximum load capacities of the selected products.
 - 2. Manufacturer's Installation Instructions: Include assembly instructions, recommended parts and special procedures as required.
- C. Design Requirements:
 - 1. Obtain the services of a specialized and qualified design firm (Seismic Design Firm) to design the overall vibration and seismic control restraint systems for the non-structural elements of this Project in accordance with the International Building Code (IBC) chapters 16 and 17.
 - 2. Duties of the Seismic Design Firm include:
 - a. Specialized design, product review, and product selection.
 - b. Site verification and written certification that the installed vibration and seismic control and related products and systems meet the specialized design requirements.
 - 3. Exceptions.
 - a. Pre-engineered seismically designed and certified assemblies in accordance with IBC, accompanied with written "Certificate of Compliance" acceptable to the Authority Having Jurisdiction, AHJ.
 - b. Pre-manufactured equipment designed in accordance with IBC accompanied with written "Certificate of Compliance" acceptable to the Authority Having Jurisdiction.
- D. Shop Drawings:
 - 1. Provide shop drawings to include the following:
 - a. Pre-engineered and field fabricated support system details for each installation location. To include but not limited to:
 - 1). Raceway and lighting fixture support.
 - 2). Conduit and control panel support.
 - 3). Trapeze hangers.
 - 4). Electrical equipment support.
 - b. Indicate hanger type/attachment method and hanger spacing intervals.

1.05 DELIVERY, STORAGE, AND HANDLING

- A. Acceptance at Site:
 - 1. Verify products are delivered in original factory packaging and are free from damage and corrosion.
 - 2. Replace equipment delivered to job site that does not comply with above requirements at no expense to the Owner.
- B. Storage and Protection:
 - 1. Handle items to avoid damage.
 - 2. Replace damaged items with same item in new condition.

1.06 WARRANTY

- A. Provide warranty in accordance with Section 260000 - General Electrical Requirements.

PART 2 PRODUCTS

2.01 PRE-ENGINEERED SUPPORT SYSTEMS

- A. Manufacturers:
 - 1. Unistrut
 - 2. Super-Strut
 - 3. B-Line
 - 4. K-Line
 - 5. Erico.
 - 6. Approved equal.
- B. Material:
 - 1. Cold worked steel.
 - 2. Type 304 stainless steel: Use for liquid-tight flex installed in outdoor, damp, corrosive or marine environments.
- C. Finish:
 - 1. Heated indoor areas: Pre-galvanized zinc coating.
 - 2. Outdoor areas: Hot dipped galvanized finish. In addition, coat hot dipped galvanized finish channel field cuts with zinc rich paint provided by the support system manufacturer.
- D. Channel:
 - 1. Standard Size: 1-5/8 inch x 1-5/8 inch. Gauge thickness as required for attached load.
 - 2. Standard Hole Pattern: Slotted. Provide solid channel in exposed public areas.
- E. Nuts and Hardware:
 - 1. Channel nuts: Hardened steel (ASTM-A675 and ASTM A36).
 - 2. Bolts, screws and nuts: Hardened steel (ASTM-A307, ASTM A563 and SAE J429).
 - 3. Finish: Electroplated zinc.
- F. Fittings: Plate steel (ASTM A635). Epoxy or electroplated zinc coating.
- G. Electrical Accessories: Provide accessories from the support system manufacturer designed for the specific equipment to be supported to include but not limited to:
 - 1. Snap-in closures.
 - 2. Conduit connection plates.
 - 3. Junction box adapters.
 - 4. Strut joiners.

PART 3 EXECUTION

3.01 GENERAL INSTALLATION

- A. Install hangers and supports in accordance with manufacturer's instructions, applicable Code requirements (NFPA 70) and approved shop drawings.

3.02 VIBRATION AND SEISMIC CONTROL PRODUCT INSTALLATION

- A. Install vibration isolators, seismic control and wind restraint systems in strict compliance with the manufacturer's written instructions and certified and approved application engineering installation drawings and details.

3.03 ATTACHMENT INSTALLATION

- A. Attach electrical equipment to structure as follows:
 - 1. Hollow masonry: Toggle bolts.
 - 2. Solid masonry and concrete: Preset inserts or expansion bolts.
 - 3. Structural steel: Beam clamps which engage both sides of structural member or have retaining clips or other approved means for positive engagement.
 - 4. Metal surfaces: Machine screws, bolts or welding.
 - 5. Wood construction: Wood or sheet metal screws. Bugle head drywall screws or deck screws are not allowed.
 - 6. Do not use powder actuated fasteners for anchorage in tension applications. Obtain written permission from the Owner prior to using any type of powder powered studs.
 - 7. Attachment to plaster or gypsum board (sheet rock) not approved. Equipment shall be attached to or supported from structure.

3.04 PENETRATIONS

- A. Coordinate electrical penetrations with architectural, structural and mechanical construction details prior to installation.
- B. Provide compatible materials, fasteners, adhesives, sealants, and other products required for proper installation.
- C. Penetrations through roof and floor shall be weather and watertight.
- D. Penetration of exterior walls is not allowed.
- E. Escutcheons: Provide escutcheons for conduit passing through walls, floors and ceilings in finished areas, below counters and inside closets and casework subject to view when doors are open. Size escutcheons to cover conduits. Secure escutcheons in position.

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SECTION 260533
RACEWAY AND BOXES FOR ELECTRICAL SYSTEMS

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes specific requirements, products, and methods of execution relating to conduit, conduit fittings, surface raceways, multi-outlet assemblies, wireways, outlet boxes, pull boxes and junction boxes approved for use on this project. Type, size and installation methods shall be as shown on Drawings, required by Code and/or specified in this Section.
- B. Related Sections
 - 1. 260526 – Grounding and Bonding for Electrical Systems
 - 2. 260529 – Hangers and Supports for Electrical Systems

1.02 REFERENCES

- A. American National Standards Institute/Underwriters Laboratory
 - 1. ANSI C80.1 – Electrical Rigid Steel Conduit
 - 2. ANSI/UL 1 – Flexible Metal Conduit
 - 3. ANSI/UL 6 – Electrical Rigid Metal Conduit – Steel
 - 4. UL 6A – Standard for Electrical Rigid Metal Conduit – Aluminum and Stainless Steel
 - 5. UL 360 – Standard for Liquid Tight Flexible Steel Conduit
 - 6. UL 514A – Metallic Outlet Boxes
 - 7. UL 514B – Conduit, Tubing and Cable Fittings
- B. National Electrical Manufacturers Association
 - 1. NEMA 250 – Enclosures for Electrical Equipment (1000 Volts Maximum)
 - 2. NEMA FB 1 – Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing and Cable
 - 3. NEMA RN 1 – Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
 - 4. NEMA WD 6 - Wiring Device Configurations.
- C. NECA (National Electrical Contractors Association) Standard of Installation.

1.03 SUBMITTALS

- A. Provide submittals for products in accordance with Section 260000 - Electrical General Requirements and Division 1.
- B. Product Data: Provide dimensions, knockout sizes and locations, materials, fabrication details, surface raceway finishes (custom factory pre-painting, color as selected by architect), and accessories.
- C. Manufacturer's Instructions: Indicate application conditions and limitations of use stipulated by Product testing agency specified under Regulatory Requirements. Include instructions for storage, handling, protection, examination, preparation, and installation of Product.

1.04 QUALITY ASSURANCE

- A. Raceways and boxes shall be standard types and sizes as manufactured by a nationally recognized manufacturer of this type of materials and be in conformity with applicable standards and UL listings.
- B. Surface raceways shall be of the latest approved design as manufactured by a nationally recognized manufacturer and shall be listed by the Underwriters' Laboratory and bear the UL label.
- C. Pull and junction boxes 50 cubic inches and smaller shall conform to specifications for outlet boxes.
- D. Pull and junction boxes larger than 50 cubic inches shall conform to U.L. Standard 50, Cabinets and Boxes.

- E. Perform work in accordance with NECA Standard of Installation.
- F. Furnish products listed and classified by Underwriters Laboratories, Inc. as suitable for purpose specified and shown.

PART 2 PRODUCTS

2.01 CONDUIT

- A. Conduit types specifically approved for use on this project shall be of the following types only:
 - 1. Galvanized rigid metal conduit - GRC or RMC.
 - 2. Extreme temperature liquid-tight flexible steel conduit - AT: Shall have temperature rating of -67 ° F to +220 ° F, Liqueflex "ATLA", or as approved.
 - 3. Types specifically identified on the Drawings or in the Specifications
 - 4. Other products not specifically approved such as EMT, IMC, PVC, ENT, MC Cable, etc., are not allowed.

2.02 CONDUIT FITTINGS

- A. Fittings utilized with rigid steel shall be galvanized steel and shall be threaded. Conduit bushings shall be provided and shall be of the insulated types. Where grounding bushings are required, provide insulated grounding bushings with integral pressure type ground lugs, Thomas & Betts "Blackjack", or as approved.
- B. Fittings for liquid-tight flexible conduit shall be steel or malleable iron, of a type incorporating a threaded grounding cone, nylon or plastic compression ring, and a tightening gland, providing a low resistance ground connection. Throats shall be insulated.

2.03 CAST BOXES

- A. Provide cast boxes with threaded hubs, external mounting brackets or holes, and gasketed covers.

2.04 INDOOR PULL AND JUNCTION BOXES

- A. Indoor pull and junction boxes shall conform to Article 314 of the NEC and the following requirements:
 - 1. Boxes installed in wet locations shall be NEMA 3R, unless otherwise noted.
 - 2. Furnish such boxes, whether shown or not, in order to conform to requirements for maximum pulling length and maximum number of bends allowed.
 - 3. Special boxes, as noted on the Drawings, shall be installed in areas of specific service and/or hazards.
- B. Junction box extension rings will not be accepted on new boxes. Appropriate size boxes shall be used for each application.

2.05 OUTDOOR ABOVE-GROUND PULL AND JUNCTION BOXES

- A. Boxes exposed to rain or installed in wet locations shall be NEMA 3R unless otherwise noted.
- B. Outdoor pull and junction boxes and conduit bodies for use with galvanized conduits shall be made of galvanized ferrous metal or cast aluminum, with integral threaded hubs or Myers-type weathertight hubs of matching composition and finish.
- C. Furnish such boxes, whether shown or not, in order to conform to requirements for maximum pulling length and maximum number of bends allowed.

PART 3 EXECUTION

3.01 CONDUIT USES PERMITTED

- A. Unless otherwise noted, conduits installed in the following locations shall be of the types specifically identified only:
 - 1. Underground or encased in concrete – RMC.
 - 2. Outdoors aboveground or damp locations – RMC or Extreme temperature liquid-tight flexible steel conduit where shown on the drawings.
 - 3. Dry indoor locations, concealed or exposed - RMC.

3.02 RACEWAY INSTALLATION METHODS - GENERAL

- A. Concealed raceways: In occupied areas, conduit and raceways shall be concealed unless specifically noted otherwise.
- B. Concealed raceways shall be routed as directly as possible with a minimum of bends.
- C. Exposed Raceways shall comply with the following:
 - 1. Exposed raceways shall be run parallel or perpendicular to building lines and bent symmetrically or made up with standard elbows or fittings.
 - 2. Connectors and fittings for raceways and conduits installed on the surface in exterior locations shall be suitable for and Listed for use in a wet location.
- D. There shall not be more than the equivalent of four quarter bends (360 degrees total) between pull points. Pull boxes added to conduit runs as a result of this requirement shall be in accordance with this Section.
- E. Each conduit shall enter and be securely connected to a cabinet, junction box, pull box or outlet box by means of a locknut on the outside and a locknut/bushing on the inside, or by means of a liquid-tight, threaded, self-locking, cold-weld type wedge adapter. Connections shall be made wrench tight. Locknuts shall be the bonding type with sharp edges and shall be installed in a manner that will assure a locking installation. Locknuts and bushings or self-locking adapters will not be required where conduits are screwed into threaded connections. Conduit runs shall be protected from the entrance of foreign material prior to the installation of conductors.
- F. Pull wires shall be provided in spare and unused conduits. (Nylon "jet-line" or as approved.)
- G. Conduits stubbed up out of floor and terminating inside of an enclosure shall have insulating grounding bushings installed.
- H. Raceways penetrating vapor barriers or traversing from warm to cold areas shall be sealed on the inside with a non-hardening duct sealing compound to prevent the accumulation of moisture, and shall be taped airtight to the vapor barrier on the outside. Refer to Section 26 0000 for additional requirements and limitations regarding penetration of vapor barriers.

3.03 CONDUIT SIZES – GENERAL

- A. Minimum size for RMC conduits shall be ¾ inch.
- B. Minimum size for flexible conduits shall be ¾ inch.

3.04 STRUCTURAL COORDINATION

- A. Structural members shall not be cut, drilled, or notched for raceways or other electrical features unless specifically accepted by the Contracting Agency.

3.05 JUNCTION BOX AND PULL BOX INSTALLATION

- A. Junction and pull boxes shall be installed so that covers are readily accessible and adequate working clearance is maintained after completion of the installation.

END OF SECTION

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SECTION 264100
FACILITY LIGHTNING PROTECTION

PART 1 GENERAL

1.01 INTRODUCTION

- A. This Section specifies the lightning protection and grounding system for the building(s) or structure(s). This system provides facility protection for the building and occupants by preventing damage to the structure caused by lightning and induced transient currents. The design of this system is to be in strict accordance with this section of the specification and all contract drawings that apply.
- B. The work covered under this section of the specifications consists of furnishing labor, materials and engineering services required for the completion of a functional and unobtrusive lightning protection and facility grounding system approved by the architect and engineer.

1.02 SYSTEM DESCRIPTION

- A. The entire lightning protection system shall be designed and installed in accordance with:
 - 1. Underwriters' Laboratories, Inc. (U/L) Std. 96A
 - 2. National Electrical Code – NFPA 70 (NEC)

1.03 SUBMITTALS

- A. A complete shop drawing shall be submitted to the architect and engineer for approval prior to commencement of the installation. The shop drawing will show the extent of the system layout designed for the structure along with details of the products to be used in the installation.

1.04 QUALITY ASSURANCE

- A. The contractor shall furnish an LPI-IP Certificate or a UL Certificate upon completion of the installation.
- B. The System Design shall be completed and the shop drawing stamped by an LPI Certified Master Installer – Designer of Lightning Protection Systems.
- C. The installing contractor shall be listed with the Lightning Protection Institute, and Underwriters' Laboratories, Inc. The installation contractor shall have personnel on staff Certified by the LPI as a Master Installer – Designer of lightning protection systems. LPI qualified staff, Journeyman or higher, shall provide on-site supervision of the installation to the Standards.

PART 2 PRODUCTS

2.01 STANDARDS

- A. All materials shall comply in weight, size, and composition with the requirements of the UL 96 Materials Standards. All equipment shall be UL listed and properly labeled. The system furnished under this specification shall be the standard product of a manufacturer regularly engaged in the production of lightning protection equipment and a member of LPI. Equipment shall be the manufacturer's latest approved design of construction to suit the application where it is to be used in accordance with accepted industry standards and with NFPA, LPI, & UL requirements.
- B. Surge Suppression products for the electrical service entrance shall comply with both UL 1449 5th Edition.

2.02 ACCEPTABLE MANUFACTURERS

- A. VFC (www.vfclp.com)
- B. ABB / Furse (www.tnb.com)
- C. ERICO, Inc. (www.erico.com)
- D. Lyncole (www.lyncole.com)

2.03 MATERIALS

A. Lightning Protection System

1. Class I materials shall be used for systems on structures not exceeding 75 feet in height and Class II materials shall be used for systems on structures exceeding 75 feet above grade.
2. Copper shall be of the grade ordinarily required for commercial electrical work, generally designated as being 95 percent conductive when annealed. Aluminum conductors shall be of electrical grade aluminum.
3. Lightning protection materials shall be coordinated with building construction materials to assure compatibility. Aluminum lightning protection materials shall not be embedded in concrete or masonry, installed on or below copper surfaces, or used for the in-ground system. Copper lightning protection materials shall not be installed on aluminum surfaces or on exterior sheet metal surfaces. Copper system components within 2 feet of chimney exhausts shall be tin coated to protect against deterioration.
4. Strike termination devices shall be provided to place the entire structure under a zone of protection as defined by the Standards. Air terminals shall project a minimum of 10 inches above protected areas or objects. Air terminals shall be located within 2 feet of exposed corners and roof edges.
5. Metallic bodies having a thickness 3/16" or greater may serve as strike termination devices without the addition of air terminals. These bodies shall be made a part of the lightning protection system by connection(s) according to the Standards using main size conductors and bonding fittings with 3 square inches of surface contact area.
6. Cable conductors shall provide a two-way path from strike termination devices horizontally and downward to connections with the ground system.
7. Cable conductors shall be free of excessive splices and sharp bends. No bend of a conductor shall form a final included angle of less than 90 degrees nor have a radius of bend less than 8 inches. Structural elements and design features shall be used whenever possible to minimize the visual impact of exposed conductors.
8. Cable down conductors may be concealed within the building construction or enclosed within PVC conduit from roof to grade level. Down conductors shall be spaced at intervals averaging not more than 100 feet around the protected perimeter of the structure. In no case shall any structure have fewer than two down conductors. Where down conductors are exposed to environmental hazards at grade level, guards shall be used to protect the conductor to a point 6 feet above grade.
9. In the case of structural steel frame construction, cable down conductors may be omitted and roof conductors shall be connected to the structural steel frame at intervals averaging not more than 100 feet around the protected perimeter of the structure.
10. Exposed cable conductors shall be secured to the structure at intervals not exceeding 3 feet – 0 inches.
11. Connectors and splicers shall be of suitable configuration and type for the intended application and of the same material as the conductors or of electrolytic compatible materials.
12. Ground terminations suitable for the soil conditions shall be provided for each down conductor. Where the structural steel framework is utilized as main conductors for the system, perimeter columns shall be connected to the grounding system at intervals averaging 60 feet or less on the protected perimeter. For any structure in excess of 60 ft. in vertical elevation above grade, a ground loop interconnecting all ground terminals and other building grounded systems shall be provided.
13. Common interconnection of all grounded systems within the building shall be accomplished using main size conductors and fittings. Grounded metal bodies located within the calculated bonding distance as determined by the formulas of the Standards shall be bonded to the system using properly sized bonding conductors.

- B. Surge Suppression
 1. Surge Suppression units shall be provided at every system entrance to the structure to prevent massive lightning overvoltage's from entering the structure.
 2. Surge Suppression units may be built directly into the switchgear by the switchgear manufacturer. The unit used by the switchgear manufacturer must meet the minimum requirements set out by UL 1449 5th edition type 1 or type 2 units.
 3. If the Surge Suppression unit is not built into the switchgear the unit must meet UL 1449 5th edition type 1 or type 2. The unit must be installed as per the manufacturer's instructions and the NEC.
 4. Surge Suppression products for the communication service shall be installed as per the manufacturer's requirements.

PART 3 EXECUTION

3.01 STANDARD

- A. The installation shall comply with the requirements of UL 96A, and the NEC.

3.02 ACCEPTABLE DESIGN AND INSTALLATION FIRMS

- A. VFC Lightning Protection www.vfclp.com. 800-825-1948 Or equal with the following requirements:
 1. 5 Years Minimum experience as a Lightning Protection Installation firm.
 2. LPI Certified Master Designer on staff at the Lightning Protection Installation firm.
 3. LPI Master and Journeyman installers on site to complete actual installation.
 4. Participate in UL's "Alternate Quality Management System Inspection Program"

3.03 INSTALLATION

- A. Lightning Protection
 1. The installation of the lightning protection system components shall be done in a neat and workmanlike manner.
 2. Roof penetrations required for down conductors or for connections to structural steel framework shall be made using through-roof assemblies with solid bars and appropriate roof flashings. The roofing contractor shall furnish the methods and materials required at roofing penetrations of the lightning protection components and any additional roofing materials or preparations required by the roofing manufacturer for lightning conductor runs to assure compatibility with the warranty for the roof.
 3. (Note: The roofing contractor will be responsible for sealing and flashing all lightning protection roof penetrations as per the roof manufacturer's recommendations.)
 4. If the protected structure is an addition to or is attached to an existing structure that does not have a lightning protection system, the contractor shall certify that the system installed complies with the requirements of the Standards, and advise the owner of the lightning protection work required on the existing structure to obtain full certification for the structure. If the existing structure does have a lightning protection system, the contractor shall advise the owner of any additional work required on the existing system to bring it into compliance with current Standards and thus qualify for LPI or U/L certification.
- B. Surge Suppression Device
 1. At locations indicated on drawing(s), install Surge Protection Devices in full accordance with manufacturer's written instructions and comply with all applicable codes.
 2. At Distribution, MCC and Branch Panels, Surge Protection Devices with a UL 1449 Edition 5, Listing as a SPD Type 2 shall have a 30-Amp Circuit Breaker or other size as recommended by the manufacturer's installation manuals. This independent Circuit Breaker will serve as a means of disconnect for servicing the Surge Protection Device with the protected panel remaining energized.
 3. Surge Protection Devices with a UL 1449 5th Edition Listing as a SPD Type 1, and an integrated disconnect can be connected directly to the buss without a designated circuit breaker.

4. The Surge Protection Device shall be installed with the shortest possible leads, or wire length maximum 18".

3.04 FINAL SYSTEM INSPECTION AND QUALITY CONTROL

- A. The contractor shall furnish an LPI-IP Certificate or a UL Certificate upon completion of the installation.
- B. LPI certification requires a signature by a representative of the owner at multiple stages of installation & by their third party field staff. UL certification requires inspection by their third-party field staff after completion of the installation.
- C. As-Built Drawings shall be completed and stamped by an LPI Certified Master Designer - Installer of Lightning Protection Systems.
- D. Final Inspection Report – A final test and inspection report shall be completed based on ANSI/TIA/EIA 607, NEC, and UL96A industry standards as applicable. The scope of the inspection and report shall include;
 1. Test and evaluation of the grounding system.
 2. Record final systems to ground resistance level.
 3. Evaluation and Testing of the internal bonding and grounding systems.
 4. Evaluation and Testing of equipment grounding.
 5. Evaluation of AC surge suppression installation.
 6. Evaluation of telco surge suppression installation.
 7. Copy of the LPI-IP or UL Lightning Protection Certification.
 8. Final As-Built Review and submission.
- E. Report shall include detailed reporting and test results with corresponding photos of each evaluation category.

END OF SECTION

**SECTION 312000
EARTH MOVING**

PART 1 GENERAL

1.01 SUMMARY

- A. This section consists of all operations pertaining to excavation, backfill, and placement of classified earthwork fill materials.
- B. The latest revision of the following standards of the American Society for Testing and Materials (ASTM) and the American Association for State Highway and Transportation Officials (AASHTO) are hereby made part of this Specification. The publications may be referred to in the text by basic designation only.

ASTM D422	Test for Particle Size Analysis of Soil
ASTM D424	TEST FOR PLASTIC LIMIT AND PLASTICITY INDEX OF SOILS
ASTM D2216	TEST FOR WATER CONTENT OF SOILS
AASHTO T-85	TEST FOR SPECIFIC GRAVITY OF SOILS
ASTM D1556	TEST METHODS FOR MOISTURE –DENSITY RELATIONS OF SOIL AND SOIL AGGREGATE MIXTURES, USING 5.5 LB RAMMER AND 12-IN DROP
ASTM D4254	TEST METHODS FOR MINIMUM INDEX DENSITY OF SOILS AND CALCULATION OF RELATIVE DENSITY
ASTM D4253	TEST METHODS FOR MINIMUM INDEX DENSITY OF SOILS AND CALCULATION OF RELATIVE DENSITY

1.03 OSHA STANDARDS

- A. The Contractor's attention is directed to the latest provisions of Subpart P, Section 1926 of the OSHA Safety and Health Standards for Construction.

1.04 SUBMITTALS

- A. Submit sieve analysis reports under provisions of Section 013323, Submittal Procedures.

1.05 TESTING

- A. Contractor shall be responsible for performing quality control and testing of all materials including, but not limited to, gradation and compaction requirements.
- B. Fill Materials: The Contractor shall provide particle size analysis (gradation test) for each classified fill borrow source according to testing frequency table below.
 - a. Sand Blanket
 - b. Gabion Basket Rock

PART 2 PRODUCTS

2.01 RECLAIMED GRAVEL SURFACING FILL AND BACKFILL:

- A. Sourced from the project site, and contains no muck, roots, frozen material, peat, organics, or other frost susceptible aggregate materials.

2.02 SAND BLANKET:

- A. Sand containing no muck, frozen material, roots, sod, or other deleterious material with a plasticity index not greater than 6 as determined by ASTM D4318.
- B. Meet the following gradation requirements as determined by ASTM D6913:

SEIVE	PERCENT PASSING BY WEIGHT
3/8-inch	100
No. 4	95-100
No. 200	0-6

2.03 GABION BASKET ROCK:

- A. Furnish angular stone from a rock quarry or cut that is hard, durable , free of organic and spoil material, and resistant to weathering and water action. Do not use crushed river or rock with rounded surfaces. Conform to the following:
 - a. Durability Index (Course), AASHTO T 210 50 min
 - b. Density of a filled basket 100 lb/ft³ min
 - c. Gradation: Furnish rock with breadth and thickness at least one-third its length

PART 3 EXECUTION

3.01 SHEETING AND BRACING OF EXCAVATION:

- A. Furnish, install, and maintain such sheeting and bracing as may be required to support the sides of excavations to prevent any movement which could in any way diminish the width of the excavation below that necessary for proper construction, and to protect adjacent structures from undermining or other damage. Prevent voids outside of the sheeting, but if voids are formed, they shall be immediately filled and rammed.
- B. Leave in place, to be embedded in the backfill, all sheeting and bracing which the Contracting Officer may require in writing, to be left in place at any time during the progress of the Work for the purpose of preventing injury to structures, utilities, or property, whether public or private. The Contracting Officer may direct that the sheeting and bracing be cut off at any specified elevation.
- C. All sheeting and bracing not left in place shall be carefully removed in such a manner as to not endanger construction or other structures, utilities, or property. All voids left or caused by withdrawal of sheeting shall be immediately refilled with sand by ramming with tools especially adapted to that purpose, by watering or otherwise as may be directed.

3.02 PREPARATION:

- A. Preparation: All depressions or holes below the general area surface level, whether caused by removal of debris or unacceptable material, or otherwise, shall be backfilled with approved material and compacted to specified density and to a level, uniform surface before the placement of other layers.

3.03 EXCAVATION:

- A. No blasting is permitted.
- B. All excavation equipment shall be capable of mobilization to the site by light-lift or medium-lift helicopter.
- C. Excavations shall be reasonably smooth and uniform to the lines, grades and cross sections shown in the Drawings or as directed by the Contracting Officer. Excavations shall be conducted to ensure that material outside of excavation limits remains undisturbed
- D. Conform to the excavation limits as shown in the Drawings. Correct unauthorized excavation or over-excavated areas.
- E. Stockpile reusable excavated material and remove excess material not being re-used on-site. Contractor shall protect all stockpiles from wind and erosion using protective covering material. The Contractor shall conduct all operations to prevent contaminating useable excavation with unsuitable material.

3.04 DISPOSAL SITES:

- A. Dispose of excess excavation spoils at an on-site location as directed by the Contracting Officer. Disposal sites shall be located and maintained in such a manner as to prevent a public nuisance.

3.05 EMBANKMENTS:

- A. Embankments shall be constructed to a reasonably smooth and uniform shape conforming to the lines, grades and cross sections indicated on the Drawings or as directed by the Contracting Officer.
- B. When embankment is to be placed on hillsides steeper than a 4:1 slope, new embankment is to be placed alongside existing embankments, or embankments are to be built half width at a time; the foundation shall first be prepared by constructing benches of sufficient width to accommodate placing and compacting equipment. Each bench shall begin at the intersection of the original ground and the vertical side of the previous cut. Material so excavated and suitable for embankment construction shall be incorporated into the new embankment. Benching is incidental to other items in the contract and no direct payment will be made therefore.
- C. Embankments over soft ground may be constructed by end dumping an initial lift of sufficient depth to support hauling and spreading equipment.
- D. Embankments shall meet the requirements as defined in Part 3.03 Excavation.

3.06 GRAVEL AGGREGATE COURSES

- A. Relates to execution of work required for placement of reuseable gravel fill materials sourced from site excavations.
- B. Any ruts or soft yielding spots that may appear shall be corrected by loosening and removing unsatisfactory material and adding approved material as required, reshaping, and recompacting the affected areas to the lines and grades indicated on the plans.

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- C. The maximum compacted thickness of any one layer shall not exceed six inches. If the required compacted depth exceeds six inches, the base shall be constructed in two or more layers of approximately equal thickness. Each layer shall be shaped and compacted before the succeeding layer is placed.
- D. The finished surface of the aggregate course, when tested using a 10-foot straightedge, shall not show any deviation in excess of 3/8-inch between two contact points. The finish surface shall not vary more than 1/2-inch from established grade.
- E. Feather finish grades to match adjacent existing topography.

3.07 COMPACTION STANDARDS

- A. Continuous visual observation method: Compaction in lieu of specific moisture and density control is acceptable per the following requirements:
 - A. A minimum of 8 passes in each direction (16 total passes) with hand operated tampers.
 - B. The compacted soil shall not exhibit visually apparent evidence of yield of pumping. Compact each layer until the embankment does not rut or deform under loads provided by excavation or compaction equipment.

3.08 FIELD QUALITY CONTROL

- A. Contractor shall be responsible for all quality control testing, inspection equipment, and inspection personnel. Notify Contracting Officer at least 48 hours in advance to allow NPS personnel opportunity to observe inspections and testing of subgrade and excavations prior to placement of gravel backfills and gabion basket units.

END OF SECTION

SECTION 313600
GABIONS

PART 1 GENERAL

1.01 SUMMARY

- A. Product and execution requirements for installation of steel wire-mesh gabion basket units for both foundation at retaining walls. SHIPPING, HANDLING, AND STORAGE
- B. All products shall be pre-staged or stored on the project site per gabion basket manufacturer recommendations.
- C. Mobilization of gabion basket units, associated stones aggregates, and miscellaneous gabion hardware the Camp Muir project site shall comply with all requirements of the General Conditions.

PART 2 PRODUCTS

- A. Gabion Basket Mesh Materials:
 - 1. Use a 9 ga (W1.7 x W1.7) galvanized wire fabric conforming to ASTM A-185. Make a weld as each connection that has minimum shear strength of 800 pounds per foot.
 - 2. A minimum galvanized coating of 0.90 oz/ft² shall be applied, conforming to the requirements of 055000 Metal Fabrications.
 - 3. Use mesh with 4-inch openings in the longest dimension.
 - 4. Use wire that is designed to be nonraveling. It must resist pulling apart at any of the connections forming the mesh when a single wire strand in a section of mesh is cut.
- B. Gabion Permanent Fasteners:
 - 1. Lacing Wire:
 - a. Furnish nominal-sized 13.5 ga diameter wire of the same type, strength, and coating as the gabion basket mesh.
 - 2. Spiral Binders:
 - a. Form with wire having at least the same diameter, type, strength, and coating as the gabion basket mesh.
 - 3. Internal Connecting Wire:
 - a. Furnish galvanized pre-formed stiffeners (9 ga minimum), as required.
 - b. Overlapping ring wire fasteners:
 - (a) Manufacture overlapping ring wire fasteners (hog rings) from 0.12 inch galvanized wire of high tensile strength equal to 260,000 to 280,000 psi. Galvanizing shall conform to ASTM A641, Class 3 coating,
 - (b) When baskets closed, the number of wires confined, the fee ends of the ring fastener shall overlap a minimum of 1-inch. The pull-apart resistance obtained along the longest axis of the ring shall be 600 pounds per foot.
- C. Gabion Permanent Fasteners:
 - 1. Lacing Wire:
 - a. Furnish nominal-sized 13.5 ga diameter wire of the same type, strength, and coating as the gabion basket mesh.
 - 2. Spiral Binders:
 - a. Form with wire having at least the same diameter, type, strength, and coating as the gabion basket mesh.
 - 3. Internal Connecting Wire:
 - a. Furnish galvanized pre-formed stiffeners (9 ga minimum), as required.
 - b. Overlapping ring wire fasteners:
 - (a) Manufacture overlapping ring wire fasteners (hog rings) from 0.12 inch galvanized wire of high tensile strength equal to 260,000 to 280,000 psi. Galvanizing shall conform to ASTM A641, Class 3 coating,

- (b) When baskets closed, the number of wires confined, the free ends of the ring fastener shall overlap a minimum of 1-inch. The pull-apart resistance obtained along the longest axis of the ring shall be 600 pounds per foot.
- D. Gabion Rock:
 - 1. Per Section 312000 Earth Moving
- E. Geotextile Separation Fabric:
 - 1. Meet AASHTO M 288 for Separation, except provide a minimum permittivity of 0.50 sec-1, and meet Class 3 Strength Property Requirements.
- F. Sand Blanket:
 - 1. Per Section 312000 Earth Moving
- G. Rigid Insulation Board:
 - 1. Meet AASHTO M 230, Type VI, extrusion is not required, and the maximum water absorption is 0.3 percent by volume, as determined by ASTM C272. Insulation board must meet or exceed the minimum thickness called out in the plans, and have a 20-year warranted thermal resistance (R-Value) @ 75°F of 4.5 per inch of thickness as determined by ASTM C177 or ASTM C518. The required thickness is shown on the Plans and is actual thickness, not nominal thickness.
- H. Geogrid Cellular Pavers:
 - 1. Rigid HDPE permeable paver matting with 2-inch depth cellular and 4-inch maximum cell length and width.
 - 2. Paver units shall be capable of interconnection between mat sections for joining with connection points located
 - 3. Color: Black with carbon additive for long-term UV stabilization.
- I. Rock Bolt Assemblies:
 - 1. Reinforcing Bars (Dowels):
 - a. Deformed, Grade 60 bars conforming to AASHTO M 31 or AAAASHTO M 322.
 - b. Diameter: 1-inch
 - c. Embedment length: 5-foot minimum embedment.
 - 2. Bearing plates, washer, nuts:
 - a. Conform to ASTM F432
 - 3. Nonshrink Grout:
 - a. Conform to ASTM C1107. Provide a minimum compressive strength of 5,000 pounds per square inch in 3 days.

PART 3 EXECUTION

- A. Basket Assembly:
 - 1. Furnish twisted wire or welded wire baskets. Do not damage wire coatings during basket assembly, structure erection, cell filling, or backfilling. Rotate the basket panels into position and join the vertical edges with fasteners according to Subsection 3.B. Temporary fasteners may be used for basket assembly if they are supplemented during structure erection with permanent fasteners according to Subsection 3.B.

Rotate the diaphragms into position and join the vertical edges according to Subsection 3.B.
- B. Structure Erection:
 - 1. Place the empty gabion baskets on the foundation and interconnect the adjacent baskets along the top and vertical edges using permanent fasteners.
 - 2. Where lacing wire is used, wrap the wire with alternating single and double loops every other mesh opening and not more than 6 inches apart. Where spiral binders are used, crimp the ends to secure the binders in place. Where alternate fasteners are used, space the fasteners in every mesh opening and not more than 6 inches apart.

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3. Interconnect each vertical layer of baskets to the underlying layer of baskets along the front, back, and sides. Stagger the vertical joints between baskets of adjacent rows and layers by at least one-half the cell length.
- C. Gabion Basket Unit Filling:
1. Remove kinks and folds in the wire mesh, and properly align the baskets. Place rock carefully in the basket cells to prevent bulging of the baskets and to minimize voids in the rock fill. Maintain the basket alignment.
 2. Place stiffeners in each unrestrained exterior basket cell greater than 12 inches in height. This includes interior basket cells left temporarily unrestrained. Place stiffeners concurrently with rock placement.
 3. Fill the cells in a row or layer to ensure no cell is filled more than 12 inches above an adjacent cell. Repeat this process until the basket is full and the lid bears on the final rock layer.
 4. Secure the lid to the sides, ends, and diaphragms according to Subsection 3.B. Make exposed basket surfaces smooth and neat with no sharp rock edges projecting through the wire mesh.
- D. Gabion Basket Backfilling:
1. Place a geotextile filter over the back face of the gabion structure. Concurrently with the cell filling operation, backfill the area behind the gabion structure in accordance with Section 312000. Compact each layer according to the requirements of Section 312000 , except use an acceptable lightweight mechanical or vibratory compactor within 36 inches of the gabion structure.
- E. Insulation Board:
1. Prior to placing the insulation board, grade, shape, and compact the area per Section 312000.
 2. Place Sand Blanket leveling course at least two inches thick below and above insulation board. Finish the leveling course surface so it does not vary more than 0.10 foot when tested using a ten-foot straightedge.
 3. Set each board accurately to the line and grade established.
 4. Place insulation to the required thickness, using a minimum of two layers. Stagger all joints between layers.
- F. Geotextile Separation Fabric:
1. Place and cut geotextile within gabion basket units as shown on the Drawings. No geotextile fabric shall be visible through the exposed face of gabion unit.
- G. Geogrid Cellular Pavers:
1. Place cellular pavement units within the gabion basket units and over geotextile separation fabric as shown on the Drawings.
 2. Cut cellular paver mat units to fit completely within the gabion basket unit and such that edge of the mat is cut back 6 inches from the visible face of the gabion unit. Cut matting units such that no individual mat shall have length or width less than 12 inches after cutting. No cellular paver matting shall be visible through the exposed face of gabion unit.
 3. Interlock cellular mat units per manufacturer's installation instructions.
 4. Fill cellular paver cell units with Reclaimed Gravel and compact to form a visually uniform gravel surface over the matting. Reclaimed gravel shall be placed to an adequate thickness such that no portion of the cellular pavement unit is visible through the gabion.
 5. Close and fasten gabion basket top over the cellular paving matting and compacted reclaimed gravel surfacing.

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H. Rock Bolt Installation:

1. Drilling: Provide equipment capable of drilling straight and uniform-diameter holes. Do not use water or drilling slurry. Drill rock bolt and dowel hole collars and hole depths within 6 inches of the spacing and locations shown on the Drawings. Drill holes to achieve specified minimum rock bolt embedment depth. Size drill holes to provide at least 0.5-inches of grout cover over the bond zone portion of the dowel. Clean drill hole of cuttings or other debris.
2. Grouting:
 - a. Install the rock bolt within 24 hours of completing drill hole.
 - b. Use positive displacement grout pump to perform grout injection. Perform grout injection per manufacturer's recommendations. Place grout in one continuous operation. Control grout pressures to avoid ground heaving of fractures.
3. Install bearing plate, washer, and nut.
 - a. Install the bearing plate so that it is contact with the gabion basket mesh material.
 - b. When grout has reached final set per manufacturer's recommendations or a minimum compressive strength of 60 percent of the 7-day compressive strength, install the bearing plate, washer and nut and apply a torque of 100 foot-pounds to ensure proper seating.
 - c. Provide at least 4 inches of dowel length beyond the top of nut.

END OF SECTION